

Huawei Access Certification Training

# HCIA-Access

# Lab Guide for Access Engineers

ISSUE:2.5



HUAWEI TECHNOLOGIES CO., LTD.

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# About This Document

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## Overview

This document is a training course for HCIA-Access certification. It is intended for trainees who are about to take the HCIA-Access exam or who want to learn about the basic knowledge of access networks, xPON/xDSL technologies, protocols such as DHCP/PPPoE, access networking, features of Huawei access network products, basic device configurations, and data service configurations, this course is intended for readers of access network technologies, such as ODN basics and NMS basics.

## Description

This document consists of four modules, including six experiments. It describes the basic configurations of the OLT, ONU, GPON data service, and xDSL data service in the access networking.

- Module 1 is the basic operation of the xPON device (including experiment 1), which is mainly about the OLT initialization configuration. Through the basic operation and configuration, the readers can get familiar with the OLT and understand and master the related basic operation functions.
- Module 2 is the basic configuration of the xPON terminal (including experiment 2). This module describes how to create a profile on the OLT and add different types of terminals, helping readers to be familiar with and understand the process of adding terminals on the OLT and the common query methods.
- Module 3 is the GPON data service configuration (including experiment 3 and experiment 4), which describes the data service provisioning mode in the GPON access.
- Module 4 is the xDSL&Vectoring&G.fast data service configuration (including experiment 5 and experiment 6 and experiment 7), which describes the method of provisioning the data service in the xDSL device. Module 3 and module 4 describe how to provision data services in different scenarios. This chapter helps you to create and reference profiles and how to provision broadband data services in different scenarios.

## Background Knowledge Required

The intended audience should know basic WLAN knowledge, Huawei switching devices, and basic datacom knowledge. This course is a basic course for Huawei certification. To

better understand this document, the readers of this document must meet the following requirements:

- Have basic knowledge about access networks, be familiar with Huawei switches, and have basic knowledge about datacom.

## Common Icons



OLT



ONU or MDU



ONT



optical splitter



PC



Telephone set



Switch



Softswitch



BRAS/Router

## Experiment Environment Overview

### Networking Introduction

This lab environment is intended for access network engineers who are preparing for the HCIA-Access exam. Each lab environment consists of one OLT, 2 to 16 ONUs, one core switch, one eSight server, one AR router (simulating the BRAS to terminate PPPoE dialup and function as the network gateway), one to two optical splitter, and several PCs. Each lab environment is suitable for 1 to 16 trainees.

### Version Description

This document is also compatible with the MA5680T, MA5800, and EA5800. The CLI and NMS operations in this document are based on the MA5680T. If the MA5800 or EA5800 is used, perform the operations according to the instructions of the trainer.

### Device Introduction

To meet the HCIA-Access lab requirements, you are advised to configure each lab environment as follows:

| Device Name      | Device Model                                | Software Version                                                                                    |
|------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Core switch      | S3700-28TP-PWR-EI<br>or<br>S5700-28C-PWR-EI | Version 5.70 (S3700 V100R005C01SPC100) or later<br>Version 5.130 (S5700 V200R003C00SPC300) or later |
| OLT device       | MA5680T or<br>MA5800 or EA5800              | MA5680T: V800R013C00 and later/EA5800: V100R019C20                                                  |
| ONT device       | HG8247H9<br>(EG8145V5)                      | AP4030DN V200R007C10SPC100                                                                          |
| ONU device       | MA5822S、MA5616<br>(P612E/ P805E)            | V800R018C00                                                                                         |
| optical splitter | 1:16 or higher; 2:16<br>or higher           |                                                                                                     |
| NMS server       | eSight PON/NCE-<br>FAN                      | V300R010C00/R19C00                                                                                  |
| AR router        | AR1220                                      | V200R005                                                                                            |

## Experiment Environment Preparation

### Checking Whether All Devices Are Available

Before starting the experiment, check whether all required devices are ready. The following table lists the required devices.

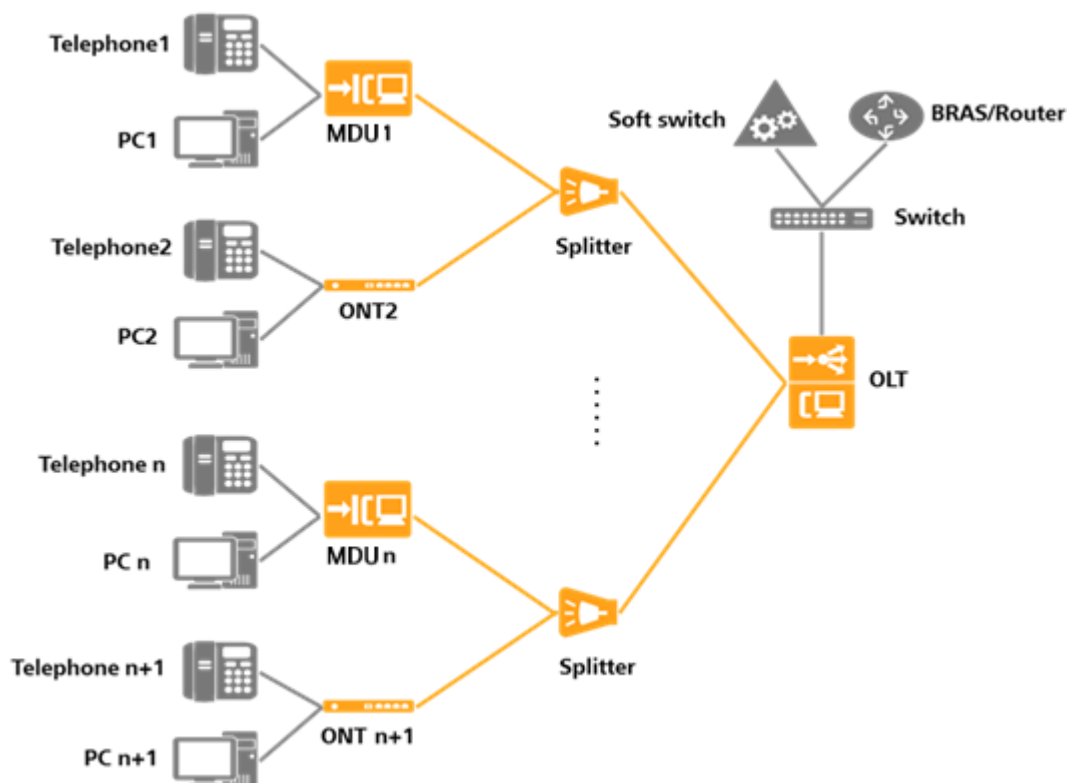
| Device Name                           | Quantity             | Remarks                           |
|---------------------------------------|----------------------|-----------------------------------|
| eSight&NCE-FAN                        | 1 PCS                | Shared by all experimental groups |
| Huawei S3700<br>or Huawei 5700 switch | All groups share one | Shared by all experimental groups |
| AR1220 router                         | 1 for all groups     | Shared by all experimental groups |
| OLT                                   | 1 for all groups     | Shared by all experimental groups |
| ONU                                   | Two for each group   | One desktop ONT and one ONU       |
| optical splitter                      | 1-2                  | Shared by all experimental groups |

|                              |                    |                                                    |
|------------------------------|--------------------|----------------------------------------------------|
|                              |                    | Ensure that the split ratio is proper.             |
| Notebook or desktop computer | One for each group | A desktop computer must have two network adapters. |
| Twisted pair&Modem           | One for each group | Used for data service tests                        |
| Console cable                | One for each group | Alternatively, use Telnet.                         |

Each group must check whether the following devices are ready:

- Prepare one desktop ONT and one ONU.
- Prepare one laptop or desktop computer.
- One twisted pair cable.
- One console cable (or a remote Telnet device)

## Experiment Topology



## Clearing OLT Configurations

Previous configurations may affect experiment results. Therefore, clear all OLT configurations before shutting down the OLTs. Before starting a new experiment, ensure that the OLTs do not have configurations. If the devices have previous configurations, clear the configurations and restart the OLTs.

### NOTE

- If you log in to an OLT in Telnet mode, do not perform this operation. Otherwise, the original account data of the OLT will be lost, and the login will fail.

To log in to the OLT, you must enter the user name and password. In this experiment, the default user name and password are **root** and **admin** respectively.

```
>>User name:root
>>User password:
Huawei Integrated Access Software (MA5600T).
Copyright(C) Huawei Technologies Co., Ltd. 2002-2013. All rights reserved.
-----
User last login information:
-----
Access Type : Serial
IP-Address : --
Login Time : 2019-04-17 10:30:32+08:00
```

Logout Time : 2019-04-17 11:00:51+08:00

-----  
All user fail login information:

| Access Type | IP-Address     | Time                      | Login Times |
|-------------|----------------|---------------------------|-------------|
| Telnet      | 10.175.195.116 | 2019-04-17 10:46:06+08:00 | 1           |
| Serial      | --             | 2019-04-17 10:30:28+08:00 | 1           |

-----  
User root has used a default password. Change the password in time.  
huawei>

To clear the configurations, run the following command:

```
huawei>enable
huawei#
huawei#erase
{ flash<K> }:flash
{ data<K> }:data
{ Options<E><active,standby> }:active
Command:
    erase flash data active
This command will clear the active board data that has been loaded or saved
Please remember to backup the system configuration data
Are you sure to continue? (y/n)[n]:y
Command executes successfully, and will take effect after active board
rebooted
```

To restart an OLT, run the following command:

```
huawei#reboot
{ Options<E><system,active,standby> }:active
Command:
    reboot active
Please check whether data has saved, the unsaved data may lose if reboot
active board, are you sure to reboot active board? (y/n)[n]:y
Standby board failure or not exist, reboot active will cause system reboot,
are you sure to reboot active board? (y/n)[n]:y
huawei#
Platform System start ...
Memory Data Bus Test .....pass.
Memory Address Bus Test .....pass.
Press 'Ctrl+T' To Start Memory Complete Test
Trace code : 0x00000000
Copy from ROM to RAM ..... OK!
Uncompress from ROM to RAM ..... OK!
Update fpga...OK.
Copyright (c) 1998 - 2013 by Huawei Technologies Co.,Ltd.
All Rights Reserved.
The last update date of base BIOS B is : Jan 15 2013   16:39:10
=====
BOARD INFORMATION :
```

CPU: MPC8548E - Security Engine. Processor #1.(Rev 2.02) running at 1333Mhz  
512M bytes DDR(ECC), 128M bytes flash installed on board  
Baudrate of serial channel is 9600bps  
Board's LAN IP address is 10.11.104.6  
Board's LAN Subnet mask is 255.255.255.0  
Board's LAN MAC address is 08:19:A6:C1:6A:52

```
=====
Base BIOS version is 307
Save extended BIOS disable start flag...OK!
Transferring control to the extended BIOS ...OK!
System Vfs init...OK
The last update date of extended BIOS is : Feb 28 2013 16:23:52
System is booting from extended BIOS...
Extended BIOS version is 311
Press <D> key to stop auto-boot      0
Now system will boot from flash memory.
Testing Flash memory...OK!
Select area information : Program Area A ,Data Area A .
Begin to expand program... OK!
Save extended BIOS enable start flag...OK!
Transferring control to the loaded program...OK!
Starting system application init.....successfully!
Starting system configuration data init...successfully!
Press any key to get started
>>User name:
```

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# 1 Basic Operations on xPON Devices

## 1.1 Experiment Introduction

### 1.1.1 About This Experiment

This guide describes the basic operations, system operations, and network management configurations of the GPON broadband access device. After studying this chapter, you can understand how to maintain and manage an OLT.

### 1.1.2 Objectives

- Understand environment setup.
- Be familiar with the command line mode and features.
- Master system user management methods.
- Be familiar with device management.

### 1.1.3 Networking

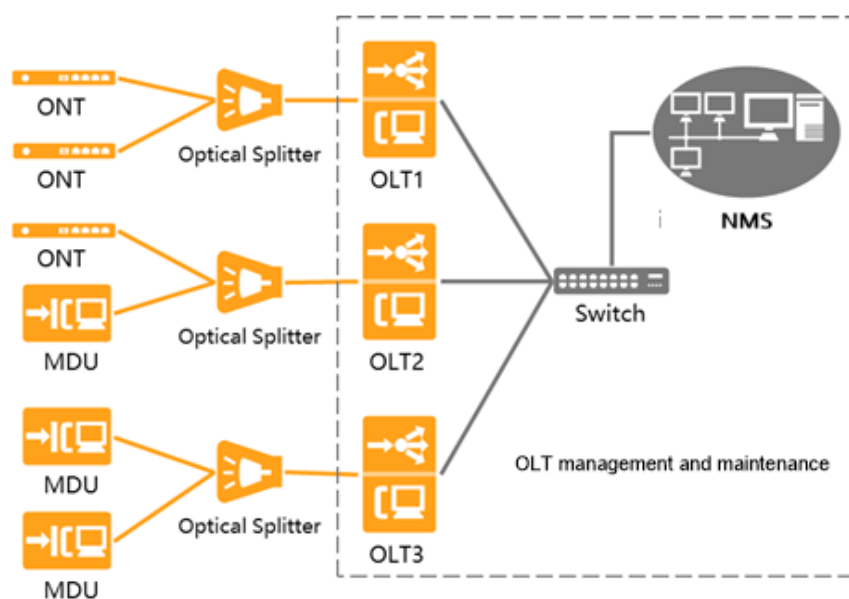
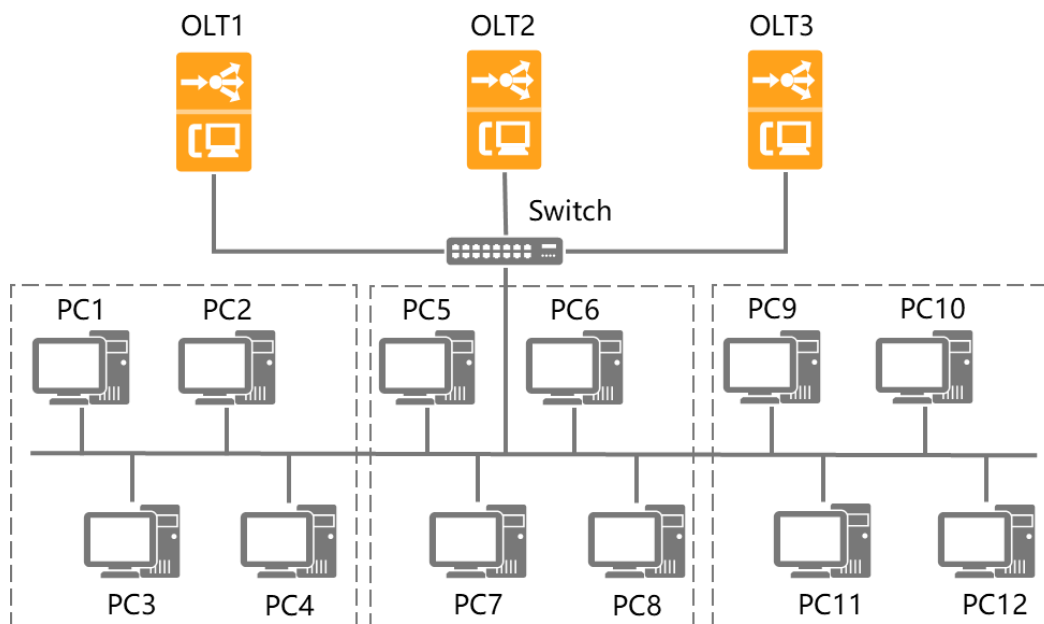


Figure 1-1 GPON maintenance environment topology

This experiment aims to perform basic operations on OLTs. PCs are used to log in to and manage the OLTs in outband management mode. Figure 1-2 shows the layout of the involved devices.



**Figure 1-2 Layout of the devices involved in an experiment for xPON basic operations**

## 1.1.4 Data Plan

Device maintenance environment:

According to Figure 1-2, find your group, specify your group members and required devices, write down the management IP addresses of the devices, and log in to the devices.

ETH: outband maintenance IP address

OLT1 (for users 1–4)

IP:

User name:

Password:

OLT2 (for users 5–8)

IP:

User name:

Password:

OLT3 (for users 9–12)

IP:

User name:

Password:

When creating a terminal user and a user profile, you are advised to name the profile and user by your own name. This avoids name conflicts and facilitates the query and deletion after the configuration.

User profile name: \_\_\_\_\_

User name:

Password:

## 1.2 Configuration Environment Setup

### 1.2.1 Serial Port Maintenance Mode

#### 1.2.1.1 Connection Topology

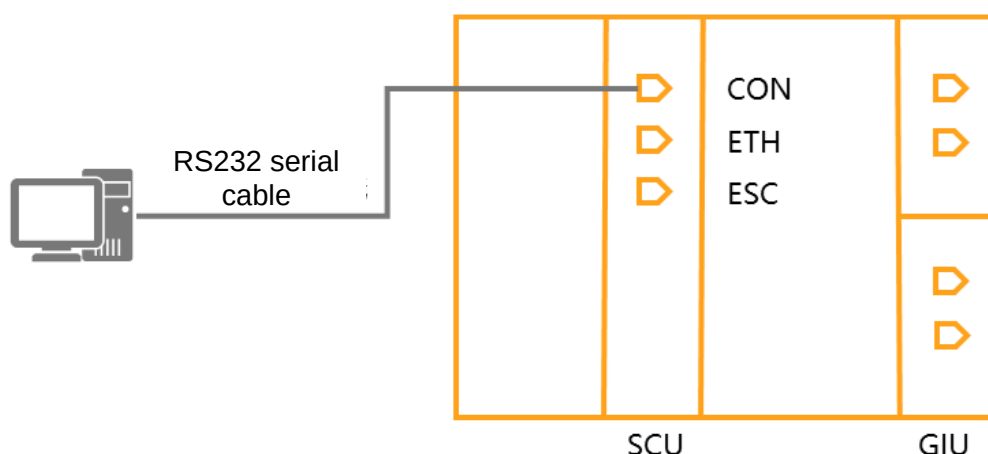


Figure 1-3 Topology diagram in serial port maintenance mode

#### 1.2.1.2 Configuration Procedure

Set up physical connections.

Use an RS232 serial cable to connect the serial port of a PC to the serial port of a control board on an OLT.

Set up a connection through CRT and set parameters.

Start the CRT program (or HyperTerminal) on the PC. Double-click the CRT program, choose **File > Connect in Tab**, and click the icon of **New Session**. In the dialog box that is displayed, select **Serial** from the **Protocol** drop-down list, select the COM port, retain the default baud rate 9600 bit/s, and click **OK**.

Log in to the OLT.

Enter the user name and password as prompted. The default user name and password are **root** and **admin** respectively.

Press **Enter**. You can successfully log in to the OLT if the login information is correct.

## 1.2.2 Outband Maintenance Mode

### 1.2.2.1 Connection Topology

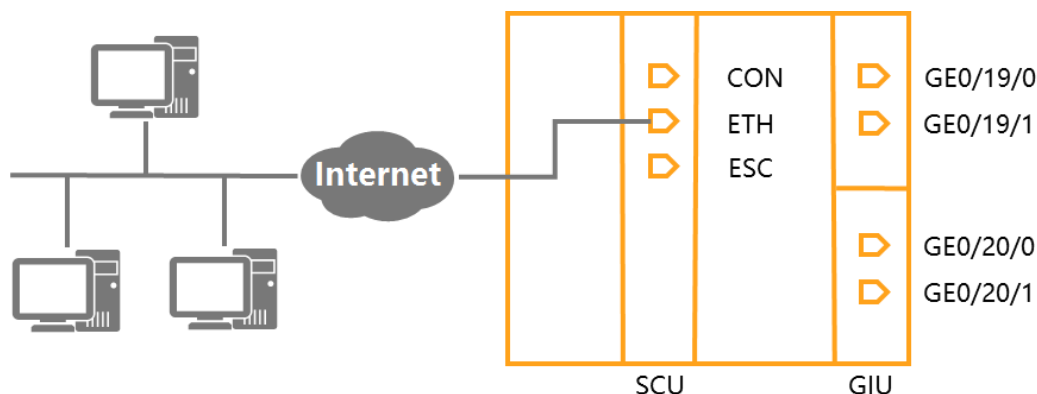


Figure 1-4 Topology for outband maintenance

### 1.2.2.2 Configuration Procedure

#### Step 1 Configure management ports.

After logging in to the OLT using a console port, check the IP address of the Meth 0 port. If the IP address does not exist, configure the IP address of the maintenance network, add an outband management route, save the configuration, and exit the login.

#### Step 2 Set up physical connections.

Use a network cable to connect the PC to the ETH management network port on the control board of the OLT.

#### Step 3 Test the network connectivity.

Open the command prompt on the PC, run the **ping** command to test the network connectivity between the PC and the ETH management network port of the OLT. If the ping operation fails, run the **ipconfig** command to check whether the IP address of the host is correct. If the IP address can be pinged, log in to the OLT in telnet mode.

#### Step 4 Log in to the OLT.

Start the CRT program on the PC, set up a telnet connection, enter the IP address of the ETH management network port in the telnet address bar, and enter the user name and password. Then, you can log in to the OLT and perform operations on the OLT.

## 1.2.3 Inband Maintenance Mode

### 1.2.3.1 Connection Topology

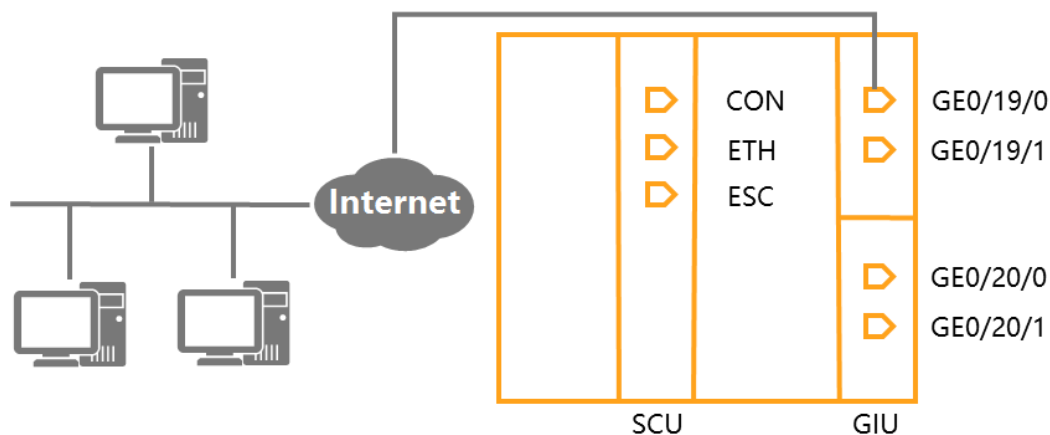


Figure 1-5 Topology for inband maintenance

### 1.2.3.2 Configuration Procedure

Step 1 Configure the management VLAN and management IP address on an OLT.

Log in to the OLT in serial port mode, and configure a VLAN port and IP address on the OLT for inband management. After the configuration, you can log in to the OLT using the IP address.

```
huawei(config)#vlan 500 smart
huawei(config)#port vlan 500 0/19 0
huawei(config)#interface vlanif 500
huawei(config-if-vlanif500)#ip address 10.175.192.36 22
```

Step 2 huawei(config-if-vlanif500)#quit

Step 3 Set up physical connections.

Use a network cable to connect the PC to the upstream service port of the upstream board on the OLT.

Step 4 Remotely log in to the OLT.

# Check the IP address and gateway address automatically obtained by the PC. Ensure that the IP address and the management IP address of the OLT are in the same network segment. In this experiment, the gateway address is 10.175.192.1.

# Run the **ping 10.175.192.36** command to test the connectivity between the PC and the OLT.

# Open the command prompt on the PC, run the **telnet 10.175.192.36** command to log in to the OLT, and enter the user name and password to log in to the OLT.

## 1.3 Exercise for Basic Operations

### 1.3.1 Task Requirements

Table 1-1 Requirements for experiment tasks

| Task                                          | Requirement                                                                                                        | Completion Status |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------|
| Be familiar with command lines.               | Be familiar with basic operation commands and query commands.                                                      |                   |
| Add a user profile.                           | The login is allowed only in the period of 08:00–18:00 p.m. The profile name is mandatory.                         |                   |
| Add a user.                                   | Add a user to use the added profile, and write down the user name.                                                 |                   |
| Query the version of the OLT.                 | Query the versions of the control board and service board.                                                         |                   |
| Query the upstream port of an upstream board. | Query the duplex mode and rate.                                                                                    |                   |
| Create a Layer 3 interface.                   | Create a VLAN, create a Layer 3 interface, and write down the IP address and MAC address of the Layer 3 interface. |                   |
| Delete a Layer 3 interface.                   | Delete the Layer 3 interface that you add.                                                                         |                   |

### 1.3.2 Login to an OLT and System Information Query

#### Step 1 Log in to an OLT.

In this experiment, the outband maintenance mode is adopted. A network cable is used to connect a PC and the OLT management port on the same local area network (LAN). Start the CRT program of a PC and connect a telnet connection.

# Choose **Connect > New Session**. On the page that is displayed, select the telnet protocol, click **Next**, enter the host name **10.175.192.36**, and click **Next** until the operation is complete.

# On the connection page, select the created session 10.175.192.36, click **Connect**, and enter the user name and password on the login page to log in to the OLT. (The user name and password are allocated based on the experiments. In this experiment, **root** and **admin** are used.)

```
>>User name:root
>>User password:      // The password is admin, which is not displayed.
    Huawei Integrated Access Software (MA5600T).
    Copyright(C) Huawei Technologies Co., Ltd. 2002-2013. All rights reserved.
    User root has used a default password. Change the password in time.
MA5600T>
```

## Step 2 Be familiar with basic command line operations.

Learn about the command line modes and understand how to switch the configuration modes.

# Enter privileged mode from user mode:

```
MA5600T>enable
MA5600T#
```

# Enter global config mode.

```
MA5600T#config
MA5600T(config)#
```

# Enter interface mode (for example, meth 0).

```
MA5600T(config)#interface meth 0
MA5600T(config-if-meth0)#
```

# Return to the upper mode from the current mode.

```
MA5600T(config-if-meth0)#quit
MA5600T(config)#quit
MA5600T#
```

# Exit privileged mode.

```
MA5600T#disable
MA5600T>
```

# Log out.

```
MA5600T#quit
```

Check whether system data has been changed. Please save data before logout.

Are you sure to log out? (y/n)[n]:y

## Step 3 Perform operations related to the system name and time.

Change the OLT name to **huawei**.

# Change the system name.

```
MA5600T#sysname huawei
huawei#
```

# Query the system language.

```
huawei#display language
Local:
  Description: CHINESE SIMPLIFIED (DEFAULT LANGUAGE)
  Version:    MA5600V800R013C00    // Query the NE software version information.
  Encoding:   GBK
General:
  Description: ENGLISH (DEFAULT LANGUAGE)
  Version:    MA5600V800R013C00
  Encoding:   ANSI
```

# Query system time.

```
huawei#display time
{ <cr>|date-format<K>|dst<K>|time-stamp<K> }:
Command:
  display time
2018-04-10 16:48:15+08:00
```

# Change the local system time.

```
huawei#time 16:00:00 2018-04-10
huawei#
```

#### Step 4 Perform operations related to subracks and boards.

Query the detailed information about subracks and boards.

# Query the subrack information. The subrack ID is 0.

```
huawei(config)#display frame info 0
-----
Type:      H801MABC
State:     Normal
Desc:      H801MABC_0
EMU ID:    0   Subnode:1   State:Communications normal
-----
```

// If 0 is not entered, the IDs, types, and status of all subracks in the system can be queried.

# Query the information about all boards in a subrack.

```
huawei(config)#display board 0
-----
SlotID BoardName Status      SubType0 SubType1 Online/Offline
-----
```

```

0
1
2
3
4
5      H802GPBD   Auto_find    // The board is in auto-find state and can be used only after
being confirmed.
6
7
8
9      H801SCUN   Active_normal // The board is in the Normal state.
10
11
12
13
14     H802EPBD   Auto_find
15
16
17
18
19     H801GICG   Auto_find
20
21
22
-----

```

After a board is inserted into a subrack, the OLT automatically detects the board. However, the board in the **Auto-find** state cannot provide services. You need to confirm the board and ensure that the board status is normal.

# Confirm all boards in the **Auto-find** state.

```
huawei(config)#board confirm 0
```

# Query the status of the confirmed boards.

```
huawei(config)#display board 0
```

```

-----
SlotID  BoardName  Status          SubType0 SubType1  Online/Offline
-----
0
1
2
3
4
5      H802GPBD   Normal          // The board is in the Normal state.
6
7
8
9      H801SCUN   Active_normal
10
11
12
13

```

```

14      H802EPBD  Normal
15
16
17
18
19      H801GICG  Normal
20
21
22

```

# Query the information about a board in the specified slot.

```
huawei(config)#display board 0/5
```

```

-----
Board Name       : H802GPBD
Board Status     : Normal
-----

```

```

-----
Power Status      Power-off cause                Power-off Time
-----

```

```

POWER-ON         -                             -
-----

```

| Port | Port type | min-distance (km) | max-distance (km) | Optical-module status |
|------|-----------|-------------------|-------------------|-----------------------|
| 0    | GPON      | 0                 | 20                | Online                |
| 1    | GPON      | 0                 | 20                | Online                |
| 2    | GPON      | 0                 | 20                | Offline               |
| 3    | GPON      | 0                 | 20                | Offline               |
| 4    | GPON      | 0                 | 20                | Offline               |
| 5    | GPON      | 0                 | 20                | Offline               |
| 6    | GPON      | 0                 | 20                | Offline               |
| 7    | GPON      | 0                 | 20                | Offline               |

```

-----
In port 0/ 5/0 , the total of ONTs are: 0, online: 0
In port 0/ 5/1 , the total of ONTs are: 0, online: 0
In port 0/ 5/2 , the total of ONTs are: 0, online: 0
In port 0/ 5/3 , the total of ONTs are: 0, online: 0
In port 0/ 5/4 , the total of ONTs are: 0, online: 0
In port 0/ 5/5 , the total of ONTs are: 0, online: 0
In port 0/ 5/6 , the total of ONTs are: 0, online: 0
In port 0/ 5/7 , the total of ONTs are: 0, online: 0

```



#### NOTE

GPON service boards are installed in slot 0/5. The information shows that the GPON 0 port used in this experiment is normal.

Step 5 Query system version information.

# Query the system software version.

```

huawei#config
huawei(config)#display version
{ <cr>|backplane<K>|frameid/slotid<S><Length 1-15> }:
Command:
    display version
VERSION : MA5600V800R013C00
PRODUCT : MA5600T
Active Mainboard Running Area Information:
-----
Current Program Area : Area A
Current Data Area : Area A
Program Area A Version : MA5600V800R013C00
Program Area B Version : MA5600V800R013C00
Data Area A Version : MA5600V800R013C00
Data Area B Version : MA5600V800R013C00
-----
Uptime is 0 day(s), 0 hour(s), 26 minute(s), 34 second(s)

```

# Query the software version of a service board.

```

huawei(config)#display version 0/5          // Query the software versions of the service boards in
slots 0/5.
Send message for inquiring board version successfully, board executing...
Main Board: H802GPBD
-----
Pcb    Version: H802GPBD VER B
Mab    Version: 0000
Logic Version: (U50)001(U25)008(U26)008(U49)023
Main CPU :
CPU    Version: (U48)MPC8349
APP    Version: 405(2013-4-25)
BIOS   Version: (U57)310

```

# Query the software version of a control board.

```

huawei(config)#display version 0/9
Main Board: H801SCUN
-----
PCB          Version: H801SCUN  VER B
Base        BIOS Version: 307
Extended BIOS Version: 311
Software     Version: MA5600V800R013C00
Logic       Version: (U48)118
MAB         Version: 0002

```

### 1.3.3 System User Management.

Step 1 Add a maintenance user profile.

The following lists the maintenance user profile information:

Profile name: weihuzu

Validity period of the user name: 30 days

Validity period of the user password: 30 days

Login start time: 08:00

Login end time: 18:00

```
huawei(config)#terminal user-profile add
User profile name(<=15 chars):weihuzu
Validity period of the user name(0--999 days)[0]:30
Validity period of the password(0--999 days)[0]:30
Permitted start time of logon by a user(hh:mm)[00:00]:08:00
Permitted end time of logon by a user(hh:mm)[00:00]:18:00
Adding user-profile successfully
Repeat this operation? (y/n)[n]:n
```

## Step 2 Add a maintenance user account.

When adding a maintenance user, you can set your name as the user name and specify a password as you want to facilitate further query.

# Add two maintenance users. The following lists the information about the first maintenance user account:

User name: huawei

Password: huawei123

Bound user profile: weihuzu

Permission: administrator

Number of concurrent login times: 2

```
huawei(config)#terminal user name
User Name(length<6,15>):huawei
User Password(length<6,15>):
Confirm Password(length<6,15>):
User profile name(<=15 chars)[root]:weihuzu
User's Level:
  1. Common User  2. Operator  3. Administrator:3
Permitted Reenter Number(0--4):2
User's Appended Info(<=30 chars):
Adding user successfully
Repeat this operation? (y/n)[n]:y
```

The following lists the information about the second maintenance user account:

User name: huawei1

Password: huawei123

Bound user profile: weihuzu

Permission: operator

Number of concurrent login times: 2

```
User Name(length<6,15>):huawei1
User Password(length<6,15>):           //The password is huawei123.
Confirm Password(length<6,15>):
User profile name(<=15 chars)[root]:weihuzu
User's Level:
  1. Common User  2. Operator  3. Administrator:2
Permitted Reenter Number(0--4):2
```

```
User's Appended Info(<=30 chars):
Adding user successfully
Repeat this operation? (y/n)[n]:n
```

### Step 3 Modify the maintenance account.

Modify other attributes of user **huawei1**. For example, change the user level, login password, and number of concurrent login times.

# Change the user level to a common user. (Only a higher-level user can change the level of another user.)

```
huawei(config)#terminal user level
User Name(<=15 chars):huawei1
1. Common User 2. Operator 3. Administrator:
User's Level:1
Confirm Level: 1
Information will take effect when this user logs on next time
Repeat this operation? (y/n)[n]:n
```

# Change the user password to huawei1234.

```
huawei(config)#terminal user password
```

```
User Name(<=15 chars):huawei1
New Password(length<6,15>): //The password is huawei1234.
Confirm Password(length<6,15>):
Information takes effect
Repeat this operation? (y/n)[n]:n
```

# Change the number of concurrent login times to 4.

```
huawei(config)#terminal user reenter
User Name(<=15 chars):huawei1
Permitted Reenter Number(0--4):4
Confirm Reenter Number(0--4):4
Information will take effect when this user logs on next time
Repeat this operation? (y/n)[n]:n
```

### Step 4 Maintain an online user.

If the operation of an online terminal user threatens system security, you can force the user to go offline so that the user cannot perform any operations on the system.

# Query the client ID of the online user.

```
huawei(config)#display client
```

```
-----
ID Client name   Domain name   IP Address    Login Time
-----
1  root          --           10.175.197.60 2018-04-10 09:20:23+08:00
2  huawei1        --           10.175.197.116 2018-04-10 09:40:49+08:00
-----
```

# Force **huawei1** to log out.

```
huawei(config)#client kickoff
{ clientID<U><1,22> }2          // Find that the client ID of user huawei1 is 2.
Command:
      client  kickoff 2
Are you sure to kick the user off?(y/n)[n]:y
The user has been kicked off successfully
```

#### NOTE

When you run a command to force a user to go offline, exercise caution based on the experiment environment to prevent other trainees from being logged off. You are advised to perform this operation at the end of the experiment and delete the users created by yourself.

# Query online users again and find that user **huawei1** is offline.

```
huawei(config)#display client
```

| ID | Client name | Domain name | IP Address    | Login Time                |
|----|-------------|-------------|---------------|---------------------------|
| 1  | root        | --          | 10.175.197.60 | 2018-04-10 09:20:23+08:00 |

Step 5 Delete the maintenance account.

In general, only administrators have the right to delete users whose levels are lower than their own levels and cannot delete online users.

# View all users in the system.

```
huawei(config)#display terminal user all
```

| Name    | Level | Status  | Reenter Num | Profile | Append Info |
|---------|-------|---------|-------------|---------|-------------|
| root    | Super | Online  | 1           | root    | ----        |
| huawei  | Admin | Offline | 2           | weihuzu | ----        |
| huawei1 | User  | Offline | 4           | weihuzu | ----        |

Total record(s) number: 3

# Delete the maintenance user **huawei1**.

```
huawei(config)#undo terminal user name
User Name(<=15 chars):huawei1
Are you sure to delete the user?(y/n)[n]:y
Deleting user successfully
Repeat this operation? (y/n)[n]:n
```

# Query all the users in the system again. User **huawei1** is deleted.

```
huawei(config)#display terminal user all
```

| Name   | Level | Status  | Reenter<br>Num | Profile | Append<br>Info |
|--------|-------|---------|----------------|---------|----------------|
| root   | Super | Online  | 1              | root    | ----           |
| huawei | Admin | Offline | 2              | weihuzu | ----           |

Total record(s) number: 2

### 1.3.4 Inband Management Environment Configuration

**Step 1** Create management VLAN 500.

# Create management VLAN 500 and set the VLAN type to **smart**.

```
huawei(config)#vlan 500
{ <cr>|to<K>|vlantype<E><mux,standard,smart,super> }:smart
Command:
    vlan 500 smart
```

**Step 2** Add an upstream port.

# Add upstream GIU port 0/19/0 to VLAN 500.

```
huawei(config)#port vlan
{ name<K>|vlan-list<S><Length 1-256>|vlanid<U><1,4093> }:500
{ frameid/slotid<S><Length 3-15>|inner-vlan-list<K>|to<K> }:0/19
{ portlist<S><Length 1-256> }:0
Command:
    port vlan 500 0/19 0
```

**Step 3** Configure the IP address of the management VLAN interface.

# Enter the management VLAN interface.

```
huawei(config)#interface vlanif 500
```

# Configure the IP address of the management VLAN interface.

```
huawei(config-if-vlanif500)#ip address 10.175.192.36 22
{ <cr>|description<K>|sub<K> }:  
Command:  
    ip address 10.175.192.36 22  
huawei(config-if-vlanif500)#quit
```

**Step 4** Configure the default VLAN of the upstream port.

```
huawei(config)#interface giu 0/19
huawei(config-if-giu-0/19)#native-vlan
{ all<K>|portid<U><0,1> }:0
{ vlan<K> }:vlan
{ vlanid<U><1,4093> }:500
Command:
```

```
native-vlan 0 vlan 500
huawei(config-if-giu-0/19)#
```

### Step 5 Configure the management route.

# Configure the management route with the next hop pointing to the gateway address.

```
huawei(config)#ip route-static 0.0.0.0 0.0.0.0 10.175.192.1
```

### Step 6 Query related configurations.

# Query all IP addresses of the OLT.

```
huawei(config)#display ip interface brief
{ <cr>|meth<K>|null<K>|vlanif<K> }:
```

Command:

```
display ip interface brief
```

\*down: administratively down  
(l): loopback  
(s): spoofing

The number of interface that is **UP in Physical** is 1  
The number of interface that is DOWN in Physical is 2  
The number of interface that is **UP in Protocol** is 1  
The number of interface that is DOWN in Protocol is 2

| Interface | IP Address/Mask  | Physical | Protocol |
|-----------|------------------|----------|----------|
| meth0     | 10.11.104.6/24   | down     | down     |
| null0     | unassigned       | up       | up(s)    |
| vlanif500 | 10.175.192.36/22 | down     | down     |

# Query the inband management address of the OLT.

```
huawei(config)#display interface vlanif 500
vlanif500 current state : DOWN
Line protocol current state : DOWN
Description : HUAWEI, SmartAX Series, vlanif500 Interface
The Maximum Transmit Unit is 1500 bytes
Forward plane MTU: -
Internet Address is 10.175.192.36/22
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 0819-a6c1-6a53
VLAN Encap-mode : single-tag
```

# Query the outband management address of the OLT.

```
huawei(config)#display interface meth 0
meth0 current state : UP
Line protocol current state : UP
Description : HUAWEI, SmartAX Series, meth0 Interface
The Maximum Transmit Unit is 1500 bytes
Internet Address is 10.11.104.6/24
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 0819-a6c1-6a52
Auto-duplex(Full), Auto-speed(100M)
5 minutes input rate 2190 bytes/sec, 35 packets/sec
5 minutes output rate 14 bytes/sec, 0 packets/sec
```

142413 packets input, 8647104 bytes  
2043 packets output, 130601 bytes

# Display the port information of the upstream interface board.

```
huawei(config)#interface giu 0/19
```

```
huawei(config-if-giu-0/19)#display port state all
```

| Port | Port Type | Optic Status | Native VLAN | MDI  | Speed (Mbps) | Duplex | Flow-Ctrl | Active State | Link    |
|------|-----------|--------------|-------------|------|--------------|--------|-----------|--------------|---------|
| 0    | GE        | -            | 500         | auto | auto         | auto   | off       | active       | offline |
| 1    | GE        | -            | 1           | auto | auto         | auto   | off       | active       | offline |

| Port | Detecting Time(Min) | TX State | Auto-sensing |
|------|---------------------|----------|--------------|
| 0    | -                   | -        | -            |
| 1    | -                   | -        | -            |

| Port | 15Min AlmProf | 15Min WarnProf | 24Hour AlmProf | 24Hour WarnProf |
|------|---------------|----------------|----------------|-----------------|
| 0    | -             | -              | -              | -               |
| 1    | -             | -              | -              | -               |

Note: For any port, setting MDI to auto-negotiation is invalid, For a 1000 M Electrical port in the full-duplex mode, setting MDI to any value is invalid

# Display route information.

```
huawei(config)#display ip routing-table
```

```
{ <cr>|acl<K>|all-vpn-instance<K>|ip-prefix<K>|ip_addr<I><X.X.X.X>|limit<K>|protocol<K>|statistics<K>|time-range<K>|verbose<K>|vpn-instance<K>||<K> }:
```

Command:

```
display ip routing-table
```

Route Flags: R - relay, D - download to fib

Routing Tables: Public

| Destinations : 5 |        |     |      | Routes : 5 |               |             |  |
|------------------|--------|-----|------|------------|---------------|-------------|--|
| Destination/Mask | Proto  | Pre | Cost | Flags      | NextHop       | Interface   |  |
| 0.0.0.0/0        | Static | 60  | 0    | RD         | 10.175.192.1  | vlanif500   |  |
| 10.175.192.0/22  | Direct | 0   | 0    | D          | 10.175.192.36 | vlanif500   |  |
| 10.175.192.36/32 | Direct | 0   | 0    | D          | 127.0.0.1     | vlanif500   |  |
| 127.0.0.0/8      | Direct | 0   | 0    | D          | 127.0.0.1     | InLoopBack0 |  |
| 127.0.0.1/32     | Direct | 0   | 0    | D          | 127.0.0.1     | InLoopBack0 |  |

Step 7 Save configuration data.

```
huawei(config)#save
```

```
{ <cr>|configuration<K>|data<K> }:
```

Command:

```
save
```

It will take several minutes to save configuration file, please wait...

Configuration file had been saved successfully

Note: The configuration file will take effect after being activated

The data is being saved, please wait a moment...

## 1.4 Result Verification

### 1.4.1 Viewing Terminal User Information

# Display information about terminal users on an OLT.

```
huawei(config)#display terminal user all
```

| Name   | Level | Status  | Reenter Profile Num | Append Info |
|--------|-------|---------|---------------------|-------------|
| root   | Super | Online  | 1                   | root        |
| huawei | Admin | Offline | 2                   | weihuzu     |

Total record(s) number: 2

### 1.4.2 Logging In as a Terminal User (in Inband Management Mode)

# Use a network cable to connect the ETH port of a PC to the upstream port giu 0/19/0 of the upstream board on the MA5600T.

# Enable the network configuration on the PC. Ensure that the IP address automatically obtained by the PC is in the same network segment as the IP address of the upstream port 0/19/0 of the OLT.

# Open the command prompt on the PC, run the **Ping 10.175.192.36** command to test the connectivity between the PC and OLT. If they are not properly connected, check for the fault and rectify it.

# If they are properly connected, run the **telnet 10.175.192.36** command, and enter the user name and password (**huawei** and **huawei123** respectively) to check whether the login is successful and whether operations can be performed.

```
telnet 10.175.192.36
>>User name:huawei
>>User password:
Huawei Integrated Access Software (MA5600T).
Copyright(C) Huawei Technologies Co., Ltd. 2002-2013. All rights reserved.
-----
User last login information:
-----
Access Type : Serial
```

```

IP-Address : --
Login Time : 2018-04-12 14:46:37+08:00
Logout Time : 2018-04-12 14:58:06+08:00
-----
-----
User fail login information:
-----
Last Access Type : Serial
Last IP-Address : --
Last Login Time : 2018-04-12 15:00:28+08:00
Login Failure Times : 1
-----
-----
All user fail login information:
-----
Access Type IP-Address Time Login Times
-----
Serial -- 2018-04-12 15:00:28+08:00 1
-----
User root has used a default password. Change the password in time.
huawei>

```

According to the command output, the login is successful.

## 1.5 Configuration Reference

### 1.5.1 Configurations on the OLT Side

```

#
>>User name:root
>>User password:admin
#
enable
config
sysname huawei
switch language-mode
display language
display time
time 16:00:00 2018-04-10
#
interface meth 0
quit
#
board confirm 0
display board 0
display board 0/5
#
terminal user-profile add
terminal user name huawei huawei123 weihuzu 3 2
terminal user name huawei1 huawei123 weihuzu 2 2

```

```
#
terminal user level huawei1 1 1
terminal user password huawei1 huawei1234
terminal user reenter huawei1 4 4
#
display client
client kickoff 2
display terminal user all
undo terminal user name huawei1
#
vlan 500 smart
port vlan 500 0/19 0
#
interface vlanif 500
ip address 10.175.192.36 22
quit
#
ip route-static 0.0.0.0 0.0.0.0 10.175.192.1
display ip routing-table
#
interface giu 0/19
native-vlan 0 vlan 500
quit
#
interface giu 0/19
display port state all
#
save
#
display terminal user all
telnet 10.175.192.36
>>User name:huawei
>>User password:huawei123
```

## 1.5.2 Common Commands

The following table describes common commands.

| Common Command          | Function                                                                                                                                                                                                                                                             |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| terminal user name      | Adds a device operation user. To add a user and authorize the user to access and operate the device, run this command. After a user is added successfully, the user can access the device and perform operations such as configuration and management on the device. |
| undo terminal user name | Deletes a device operation user. After a user is deleted successfully, the user cannot access the device or perform operations such as configuration and management on the device.                                                                                   |
| terminal user level/    | Changes the user level, number of concurrent login times,                                                                                                                                                                                                            |

|                  |                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reenter/password | and password.                                                                                                                                                                                                                                                                                                                                                         |
| display client   | Queries the information about the current login user. To query the information about the current login user, such as the user ID, IP address, and login time, for security management, run this command.                                                                                                                                                              |
| client kickoff   | Disconnects a client. To forcibly disconnect a client or a type of clients whose operation threatens the system security, run this command. After being disconnected successfully, the user cannot perform any operation on the system.                                                                                                                               |
| time             | Sets the system time. To set the system time, run this command. After the command is executed successfully, the system time changes to the set time.                                                                                                                                                                                                                  |
| board add        | Adds a board offline. When the system needs to implement the xDSL service and the device subtending service, you can run this command to add the service board, Ethernet subtending board or power interface board to the vacant slot of the subrack. After a board is added successfully, you can configure and query the board.                                     |
| board confirm    | Confirms an auto-found board or boards in an entire subrack. After inserting a service board to a vacant slot, you need to confirm the board for implementing the xDSL service or device subtending. In this scenario, run this command. You can configure and query the board after it is confirmed successfully.                                                    |
| display board    | Queries the information about the boards in an entire subrack or the details of a specified board. After performing an operation on a board, and you need to query the slot ID, name, status, daughter board information, port information or online/offline state of the board, run this command.                                                                    |
| board delete     | Deletes a board in a subrack. When a board becomes faulty or is removed and does not process services, and no other board registers again in that slot, the slot will be in the board-faulty state all the time. To change the slot state to the vacant state, run this command. After the board is deleted successfully, you can add a new board to the vacant slot. |
| undo ip address  | Deletes the IP address and subnet mask of a VLAN interface. When the IP address of an interface is not required, run this command. After the IP address and subnet mask of the interface are deleted successfully, this IP address cannot be used by the VLAN interface for L3 forwarding.                                                                            |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ip route-static      | Configures a unicast static route. Static routes, instead of dynamic routes, are sufficient for a simple network to function. After a static route is created, the network devices can implement L3 interworking.                                                                                                                                                                                                                                                                |
| undo ip route-static | Deletes a unicast static route. After a network fault occurs or the topology changes, a static route does not change automatically. Therefore, you must run this command to delete the static route. After a static route is deleted, the network devices need to select other routes to forward packets.                                                                                                                                                                        |
| display ip interface | Queries the IP-related configuration and statistics of an interface. To query the numbers of packets, bytes and multicast packets received/transmitted by an interface, or query the number of broadcast packets received, transmitted, forwarded and discarded by an interface, run this command.                                                                                                                                                                               |
| display interface    | Queries the current state and statistics of an interface. To query the interface running state and the packet forwarding state through the interface, run this command.                                                                                                                                                                                                                                                                                                          |
| display frame info   | Queries the basic information about a subrack. To query the IDs, types, statuses, and cascade port information of all subracks, query the IDs, types, statuses, and cascade ports that satisfy the specified search criteria such as the subrack description, MAC address, sequence number, subrack ID or slot ID of a cascade board, or subrack ID, slot ID, and port ID of a cascade port, or query the subrack type, status, and description by subrack ID, run this command. |
| display version      | Queries the versions of the NE software and board software.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| port vlan            | Adds an upstream port to a VLAN. To enable the user packets of a VLAN to be transmitted upstream using an upstream port, run this command to add the upstream port to the VLAN. After the upstream port is added to the VLAN, the packet of this VLAN can be transmitted upstream using this port.                                                                                                                                                                               |
| undo port vlan       | Deletes an upstream port from a VLAN. To stop the user packets of a VLAN from being transmitted upstream using an upstream port, run this command to delete the upstream port from the VLAN. After the upstream port is deleted from the VLAN, the packet of this VLAN cannot be transmitted upstream on this port.                                                                                                                                                              |
| undo vlan            | Deletes one VLAN or multiple VLANs of the same type in                                                                                                                                                                                                                                                                                                                                                                                                                           |

|  |                                                                                                                                                |
|--|------------------------------------------------------------------------------------------------------------------------------------------------|
|  | batches. When you no longer need a VLAN, you can run this command to delete it. After a VLAN is deleted successfully, the VLAN is unavailable. |
|--|------------------------------------------------------------------------------------------------------------------------------------------------|

## 1.6 Quiz

When I run the **telnet 10.175.192.36** command on a PC to log in to an OLT in inband management mode, the login fails. What should I do?

# 2 xPON Terminal Basic Configuration

---

## 2.1 Experiment Introduction

### 2.1.1 About This Experiment

This experiment is to perform basic configurations on GPON devices so that the same OLT can manage ONUs and provide services for them.

### 2.1.2 Objectives

- Understand the principles of configuring an ONU on the MA5600T.

- Understand the similarities and differences between ONUs and their management methods.
- Master the configuration procedures for adding and managing various types of ONUs.

### 2.1.3 Networking

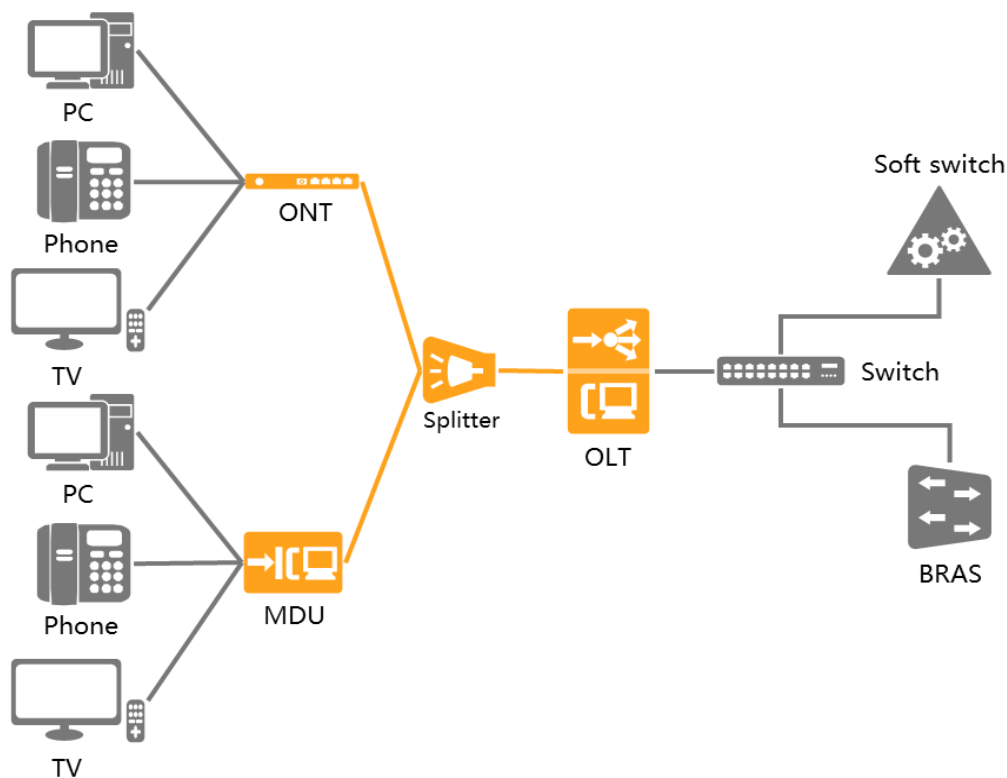


Figure 2-1 Topology for xPON basic configuration experiment

### 2.1.4 Data Plan

The following lists the configuration resources planned for each group based on the lab networking.

GPON OLT data plan for each group:

OLT maintenance mode:

OLT outband maintenance IP address:

User name and password for login:

DBA profile ID:

DBA bandwidth planning:

ONT line profile:

OLT inband management VLAN ID:

OLT inband management IP address:

The parameters in the following data plan table are for reference only. You need to plan the parameters based on the actual networking environment to avoid conflicts and service configuration failures.

**Table 2-1 ONT data plan for xPON configurations**

| ONT Data Plan                | ONT                                                                       |                  | MDU                                                                                                                                                                      |
|------------------------------|---------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device type                  | HG824x                                                                    | HG824x           | MA5822S                                                                                                                                                                  |
| OLT PON port                 | 0/5/0                                                                     | 0/5/0            | 0/5/0                                                                                                                                                                    |
| ONT_ID                       | 1                                                                         | 5                | 2                                                                                                                                                                        |
| ONT SN                       | 48575443D8EAC605                                                          | 48575443D8F92405 | 4857544329518907                                                                                                                                                         |
| Management mode              | OMCI                                                                      | OMCI             | SNMP                                                                                                                                                                     |
| ont-srvprofile               | ID: 31; name: HG824x<br>Ports: 4 ETH ports, 2 POTS ports, and 1 CATV port |                  | /                                                                                                                                                                        |
| ont-lineprofile              | ID: 31; name: HG824x                                                      |                  | <ul style="list-style-type: none"> <li>ID: 32</li> <li>name: MA5822S</li> </ul>                                                                                          |
|                              |                                                                           |                  | <ul style="list-style-type: none"> <li>Inband management:</li> <li>T-CONT ID: 0</li> <li>GEM port 0 cached to T-CONT 0</li> <li>GEM port 0 mapped to VLAN 500</li> </ul> |
| Inband management IP address | /                                                                         |                  | <ul style="list-style-type: none"> <li>OLT management VLAN: 500</li> <li>Upstream port: 0/19/0</li> <li>Static IP address: 10.175.192.36/22</li> </ul>                   |
|                              |                                                                           |                  | <ul style="list-style-type: none"> <li>ONT management VLAN: 500</li> <li>Static IP address: 10.175.192.41/22</li> </ul>                                                  |

**Table 2-2 Access service data plan for GPON configurations**

| ONU | Service | Port | VLAN | GEM Port | T-CONT | DBA |
|-----|---------|------|------|----------|--------|-----|
|-----|---------|------|------|----------|--------|-----|

| Type | Type            |        | ID  | ID | ID | ID and Name    | Bandwidth                                                      |
|------|-----------------|--------|-----|----|----|----------------|----------------------------------------------------------------|
| ONT  | Voice           | lphost | 172 | 1  | 1  | 21, ftth_voip  | Type 1 (fixed bandwidth: 1 Mbit/s)                             |
|      | Internet access | ETH1   | 11  | 2  | 2  | 22, ftth_hsi   | Type 2 (assured bandwidth: 2 Mbit/s)                           |
|      | Multicast       | ETH2   | 200 | 3  | 3  | 23, ftth_ipvtv | Type 3 (assured bandwidth: 4 Mbit/s; max. bandwidth: 8 Mbit/s) |
| ONU  | Voice           | All    | 172 | 4  | 1  | 21, ftth_voip  | Type 1 (fixed bandwidth: 1 Mbit/s)                             |
|      | Internet access | ETH1   | 11  | 5  | 2  | 22, ftth_hsi   | Type 2 (assured bandwidth: 2 Mbit/s)                           |
|      | Multicast       | ETH2   | 200 | 6  |    | 23, ftth_ipvtv | Type 3 (assured bandwidth: 4 Mbit/s; max. bandwidth: 8 Mbit/s) |

## 2.2 Experiment Configuration

### 2.2.1 Configuration Tasks

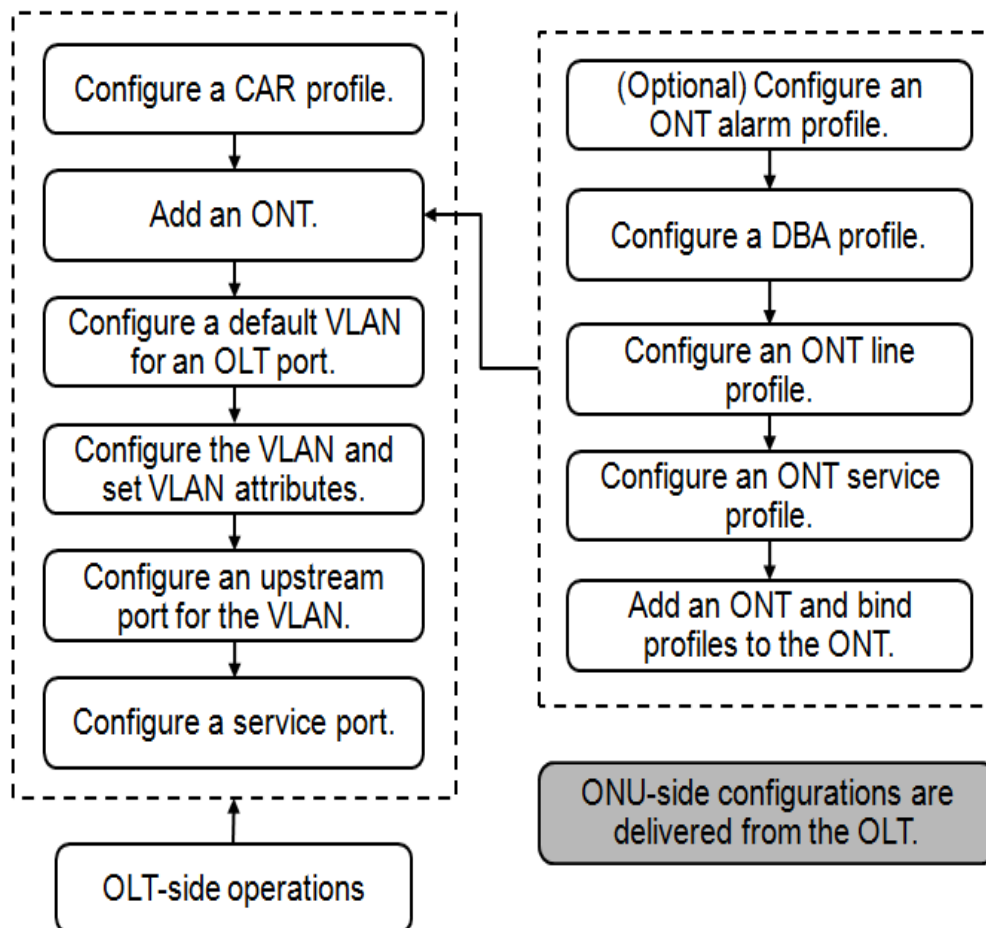
Table 2-3 describes the tasks to be completed during the experiment.

**Table 2-3 Tasks required during xPON basic configurations**

| Task | Requirement | Completion Status |
|------|-------------|-------------------|
|------|-------------|-------------------|

|                                                        |                                                                                                                                                                          |  |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Add a DBA profile.                                     | Add a profile using an allocated profile ID. Type 2 bandwidth is used (2 Mbit/s or other bandwidth for upstream transmission).                                           |  |
| Add an ONT line profile.                               | Add a line profile in non-encryption mode using the allocated profile ID and the added DBA profile.                                                                      |  |
| Add an ONT service profile.                            | Add a service profile based on the allocated ONT and profile ID.                                                                                                         |  |
| Enable the ONT auto-find function of a PON port.       | Connect an optical fiber to the ONT and check whether the ONT can be automatically discovered.                                                                           |  |
| Add an ONT.                                            | The ONT can be automatically or manually added. Record the ONT information, including the status, control flag, running flag, configuration status, and matching status. |  |
| Add an MDU.                                            | Write down the MDU status, including the control flag, running flag, configuration status, and matching status.                                                          |  |
| Configure the MDU management channel and IP address.   | Configure the channel and IP address based on the lab data.                                                                                                              |  |
| (Optional) Log in to the ONU in telnet mode on an OLT. | Be familiar with the command environment.                                                                                                                                |  |

## 2.2.2 Configuration Process of Adding an ONT



## 2.2.3 Configuration Procedure for Adding an ONT

### Step 1 Log in to an OLT.

Use a network cable to connect a PC to the management port of an OLT. Ensure that the IP address automatically obtained by the PC is in the same network segment as the management port IP address of the OLT. Then, log in to the OLT in telnet mode and enter the user name and password. (Enter the user name and password as planned. In this experiment, the user name and password are **root** and **admin**.)

# Enter the user name and password to log in to the OLT.

```

>>User name:root
>>User password: // Enter the password admin, which is not displayed.
Huawei Integrated Access Software (MA5600T).
Copyright(C) Huawei Technologies Co., Ltd. 2002-2013. All rights reserved.
  
```

User last login information:

```
-----
Access Type : Serial
IP-Address  : --
Login Time  : 2018-04-17 17:08:55+08:00
Logout Time : 2018-04-17 17:10:46+08:00
-----
```

User root has used a default password. Change the password in time.

## Step 2 Configure board information.

# Enter the configuration mode.

huawei#**config**

Configure board information and ensure that the boards are properly running.

# In global config mode, check for boards. The boards are to be confirmed.

huawei(config)#display board 0

```
-----
SlotID BoardName Status SubType0 SubType1 Online/Offline
-----
0
1
2
3
4
5 H802GPBD Auto_find // The board is in auto-find state and can be used only after
being confirmed.
6
7
8
9 H801SCUN Active_normal // The board is in the Normal state.
10
11
12
13
14 H802EPBD Auto_find
15
16
17
18
19 H801GICG Auto_find
20
21
22
-----
```

# Check all boards in subrack 0 and check the running status of the boards.

huawei(config)#board confirm 0

huawei(config)#**display board 0**

```
-----
SlotID BoardName Status SubType0 SubType1 Online/Offline
-----
0
```

```

1
2
3
4
5      H802GPBD   Normal
6
7
8
9      H801SCUN   Active_normal
10
11
12
13
14     H802EPBD   Normal
15
16
17
18
19     H801GICG   Normal
20
21
22

```

### Step 3 Configure DBA profiles.

In the global view, create 3 DBA profiles and record the DBA-profile indexes that are allocated.

# In the global view, check the existing DBA profiles in the system.

```

huawei(config)#display dba-profile all
huawei(config)#display dba-profile all

```

| Profile-ID | type | Bandwidth<br>compensation | Fix<br>(kbps) | Assure<br>(kbps) | Max<br>(kbps) | Bind<br>times |
|------------|------|---------------------------|---------------|------------------|---------------|---------------|
| 0          | 3    | No                        | 0             | 8192             | 20480         | 1             |
| 1          | 1    | No                        | 5120          | 0                | 0             | 0             |
| 2          | 1    | No                        | 1024          | 0                | 0             | 1             |
| 3          | 4    | No                        | 0             | 0                | 32768         | 0             |
| 4          | 1    | No                        | 1024000       | 0                | 0             | 0             |
| 5          | 1    | No                        | 32768         | 0                | 0             | 0             |
| 6          | 1    | No                        | 102400        | 0                | 0             | 0             |
| 7          | 2    | No                        | 0             | 32768            | 0             | 0             |
| 8          | 2    | No                        | 0             | 102400           | 0             | 0             |
| 9          | 3    | No                        | 0             | 32768            | 65536         | 0             |

# Add the first DBA profile. Set the profile ID to **21**, profile name to **ftth\_voip**, broadband type to **type1**, and fixed bandwidth to **1 Mbit/s**.

```

huawei(config)# dba-profile
{ add<K>|delete<K>|modify<K> }:add

```

```
{ profile-id<K>|profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-id
{ profile-id<U><10,512> }:21
{ profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-name
{ profile-name<S><Length 1-33> }:ftth_voip
{ type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:type1
{ fix<K> }:fix
{ fix-bandwidth<U><128,10000000> }:1024
{ <cr>|bandwidth_compensate<K> }:  
Command:  
    dba-profile add profile-id 21 profile-name ftth_voip type1 fix 1024  
Adding a DBA profile succeeded  
Profile ID   : 21  
Profile name: ftth_voip
```

# Add the second DBA profile. Set the profile ID to **22**, profile name to **ftth\_hsi**, broadband type to **type2**, and fixed bandwidth to **2 Mbit/s**.

```
huawei(config)#dba-profile
{ add<K>|delete<K>|modify<K> }:add
{ profile-id<K>|profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-id
{ profile-id<U><10,512> }:22
{ profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-name
{ profile-name<S><Length 1-33> }:ftth_hsi
{ type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:type2
{ assure<K> }:assure
{ assure-bandwidth<U><128,10000000> }:2048
Command:  
    dba-profile add profile-id 22 profile-name ftth_hsi type2 assure 2048  
Adding a DBA profile succeeded  
Profile ID   : 22  
Profile name: ftth_hsi
```

# Add the third DBA profile. Set the profile ID to **23**, profile name to **ftth\_ipvtv**, broadband type to **Type3**, assured bandwidth to **4 Mbit/s**, and maximum bandwidth to **8 Mbit/s**.

```
huawei(config)#dba-profile
{ add<K>|delete<K>|modify<K> }:add
{ profile-id<K>|profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-id 23
{ profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-name
{ profile-name<S><Length 1-33> }:ftth_ipvtv
{ type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:type3
{ assure<K> }:assure
{ assure-bandwidth<U><128,10000000> }:4096
{ max<K> }:max
{ max-bandwidth<U><128,10000000> }:8064
Command:  
    dba-profile add profile-id 23 profile-name ftth_ipvtv type3 assure 4096 max 8064  
Adding a DBA profile succeeded  
Profile ID   : 23  
Profile name: ftth_ipvtv
```

**Step 4** Configure an ONT line profile.

In the global view, create an ONT line profile (**ont-lineprofile**), set detailed parameters, bind the corresponding DBA profile, and record the index of the line profile.

# Check for the ont-lineprofile in the system.

```
huawei(config)#display ont-lineprofile
{ epon<K>|gpon<K> }:gpon
{ all<K>|bound-info<K>|profile-id<K>|profile-name<K> }:all
Command:
display ont-lineprofile gpon all
```

| Profile-ID | Profile-name           | Binding times |
|------------|------------------------|---------------|
| 0          | line-profile_default_0 | 0             |

Total: 1

# Create an ONT line profile named HG824x and enter profile configuration mode. The profile ID is 31.

```
huawei(config)#ont-lineprofile
{ epon<K>|gpon<K> }:gpon
{ <cr>|profile-id<K>|profile-name<K> }:profile-id
{ profile-id<U><0,8192> }:31
{ <cr>|profile-name<K> }:profile-name
{ profile-name<S><Length 1-32> }:HG824x
Command:
ont-lineprofile gpon profile-id 31 profile-name HG824x
```

#### NOTE

When you create a line profile or DBA profile, the system automatically allocates a profile ID if you do not specify an ID.

In profile configuration mode, bind a T-CONT to the corresponding DBA profile.

# Bind T-CONT 1 to DBA profile 21.

```
huawei(config-gpon-lineprofile-31)#tcont
{ tcont-list<S><Length 1-13> }:1
{ <cr>|dba-profile-id<K>|dba-profile-name<K> }:dba-profile-id
{ profile-id<U><0,512> }:21
Command:
tcont 1 dba-profile-id 21
```

# Bind T-CONT 2 to DBA profile 22.

```
huawei(config-gpon-lineprofile-31)#tcont
{ tcont-list<S><Length 1-13> }:2
{ <cr>|dba-profile-id<K>|dba-profile-name<K> }:dba-profile-id
{ profile-id<U><0,512> }:22
Command:
tcont 2 dba-profile-id 22
```

# Bind T-CONT 3 to DBA profile 23.

```

huawei(config-gpon-lineprofile-31)#tcont
{ tcont-list<S><Length 1-13> }:3
{ <cr>|dba-profile-id<K>|dba-profile-name<K> }:dba-profile-id
{ profile-id<U><0,512> }:23

```

Command:

```
tcont 3 dba-profile-id 23
```

In profile config mode, add a GEM port and map it to the corresponding T-CONT.

# Add GEM port 1 to carry Ethernet services and map it to T-CONT 1.

```

huawei(config-gpon-lineprofile-31)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:add
{ gem-index<U><0,1023> }:1
{ service-type<E><eth,tdm> }:eth
{ tcont<K> }:tcont
{ tcont-id<U><0,127> }:1
{ <cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K> }:

```

Command:

```
gem add 1 eth tcont 1
```

# Add GEM port 2 to carry Ethernet services and map it to T-CONT 2.

```

huawei(config-gpon-lineprofile-31)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:add
{ gem-index<U><0,1023> }:2
{ service-type<E><eth,tdm> }:eth
{ tcont<K> }:tcont
{ tcont-id<U><0,127> }:2
{ <cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K> }:

```

Command:

```
gem add 2 eth tcont 2
```

# Add GEM port 3 to carry Ethernet services and map it to T-CONT 3.

```

huawei(config-gpon-lineprofile-31)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:add
{ gem-index<U><0,1023> }:3
{ service-type<E><eth,tdm> }:eth
{ tcont<K> }:tcont
{ tcont-id<U><0,127> }:3
{ <cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K> }:

```

Command:

```
gem add 3 eth tcont 3
```

Configure the mapping modes of GEM ports.

# Configure the mapping mode between GEM ports and user data to VLAN mapping.

```

huawei(config-gpon-lineprofile-31)#mapping-mode
{iptos<K>|port-priority<K>|port-vlan-priority<K>|port-vlan<K>|port<K>|priority<K>|vlan-
priority<K>|vlan<K>}:vlan

```

Command:

```
mapping-mode vlan
```

Configure the mapping relationships between GEM ports and VLANs.

# Configure the mapping between the GEM port 1 and the user flow with the VLAN 172 tag. The mapping index is 1.

```
huawei(config-gpon-lineprofile-31)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:mapping
{ gem-index<U><0,1023> }:1
{ mapping-index<U><0,7> }:1
{ e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K> }:vlan
{ untag<K>|vlan-id<U><0,4095> }:172
{ <cr>|flow-car<K>|priority<K>|transparent<K> }:
Command:
gem mapping 1 1 vlan 172
```

# Configure the mapping between the GEM port 2 and the user flow with the VLAN 11 tag. The mapping index is 2.

```
huawei(config-gpon-lineprofile-31)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:mapping
{ gem-index<U><0,1023> }:2
{ mapping-index<U><0,7> }:2
{ e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K> }:vlan
{ untag<K>|vlan-id<U><0,4095> }:11
{ <cr>|flow-car<K>|priority<K>|transparent<K> }:
Command:
gem mapping 2 2 vlan 11
```

# Configure the mapping between the GEM port 3 and the user flow with the VLAN 200 tag. The mapping index is 3.

```
huawei(config-gpon-lineprofile-31)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:mapping
{ gem-index<U><0,1023> }:3
{ mapping-index<U><0,7> }:3
{ e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K> }:vlan
{ untag<K>|vlan-id<U><0,4095> }:200
{ <cr>|flow-car<K>|priority<K>|transparent<K> }:
Command:
gem mapping 3 3 vlan 200
```

# After completing the configuration, run the **commit** command to make the configuration take effect.

```
huawei(config-gpon-lineprofile-31)#commit
huawei(config-gpon-lineprofile-31)#quit
huawei(config)#
```

#### NOTE

All parameter settings can take effect only after the **commit** command is executed.

## Step 5 Configure ONT service profiles.

In global config mode, create a service profile named **ont-srvprofile**, match the profile with the actual ONT type, and record the allocated index of the service profile.

# Query the existing service profiles in the system.

```
huawei(config)# display ont-srvprofile
{ epon<K>|gpon<K> }:gpon
{ all<K>|bound-info<K>|profile-id<K>|profile-name<K> }:all
Command:
    display ont-srvprofile gpon all
Command:
    display ont-srvprofile gpon all
```

| Profile-ID | Profile-name          | Binding times |
|------------|-----------------------|---------------|
| 0          | srv-profile_default_0 | 0             |

Total: 1

# Create an ONT service profile named **HG824x** and enter profile configuration mode. The profile ID is 31.

```
huawei(config)#ont-srvprofile
{ epon<K>|gpon<K> }:gpon
{ <cr>|profile-id<K>|profile-name<K> }:profile-id
{ profile-id<U><0,8192> }:31
{ <cr>|profile-name<K> }:profile-name
{ profile-name<S><Length 1-32> }:HG824x
Command:
    ont-srvprofile gpon profile-id 31 profile-name HG824x
```

Configure ONT port capabilities.

# Configure the port types and number of ONTs based on the capabilities of the ONTs in the lab. In this experiment, 1 ONT is used, which provides 4 ETH ports, 2 POTS ports, and 1 CATV port.

```
huawei(config-gpon-srvprofile-31)#ont-port
{ catv<K>|eth<K>|moca<K>|pots<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K> }:eth
{ adaptive<K>|eth-port<U><0,8> }:4
{ <cr>|catv<K>|moca<K>|pots<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K> }:pots
{ adaptive<K>|pots-port<U><0,32> }:2
{ <cr>|catv<K>|moca<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K> }:catv
{ adaptive<K>|catv-port<U><0,8> }:1
{ <cr>|moca<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K> }:
Command:
    ont-port eth 4 pots 2 catv 1
```

Configure a VLAN for the user-side port of the ONT.

# Set the VLAN ID of the user-side iphost port on the ONT to **172**.

```

huawei(config-gpon-srvprofile-31)#port
{ car<K>|dscp<K>|e1<K>|eth<K>|igmp-forward<K>|priority-policy<K>|q-in-q<K>|vlan<K> }:vlan
{ eth<K>|iphost<K>|moca<K> }:iphost
{ q-in-q<K>|translation<K>|transparent<K>|vlanid<U><0,4095> }:172
{ <cr>|priority<K>|prival<U><0,7>|TLS<K> }:
Command:
port vlan iphost 172
Set ONT port(s) VLAN configuration, success: 1, failed: 0

```

# Set the VLAN ID of user-side Ethernet port 1 on the ONT to 11.

```

huawei(config-gpon-srvprofile-31)#port
{ car<K>|dscp<K>|e1<K>|eth<K>|igmp-forward<K>|priority-policy<K>|q-in-q<K>|vlan<K> }:vlan
{ eth<K>|iphost<K>|moca<K> }:eth
{ ont-portlist<S><Length 1-128> }:1
{ downstream<K>|q-in-q<K>|translation<K>|transparent<K>|vlanid<U><0,4095> }:11
{ <cr>|priority<K>|prival<U><0,7>|TLS<K> }:
Command:
port vlan eth 1 11
Set ONT port(s) VLAN configuration, success: 1, failed: 0

```

# Set the VLAN ID of user-side Ethernet port 2 on the ONT to 200.

```

huawei(config-gpon-srvprofile-31)#port
{ car<K>|dscp<K>|e1<K>|eth<K>|igmp-forward<K>|priority-policy<K>|q-in-q<K>|vlan<K> }:vlan
{ eth<K>|iphost<K>|moca<K> }:eth
{ ont-portlist<S><Length 1-128> }:2
{ downstream<K>|q-in-q<K>|translation<K>|transparent<K>|vlanid<U><0,4095> }:200
{ <cr>|priority<K>|prival<U><0,7>|TLS<K> }:
Command:
port vlan eth 2 200
Set ONT port(s) VLAN configuration, success: 1, failed: 0

```

# Commit the parameters to make them take effect.

```

huawei(config-gpon-srvprofile-31)#commit
huawei(config-gpon-srvprofile-31)#quit
huawei(config)#

```

**Step 6** Check whether the ONT is automatically added to the GPON system.

Switch from global config mode to GPON mode, and then enable the ONT auto-find function of an optical port. Wait for the system to detect the newly added ONT in the PON network, report the new ONT SN address, and copy the SN address. Set the ONT ID based on the data plan and bind the configured ont-lineprofile and ont-srvprofile. Ensure that the ONT is in the normal state.

Enable the ONT auto-find function of a PON port.

# Enter GPON mode. The port is the PON port connected to the ONT that is used, that is, port 0 in slot 5 of subrack 0.

```

huawei(config)#interface

```

```
{ adsl<K>|atm<K>|csp<K>|edt<K>|emu<K>|epon<K>|eponnni<K>|eth<K>|giu<K>|gpon<K>|gponnni<K>|
h248<K>|loopback<K>|meth<K>|null<K>|opf<K>|opg<K>|scu<K>|shl<K>|top-
stm1<K>|top<K>|tunnel<K>|vdsl<K>|vlanif<K>|vrgw<K> }:gpon
{ frameid/slotid<S><Length 3-15> }:0/5
```

Command:

```
interface gpon 0/5
```

# Enable the auto-find function of GPON port 0/5/0.

```
huawei(config-if-gpon-0/5)#port
{ dba<K>|optical-alarm-profile<K>|portid<U><0,7> }:0
{ fec<K>|laser-switch<K>|ont-auto-find<K>|ont-password-renew<K>|range<K> }:ont-auto-find
{ switch<E><enable,disable> }:enable
```

Command:

```
port 0 ont-auto-find enable
```

Query the found ONTs.

# In GPON mode, query the information about the auto-found ONT on GPON port 0/5/0, confirm the ONT version information, and copy the SN address.

```
huawei(config-if-gpon-0/5)#display ont autofind 0
```

```
-----
Number           : 1
F/S/P            : 0/5/0
Ont SN           : 4857544329518907
Password         : 0x30303030303030303030(0000000000)
Loid             : 0000000000
Checkcode        :
VendorID         : HWTC
Ont Version      : MA5822S VER.C
Ont SoftwareVersion : V8R308 C00
Ont EquipmentID   : SmartAX MA5822S
Ont autofind time : 2018-04-10 14:35:59+08:00
-----
```

```
Number           : 2
F/S/P            : 0/5/0
Ont SN           : 48575443D8EAC605
Password         : 0x00000000000000000000
Loid             :
Checkcode        :
VendorID         : HWTC
Ont Version      : 120D4600
Ont SoftwareVersion : V1R006C00S210
Ont EquipmentID   : 247
Ont autofind time : 2018-04-10 14:35:44+08:00
-----
```

```
Number           : 3
F/S/P            : 0/5/0
Ont SN           : 48575443D8F92405
Password         : 0x00000000000000000000
Loid             :
Checkcode        :
VendorID         : HWTC
```

```

Ont Version          : 120D4600
Ont SoftwareVersion : V1R006C00S210
Ont EquipmentID      : 247
Ont autofind time    : 2018-04-10 14:35:44+08:00
-----

```

Confirm the found ONT.

# In GPON mode, confirm that the ID of the found ONT is 1. The SN 48575443D8EAC605 is used for verification and registration, the management mode is OMCI. The ONT is bound to the line profile (ID: 31) and service profile (ID: 31).

```

huawei(config-if-gpon-0/5)#ont confirm
{ portid<U><0,7> }:0
{ all<K>|loid-auth<K>|ontid<K>|password-auth<K>|sn-auth<K> }:ontid
{ ontid<U><0,127> }:1
{ loid-auth<K>|password-auth<K>|sn-auth<K> }:sn-auth
{ sn<S><Length 13-16> }:48575443D8EAC605
{ omci<K>|password-auth<K>|snmp<K> }:omci
{ <cr>|desc<K>|ont-lineprofile-id<K>|ont-lineprofile-name<K>|ont-srvprofile-id<K>|ont-srvprofile-
name<K> }:ont-lineprofile-id
{ profile-id<U><0,8192> }:31
{ <cr>|desc<K>|ont-srvprofile-id<K>|ont-srvprofile-name<K> }:ont-srvprofile-id
{ profile-id<U><0,8192> }:31
{ <cr>|desc<K> }:  

Command:
      ont confirm 0 ontid 1 sn-auth 48575443D8EAC605 omci ont-lineprofile-id 31 ont-
srvprofile-id 31
Number of ONTs that can be added: 1, success: 1
PortID :0, ONTID :1

```

Check the status of the added ONT.

# Check the status of the ONT that is added to GPON port 0/5/0.

```

huawei(config-if-gpon-0/5)#display ont info
{ portid<U><0,7> }:0
{ all<K>|ontid<U><0,127> }:all
Command:
      display ont info 0 all

```

| F/S/P  | ONT ID | SN               | Control flag | Run state | Config state | Match state | Protect side |
|--------|--------|------------------|--------------|-----------|--------------|-------------|--------------|
| 0/ 5/0 | 1      | 48575443D8EAC605 | active       | online    | normal       | match       | no           |

| F/S/P  | ONT-ID | Description        |
|--------|--------|--------------------|
| 0/ 5/0 | 1      | ONT_NO_DESCRIPTION |

In port 0/ 5/0 , the total of ONTs are: 1, online: 1

**Step 7 (Optional) Manually add an ONT to the GPON system.**

# Enter GPON mode. Manually add an ONT to GPON port 0/5/0. The ONT ID is 5. The SN 48575443D8F92405 is used for verification and registration, the management mode is OMCI. The ONT is bound to the line profile (ID: 31) and service profile (ID: 31).

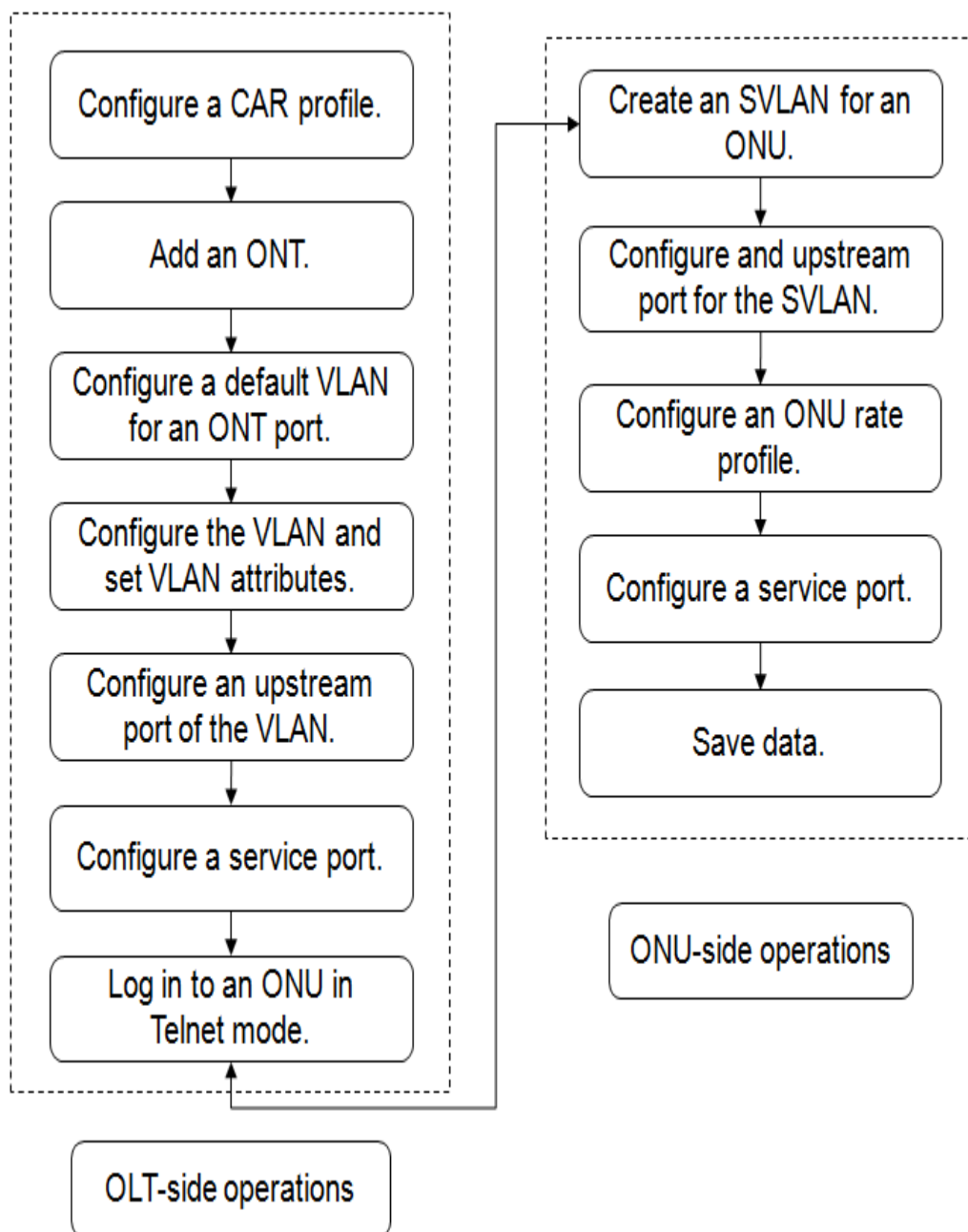
```

huawei(config)#interface gpon 0/5
huawei(config-if-gpon-0/5)#ont
{ activate<K>|add<K>|alarm-policy<K>|alarm-profile<K>|cancel<K>|confirm<K>|cos-queue-map-
profile<K>|deactivate<K>|delete<K>|factory-setting-
restore<K>|gemport<K>|ipconfig<K>|modify<K>|optical-alarm-profile<K>|port<K>|power-shedding-
config<K>|re-discovery<K>|re-register<K>|remote-ping<K>|reset<K>|snmp-profile<K>|snmp-
route<K>|statistics<K>|tdm-vcl<K>|tr069-server-config<K>|voip<K> }:add
{ portid<U><0,7> }:0
{ loid-auth<K>|ontid<U><0,127>|password-auth<K>|protect-side<K>|sn-auth<K> }:5
{ loid-auth<K>|password-auth<K>|protect-side<K>|sn-auth<K> }:sn-auth
{ sn-value<S><Length 13-16> }:48575443D8F92405
{ omci<K>|password-auth<K>|snmp<K> }:omci
{ <cr>|desc<K>|ont-lineprofile-id<K>|ont-lineprofile-name<K>|ont-srvprofile-id<K>|ont-srvprofile-
name<K> }:ont-lineprofile-id
{ profile-id<U><0,8192> }:31
{ <cr>|desc<K>|ont-srvprofile-id<K>|ont-srvprofile-name<K> }:ont-srvprofile-id
{ profile-id<U><0,8192> }:31
{ <cr>|desc<K> }:  

    Command:
        ont add 0 5 sn-auth 48575443D8F92405 omci ont-lineprofile-id 31 ont-srvprofile-id
31
    Number of ONTs that can be added: 1, success: 1
    PortID :0, ONTID :5
huawei(config-if-gpon-0/5)#quit
huawei(config)#

```

## 2.2.4 Configuration Process of Adding an ONU



## 2.2.5 Configuration Procedure for Adding an ONU

### NOTE

If an ONT has been added and the DBA profile to be bound to the ONT service profile is the same as that bound to the ONU service profile, skip steps 1 to 3.

### Step 1 Log in to an OLT.

Use a network cable to connect a PC to the management port of an OLT. Ensure that the IP address automatically obtained by the PC is in the same network segment as the management port IP address of the OLT. Then, log in to the OLT in telnet mode and enter the user name and password.

# Enter the user name and password to log in to the OLT.

```
>>User name:root
>>User password:
```

### Step 2 Configure board information.

# Enter the configuration mode.

```
huawei#config
```

Configure board information and ensure that the GPON service boards and upstream boards are properly running.

# In global config mode, check for boards. The boards are to be confirmed.

```
huawei(config)#display board 0
```

# Check all boards in subrack 0 and check the running status of the boards.

```
huawei(config)#board confirm 0
huawei(config)#display board 0
```

### Step 3 Configure DBA profiles.

In the global view, create 3 DBA profiles and record the DBA-profile indexes that are allocated.

# In the global view, check the existing DBA profiles in the system.

```
huawei(config)#display dba-profile all
```

| Profile-ID | type | Bandwidth<br>compensation | Fix<br>(kbps) | Assure<br>(kbps) | Max<br>(kbps) | Bind<br>times |
|------------|------|---------------------------|---------------|------------------|---------------|---------------|
| 0          | 3    | No                        | 0             | 8192             | 20480         | 1             |
| 1          | 1    | No                        | 5120          | 0                | 0             | 1             |
| 2          | 1    | No                        | 1024          | 0                | 0             | 1             |
| 3          | 4    | No                        | 0             | 0                | 32768         | 0             |
| 4          | 1    | No                        | 1024000       | 0                | 0             | 0             |
| 5          | 1    | No                        | 32768         | 0                | 0             | 0             |
| 6          | 1    | No                        | 102400        | 0                | 0             | 0             |
| 7          | 2    | No                        | 0             | 32768            | 0             | 0             |
| 8          | 2    | No                        | 0             | 102400           | 0             | 0             |
| 9          | 3    | No                        | 0             | 32768            | 65536         | 0             |

# Add the first DBA profile. Set the profile ID to **21**, profile name to **ftth\_voip**, broadband type to **Type1**, and fixed bandwidth to **1 Mbit/s**.

```

huawei(config)# dba-profile
{ add<K>|delete<K>|modify<K> }:add
{ profile-id<K>|profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-id
{ profile-id<U><10,512> }:21
{ profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-name
{ profile-name<S><Length 1-33> }:ftth_voip
{ type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:type1
{ fix<K> }:fix
{ fix-bandwidth<U><128,10000000> }:1024
{ <cr>|bandwidth_compensate<K> }:
Command:
    dba-profile add profile-id 21 profile-name ftth_voip type1 fix 1024
Adding a DBA profile succeeded
Profile ID : 21
Profile name: ftth_voip

```

# Add the second DBA profile. Set the profile ID to **22**, profile name to **ftth\_hsi**, broadband type to **Type2**, and fixed bandwidth to **2 Mbit/s**.

```

huawei(config)#dba-profile
{ add<K>|delete<K>|modify<K> }:add
{ profile-id<K>|profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-id
{ profile-id<U><10,512> }:22
{ profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-name
{ profile-name<S><Length 1-33> }:ftth_hsi
{ type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:type2
{ assure<K> }:assure
{ assure-bandwidth<U><128,10000000> }:2048
Command:
    dba-profile add profile-id 22 profile-name ftth_hsi type2 assure 2048
Adding a DBA profile succeeded
Profile ID : 22
Profile name: ftth_hsi

```

# Add the third DBA profile. Set the profile ID to **23**, profile name to **ftth\_ipvtv**, broadband type to **Type3**, assured bandwidth to **4 Mbit/s**, and maximum bandwidth to **8 Mbit/s**.

```

huawei(config)#dba-profile
{ add<K>|delete<K>|modify<K> }:add
{ profile-id<K>|profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-id 23
{ profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-name
{ profile-name<S><Length 1-33> }:ftth_ipvtv
{ type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:type3
{ assure<K> }:assure
{ assure-bandwidth<U><128,10000000> }:4096
{ max<K> }:max
{ max-bandwidth<U><128,10000000> }:8064
Command:
    dba-profile add profile-id 23 profile-name ftth_ipvtv type3 assure 4096 max 8064
Adding a DBA profile succeeded

```

Profile ID : 23  
Profile name: ftth\_ip tv

#### Step 4 Configure an ONT line profile.

In the global view, create an ONT line profile (**ont-lineprofile**), set detailed parameters, bind the corresponding DBA profile, and record the index of the line profile.

# Create an ONT line profile named **MA5822S** and enter profile configuration mode. The profile ID is 32.

```
huawei(config)#ont-lineprofile
{ epon<K>|gpon<K> }:gpon
{ <cr>|profile-id<K>|profile-name<K> }:profile-id
{ profile-id<U><0,8192> }:32
{ <cr>|profile-name<K> }:profile-name
{ profile-name<S><Length 1-32> }:MA5822S
Command:
ont-lineprofile gpon profile-id 32 profile-name MA5822S
```

In profile configuration mode, bind a T-CONT to the corresponding DBA profile.

# Bind T-CONT 1 to DBA profile 21.

```
huawei(config-gpon-lineprofile-32)#tcont
{ tcont-list<S><Length 1-13> }:1
{ <cr>|dba-profile-id<K>|dba-profile-name<K> }:dba-profile-id
{ profile-id<U><0,512> }:21
Command:
tcont 1 dba-profile-id 21
```

# Bind T-CONT 2 to DBA profile 22.

```
huawei(config-gpon-lineprofile-32)#tcont 2
{ <cr>|dba-profile-id<K>|dba-profile-name<K> }:dba-profile-id
{ profile-id<U><0,512> }:22
Command:
tcont 2 dba-profile-id 22
```

# Bind T-CONT 3 to DBA profile 23.

```
huawei(config-gpon-lineprofile-32)#tcont
{ tcont-list<S><Length 1-13> }:3
{ <cr>|dba-profile-id<K>|dba-profile-name<K> }:dba-profile-id
{ profile-id<U><0,512> }:23
Command:
tcont 3 dba-profile-id 23
```

In profile config mode, add a GEM port and map it to the corresponding T-CONT.

# Add GEM port 4 to carry Ethernet services and map it to T-CONT 1.

```
huawei(config-gpon-lineprofile-32)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:add
{ gem-index<U><0,1023> }:4
```

```
{ service-type<E><eth,tdm> }:eth
{ tcont<K> }:tcont
{ tcont-id<U><0,127> }:1
{ <cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K> }:
  Command:
    gem add 4 eth tcont 1
```

# Add GEM port 5 to carry Ethernet services and map it to T-CONT 2.

```
huawei(config-gpon-lineprofile-32)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:add
{ gem-index<U><0,1023> }:5
{ service-type<E><eth,tdm> }:eth
{ tcont<K> }:tcont
{ tcont-id<U><0,127> }:2
{ <cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K> }:
  Command:
    gem add 5 eth tcont 2
```

#Add GEM port 6 to carry Ethernet services and map it to T-CONT 3.

```
huawei(config-gpon-lineprofile-32)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:add
{ gem-index<U><0,1023> }:6
{ service-type<E><eth,tdm> }:eth
{ tcont<K> }:tcont
{ tcont-id<U><0,127> }:3
{ <cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K> }:
  Command:
    gem add 6 eth tcont 3
```

Configure the mapping modes of GEM ports.

# Configure the mapping mode between GEM ports and user data to VLAN mapping.

```
huawei(config-gpon-lineprofile-32)#mapping-mode
{iptos<K>|port-priority<K>|port-vlan-priority<K>|port-vlan<K>|port<K>|priority<K>|vlan-
priority<K>|vlan<K>}:vlan
  Command:
    mapping-mode vlan
```

Configure the mapping relationships between GEM ports and VLANs.

# Configure the mapping between GEM port 4 and the user flow with the VLAN 172 tag. The mapping index is 4.

```
huawei(config-gpon-lineprofile-32)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:mapping
{ gem-index<U><0,1023> }:4
{ mapping-index<U><0,7> }:4
{ e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K> }:vlan
{ untag<K>|vlan-id<U><0,4095> }:172
{ <cr>|flow-car<K>|priority<K>|transparent<K> }:
```

Command:

```
gem mapping 4 4 vlan 172
```

# Configure the mapping between GEM port 5 and the user flow with the VLAN 11 tag. The mapping index is 5.

```
huawei(config-gpon-lineprofile-32)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:mapping
{ gem-index<U><0,1023> }:5
{ mapping-index<U><0,7> }:5
{ e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K> }:vlan
{ untag<K>|vlan-id<U><0,4095> }:11
{ <cr>|flow-car<K>|priority<K>|transparent<K> }:
```

Command:

```
gem mapping 5 5 vlan 11
```

# Configure the mapping between GEM port 6 and the user flow with the VLAN 200 tag. The mapping index is 6.

```
huawei(config-gpon-lineprofile-32)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:mapping
{ gem-index<U><0,1023> }:6
{ mapping-index<U><0,7> }:6
{ e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K> }:vlan
{ untag<K>|vlan-id<U><0,4095> }:200
{ <cr>|flow-car<K>|priority<K>|transparent<K> }:
```

Command:

```
gem mapping 6 6 vlan 200
```

# Commit the parameters to make them take effect.

```
huawei(config-gpon-lineprofile-32)#commit
huawei(config-gpon-lineprofile-32)#quit
```

**Step 5** Check whether the ONT is automatically added to the GPON system.

Query the found ONT.

# In GPON mode, query the information about the auto-found ONT on GPON port 0/5/0, confirm the ONT version information, and copy the SN address.

```
huawei(config)#interface gpon 0/5
huawei(config-if-gpon-0/5)#display ont autofind
{ portid<U><0,7> }:0
```

Command:

```
display ont autofind 0
```

```
-----
Number          : 1
F/S/P           : 0/5/0
Ont SN          : 4857544329518907
Password        : 0x30303030303030303030(0000000000)
```

```
Loid : 0000000000
Checkcode :
VendorID : HWTC
Ont Version : MA5822S VER.C
Ont SoftwareVersion : V8R308 C00
Ont EquipmentID : SmartAX MA5822S
Ont autofind time : 2018-04-10 14:35:59+08:00
-----
```

Confirm the found ONT.

# In GPON mode, confirm that the ID of the found ONU is 2. The SN 4857544329518907 is used for verification and registration, the management mode is SNMP. The ONT is bound to the line profile (ID: 32).

```
huawei(config-if-gpon-0/5)#ont confirm
{ portid<U><0,7> }:0
{ all<K>|loid-auth<K>|ontid<K>|password-auth<K>|sn-auth<K> }:ontid
{ ontid<U><0,127> }:2
{ loid-auth<K>|password-auth<K>|sn-auth<K> }:sn-auth
{ sn<S><Length 13-16> }:4857544329518907
{ omci<K>|password-auth<K>|snmp<K> }:snmp
{ <cr>|desc<K>|ont-lineprofile-id<K>|ont-lineprofile-name<K> }:ont-lineprofile-id
{ profile-id<U><0,8192> }:32
{ <cr>|desc<K> }:
Command:
ont confirm 0 ontid 2 sn-auth 4857544329518907 snmp ont-lineprofile-id 32
Number of ONTs that can be added: 1, success: 1
PortID :0, ONTID :2
```

## Step 6 Configure the inband management VLAN and IP address of the OLT.

To remotely log in to an ONT from an OLT, configure the management VLANs and IP addresses and ensure that the management VLANs of the OLT and ONT are the same and the IP addresses of the OLT and ONT are in the same network segment.

# Create inband management VLAN 500 on an OLT and add upstream port 0/19/0 to the VLAN.

```
huawei(config)#vlan
{ attrib<K>|bind<K>|desc<K>|forwarding<K>|loop<K>|name<K>|packet-
policy<K>|priority<K>|reserve<K>|service-profile<K>|vlan-list<S><Length 1-
256>|vlanid<U><2,4093> }:500
{ <cr>|to<K>|vlantype<E><mux,standard,smart,super> }:smart
Command:
vlan 500 smart
huawei(config)#port vlan 500
{ frameid/slotid<S><Length 3-15>|inner-vlan-list<K>|to<K> }:0/19
{ portlist<S><Length 1-256> }:0
Command:
port vlan 500 0/19 0
```

# Enter the VLAN interface 500 and set the inband management IP address to **10.175.192.36/22**.

```
huawei(config)#interface vlanif 500
huawei(config-if-vlanif500)#ip address 10.175.192.36 22
{ <cr>|description<K>|sub<K> }:
Command:
    ip address 10.175.192.36 22
huawei(config-if-vlanif500)#quit
huawei(config)#
```

**Step 7** Configure the inband management VLAN and IP address of the ONT.

In GPON mode, run the **ont ipconfig** command to configure the inband NMS address of ONT 2. The system automatically sends the address to ONT 2 over the OMCI channel.

# Enter GPON 0/5/0 interface mode. Set the static IP address of ONT 2 to **10.175.192.41/22**, gateway to **10.172.192.1/22**, and management VLAN to **500** (same as the management VLAN of the OLT).

```
huawei(config)#interface gpon 0/5
huawei(config-if-gpon-0/5)#ont ipconfig
{ portid<U><0,7> }:0
{ ontid<U><0,127> }:2
{ dhcp<K>|dscp-mapping-table<K>|ip-index<K>|pppoe<K>|static<K> }:static
{ <cr>|gateway<K>|ip-address<K>|pri-dns<K>|slave-dns<K>|vlan<K> }:ip-address
{ ip-address<I><X.X.X.X> }:10.175.192.41
{ mask<K> }:mask
{ mask<M><X.X.X.X> }:255.255.252.0
{ <cr>|gateway<K>|pri-dns<K>|slave-dns<K>|vlan<K> }:gateway
{ gateway<I><X.X.X.X> }:10.175.192.1
{ <cr>|pri-dns<K>|slave-dns<K>|vlan<K> }:vlan
{ vlanid<U><0,4095> }:500
{ <cr>|pri-dns<K>|priority<K>|slave-dns<K> }:
Command:
    ont ipconfig 0 2 static ip-address 10.175.192.41 mask 255.255.252.0 gateway
10.175.192.1 vlan 500
huawei(config-if-gpon-0/5)#quit
```

**Step 8** Configure the line profile for an inband management channel.

Configure the management GEM port in the ONT line profile and map the GEM port to the management VLAN.

# Enter the line profile with ID 32.

```
huawei(config)#ont-lineprofile
{ epon<K>|gpon<K> }:gpon
{ <cr>|profile-id<K>|profile-name<K> }:profile-id
{ profile-id<U><0,8192> }:32
{ <cr>|profile-name<K> }:
Command:
    ont-lineprofile gpon profile-id 32
```

# In line profile mode, add a GEM port (index: 0) to carry the management traffic stream, carry the ETH service, and bind it to T-CONT 0.

```

huawei(config-gpon-lineprofile-32)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:add
{ gem-index<U><0,1023> }:0
{ service-type<E><eth,tdm> }:eth
{ tcont<K> }:tcont
{ tcont-id<U><0,127> }:0
{ <cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K> }:
Command:
gem add 0 eth tcont 0

```

Configure the mapping between the GEM port and the traffic stream of the management VLAN.

# Map the traffic stream of management VLAN 500 to GEM port 0. The mapping index is 0.

```

huawei(config-gpon-lineprofile-32)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:mapping
{ gem-index<U><0,1023> }:0
{ mapping-index<U><0,7> }:0
{ e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K> }:vlan
{ untag<K>|vlan-id<U><0,4095> }:500
{ <cr>|flow-car<K>|priority<K>|transparent<K> }:
Command:
gem mapping 0 0 vlan 500

```

# Commit the parameters to make them take effect.

```

huawei(config-gpon-lineprofile-32)#commit
huawei(config-gpon-lineprofile-32)#quit

```

**Step 9** Configure the downstream port for the inband management VLAN of the OLT.

Configure an inband management traffic stream.

# In global config mode, create traffic stream 0 for management ONT 2. Set the management VLAN to **500**, GEM port ID to **0**, and user-side VLAN to **500**.

```

huawei(config)#service-port 0
{ desc<K>|index<U><0,32767>|remote-desc<K>|source<K>|uplink-port<K>|vlan<K> }:vlan
{ aoe<K>|name<K>|vlanid<U><1,4093> }:500
{ adsl<K>|atm<K>|epon<K>|eth<K>|gpon<K>|port<K>|shdsl<K>|vdsl<K> }:gpon
{ frameid/slotid/portid<S><Length 5-18> }:0/5/0
{ ont<K> }:ont
{ ontid<U><0,127> }:2
{ eth<K>|gemport<K>|iphost<K> }:gemport
{ gemindex<U><0,1023> }:0
{ <cr>|bundle<K>|inbound<K>|multi-service<K>|rx-cttr<K>|tag-transform<K> }:multi-service
{ user-8021p<K>|user-encap<K>|user-vlan<K> }:user-vlan
{ other-all<K>|priority-tagged<K>|untagged<K>|user-vlanid<U><1,4095> }:500

```

```
{ <cr>|bundle<K>|inbound<K>|rx-cttr<K>|tag-transform<K>|user-encap<K> }:
```

Command:

```
service-port 0 vlan 500 gpon 0/5/0 ont 2 gemport 0 multi-service user-vlan 500
```

## Step 10 Test the inband management channel.

Check whether the management channel between the OLT and the ONT is available.

# Run the **ping 10.175.192.41** command on the OLT to check the connectivity between the OLT and ONT 2. Check whether the OLT can receive the response packets from ONT 2.

```
huawei(config)#ping 10.175.192.41
PING 10.175.192.41: 56 data bytes, press CTRL_C to break
Reply from 10.175.192.41: bytes=56 Sequence=1 ttl=255 time=22 ms
Reply from 10.175.192.41: bytes=56 Sequence=2 ttl=255 time=8 ms
Reply from 10.175.192.41: bytes=56 Sequence=3 ttl=255 time=8 ms
Reply from 10.175.192.41: bytes=56 Sequence=4 ttl=255 time=8 ms
Reply from 10.175.192.41: bytes=56 Sequence=5 ttl=255 time=8 ms
--- 10.175.192.41 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 8/10/22 ms
```

## Step 11 Save the data.

# Save the data before exiting.

```
huawei(config)#save
{ <cr>|configuration<K>|data<K> }:
Command:
save
It will take several minutes to save configuration file, please wait...
```

## 2.3 Result Verification

### 2.3.1 Checking the Status of the Added ONTs on the OLT

# Check the status of the ONT that is added to GPON port 0/5/0.

```
huawei(config-if-gpon-0/5)#display ont info 0 all
```

| F/S/P  | ONT ID | SN               | Control flag | Run state | Config state | Match state | Protect side |
|--------|--------|------------------|--------------|-----------|--------------|-------------|--------------|
| 0/ 5/0 | 1      | 48575443D8EAC605 | active       | online    | normal       | match       | no           |
| 0/ 5/0 | 2      | 4857544329518907 | active       | online    | normal       | match       | no           |
| 0/ 5/0 | 5      | 48575443D8F92405 | active       | online    | normal       | match       | no           |

```

F/S/P  ONT-ID  Description
-----

```

```
0/ 5/0      1  ONT_NO_DESCRIPTION
0/ 5/0      2  ONT_NO_DESCRIPTION
0/ 5/0      5  ONT_NO_DESCRIPTION
```

-----  
In port 0/ 5/0 , the total of ONTs are: 3, online: 3

#### NOTE

After adding an ONT, run the **display ont info** command to query the status of the ONT. Ensure that **Control flag** is **active**, **Run state** is **online**, **Config state** is **normal**, and **Match state** is **match**.

When **Config state** is **failed**, **Run state** is **offline**, or **Match state** is **mismatch**, refer to the following suggestions to rectify the fault:

If **Control flag** is **deactive**, run the **ont activate** command in GPON mode to activate the ONT.

If **Run state** is **offline**, a physical line may be broken, the ONT is not powered on, or the optical module may be damaged. Check the line and the optical module.

If **Config state** is **failed**, run the **display ont failed-configuration** command in diagnosis mode to check the failed configuration item and the failure cause. Rectify the fault accordingly.

If **Match state** is **mismatch**, run the **display ont capability** command to query the actual capability of the ONU, and use one of the following methods to modify the ONU configuration:

- Create a proper ONU profile based on the actual capability of the ONU, and run the **ont modify** command to modify the ONU configuration.
- Modify the ONU profile based on the actual capability of the ONU and save the modification. The ONU will automatically recover the configuration.

## 2.3.2 Querying Information About the ONU

Test the inband management channel and ensure that the management channel between the OLT and the ONT is available.

# Run the **ping 10.175.192.41** command on the OLT to check the connectivity between the OLT and ONT 2. Check whether the OLT can receive the response packets from ONT 2.

```
huawei(config)#ping 10.175.192.41
PING 10.175.192.41: 56 data bytes, press CTRL_C to break
  Reply from 10.175.192.41: bytes=56 Sequence=1 ttl=255 time=20 ms
  Reply from 10.175.192.41: bytes=56 Sequence=2 ttl=255 time=10 ms
  Reply from 10.175.192.41: bytes=56 Sequence=3 ttl=255 time=12 ms
  Reply from 10.175.192.41: bytes=56 Sequence=4 ttl=255 time=9 ms
  Reply from 10.175.192.41: bytes=56 Sequence=5 ttl=255 time=11 ms
--- 10.175.192.41 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
round-trip min/avg/max = 9/12/20 ms
```

If response packets from ONT 2 are received, the management channel between the OLT and the ONT is available.

Log in to OLT 2.

# Run the **telnet 10.175.192.41** command on an OLT to log in to ONT 2. The user name is **root** and password is **mduadmin**.

```

huawei(config)#telnet 10.175.192.41
{ <cr>|service-port<U><1,65535> }:
  Command:
    telnet 10.175.192.41
  Press CTRL_] to quit telnet mode
  Trying 10.175.192.41 ...
  Connected to 10.175.192.41 ...
>>User name:root
>>User password:
  Huawei Integrated Access Software (MA5822S).
  Copyright(C) Huawei Technologies Co., Ltd. 2002-2010. All rights reserved.
  -----
  User last login information:
  -----
  Access Type: Telnet
  IP-Address : 10.175.192.41
  Time      : 2018-04-11 15:54:01+08:00
  -----
MA5822S>

```

# Query the existing configurations of ONT 2.

```

MA5822S>enable
MA5822S#display current-configuration
{ <cr>|section<K>|simple<K>||<K> }:
  Command:
    display current-configuration
[MA562XV800R308: 3600]
#
[global-config]
<global-config>
trap convert event-to-alarm 0x3721000a
alarm alarmlevel 0x0a11a106 warning
alarm alarmlevel 0x0a11a109 warning
alarm alarmlevel 0x0a310112 warning
alarm alarmlevel 0x0a310142 warning
#
[device-config]
<device-config>
port port-index 0/0/2 port-type GPON-PORT
#
[public-config]
<public-config>
snmp-agent local-engineid 800007DB035C4CA929518A
snmp-agent community read public
snmp-agent community write private

```

```
snmp-agent trap enable standard
#
[vlan-config]
<vlan-config>
vlan 500 smart
port vlan 500 0/0 2
protocol-8021p-pri 5 vlan 500
#
[gponnni]
<gponnni-0/0/2>
interface gponnni 0/0/2
password 0000000000
#
[abs-config]
<abs-config>
config
#
[prevlanif]
<prevlanif>
interface vlanif500
#
[vlanif]
<vlanif500>
interface vlanif500
ip address 10.175.192.41 255.255.252.0
#
[meth]
<meth0>
interface meth0
ip address 10.11.104.2 255.255.255.0
#
[null]
<null0>
interface null0
#
[post-system]
<post-system>
ip route-static 0.0.0.0 0.0.0.0 10.175.192.1
#
return
# Exit the ONT.
MA5822S#quit
```

Check whether system data has been changed. Please save data before logout.

```
Are you sure to log out? (y/n)[n]:y
MA5822S#
Configuration console exit, please retry to log on
```

## 2.4 Configuration Reference

### 2.4.1 Configuration for Adding an ONT on an OLT

```
#
config
switch language-mode
#
board confirm 0
display board 0

display dba-profile all
dba-profile add profile-id 21 profile-name ftth_voip type1 fix 1024
dba-profile add profile-id 22 profile-name ftth_hsi type2 assure
dba-profile add profile-id 23 profile-name ftth_ipvtv type3 assure 4096 max 8064
#
display ont-lineprofile gpon all
ont-lineprofile gpon profile-id 31 profile-name HG824x
tcont 1 dba-profile-id 21
tcont 2 dba-profile-id 22
tcont 3 dba-profile-id 23
gem add 1 eth tcont 1
gem add 2 eth tcont 2
gem add 3 eth tcont 3
mapping-mode vlan
gem mapping 1 1 vlan 172
gem mapping 2 2 vlan 11
gem mapping 3 3 vlan 200
commit
quit
#
ont-srvprofile gpon profile-id 31 profile-name HG824x
ont-port eth 4 pots 2 catv 1
port vlan iphost 172
port vlan eth 1 11
port vlan eth 2 200
commit
quit
#
interface gpon 0/5
port 0 ont-auto-find enable
display ont autofind 0
ont confirm 0 ontid 1 sn-auth 48575443D8EAC605 omci ont-lineprofile-id 31 ont-srvprofile-id 31
display ont info 0 all
#
interface gpon 0/5
ont add 0 5 sn-auth 48575443D8F92405 omci ont-lineprofile-id 31 ont-srvprofile-id 31
```

## 2.4.2 Configuration for Adding an ONU on an OLT

```
#
ont-lineprofile gpon profile-id 32 profile-name MA5822S
tcont 1 dba-profile-id 21
tcont 2 dba-profile-id 22
tcont 3 dba-profile-id 23
gem add 4 eth tcont 1
gem add 5 eth tcont 2
gem add 6 eth tcont 3
gem add 0 eth tcont 0
mapping-mode vlan
gem mapping 4 4 vlan 172
gem mapping 5 5 vlan 11
gem mapping 6 6 vlan 200
gem mapping 0 0 vlan 500
commit
quit
#
interface gpon 0/5
display ont autofind 0
ont confirm 0 ontid 2 sn-auth 4857544329518907 snmp ont-lineprofile-id 32
#
vlan 500 smart
port vlan 500 0/19 0
interface vlanif 500
ip address 10.175.192.36 22
quit
#
interface gpon 0/5
ont ipconfig 0 2 static ip-address 10.175.192.41 mask 255.255.252.0 gateway 10.175.192.1
vlan 500
quit
#
service-port 0 vlan 500 gpon 0/5/0 ont 2 gemport 0 multi-service user-vlan 500
#
ping 10.175.192.41
#
telnet 10.175.192.41
root
mduadmin
```

## 2.5 Quiz

- If the ONT auto-find function is enabled but no ONT is automatically found, what should I do?
- When an ONU is added and the management channel between the OLT and the ONT is set up, no response packet is received after I run a ping command on the OLT to ping the connectivity. What should I do?

# 3 GPON ONT BIAS Configuration

## 3.1 Experiment Introduction

### 3.1.1 About This Experiment

In this experiment, you can configure OLT data and ONT data to provision the dial-up access service for a user using an ONT.

### 3.1.2 Objectives

- Understand the HSI service process.
- Understand the functions of VLANs.
- Be able to perform data configuration based on a network solution and data plan as guided.
- Be able to query and modify service data.

### 3.1.3 Networking

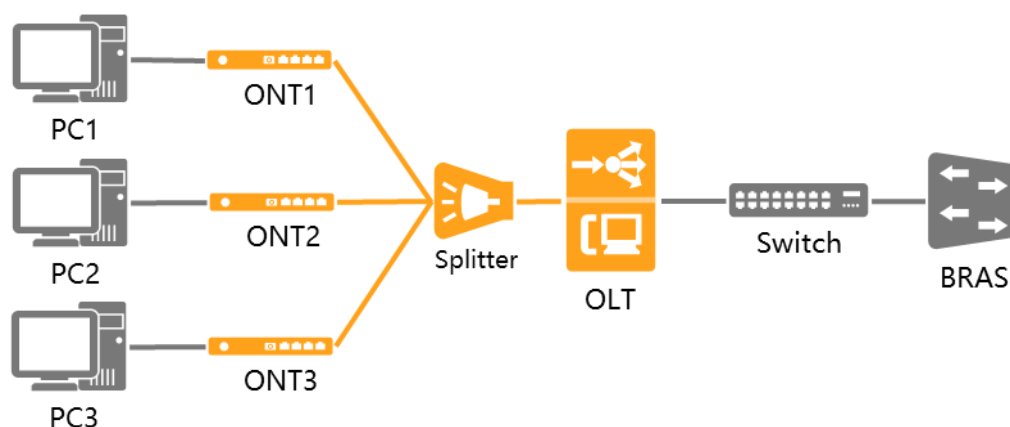


Figure 3-1 Topology for configuring the GPON ONT BIAS service

### 3.1.4 Data Plan

The following table provides a data plan reference. The data varies depending on PCs to avoid a data conflict. Some parameters such as the port ID and profile ID need to be planned based on actual situations.

**Table 3-1 Data plan for the GPON ONT BIAS service**

| Parameter          | Value            | Parameter              | Value                              |
|--------------------|------------------|------------------------|------------------------------------|
| OLT PON port       | 0/5/0            | CAR profile ID         | 10                                 |
| ONT ID             | 3                | CAR profile bandwidth  | CIR: 4096 kbit/s; PIR: 8192 kbit/s |
| ONT SN             | 48575443E1F0A409 | DBA profile ID         | 15                                 |
| Outer VLAN ID      | 2000             | ONT line profile ID    | 10                                 |
| Inner VLAN ID      | 35               | ONT service profile ID | 10                                 |
| User ETH port      | 1                | T-CONT ID              | 10                                 |
| C-VLAN             | 35               | GEM port ID            | 10                                 |
| DBA bandwidth type | Type4            | DBA bandwidth          | 4096 kbit/s                        |

## 3.2 Experiment Configuration

### 3.2.1 Configuration Tasks

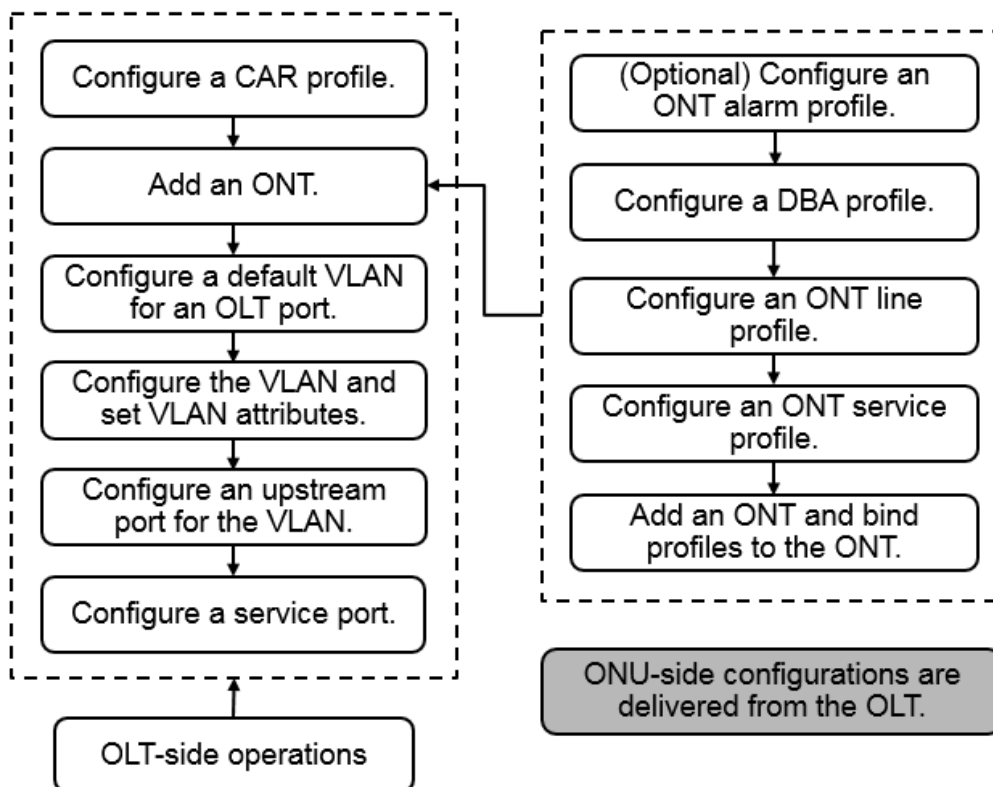
**Table 3-2 Tasks for configuring the GPON ONT BIAS service**

| Task                  | Requirement                                                                                                                                             | Completion Status |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Add an ONT.           | An ONT is automatically or manually added. Record the following ONT status data: control flag, running flag, configuration status, and matching status. |                   |
| Create an SVLAN.      | According to experiment data, create a VLAN and record the VLAN ID.                                                                                     |                   |
| Set VLAN attributes.  | Set the attributes so that double VLAN tags are carried. Record VLAN attributes.                                                                        |                   |
| Configure an upstream | Record the upstream port and                                                                                                                            |                   |

|                                                        |                                                                                                                                                                                                                                           |  |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| port of the VLAN.                                      | VLAN IDs that are allowed to pass through the upstream port.                                                                                                                                                                              |  |
| Configure service ports of the VLAN.                   | Create mapping between the inner VLAN ID and outer VLAN ID of a specified ONT, and record the inner VLAN ID and outer VLAN ID separately.                                                                                                 |  |
| Configure a native VLAN ID for the FE port of the ONT. | In the receive direction of user data, the inner VLAN ID is stripped, and Ethernet frames not carrying a tag are forwarded. Record the FE port ID of the ONT, as well as the VLAN ID and native VLAN ID that are allowed to pass through. |  |
| Perform dial-up access.                                | Run dial-up software, enter the user name and password, and try dial-up authentication.                                                                                                                                                   |  |

### 3.2.2 Service Configuration Process

Figure 3-2 Flow for configuring the GPON ONT broadband internet access service



### 3.2.3 Configuration Procedure (on the OLT Side)

#### NOTE

In this example, the HG824x is connected to OLT PON port 0/5/0, the SN of the HG824x is 48575443E1F0A409, the ONT ID is 3, the inner VLAN (C-VLAN) ID is 35, and the outer VLAN (SVLAN) ID is 2000. Parameters such as the profile name, profile ID, VLAN ID, and ONT ID need to be modified based on the actual data plan.

**Step 1** Log in to the OLT and prepare the configuration environment.

Specifically, log in to the OLT, enter configuration mode, and change the device name.

# Log in to the OLT in Telnet mode, and enter the user name and password (**huawei** and **huawei123** respectively).

```
>>User name:huawei
```

```
>>User password:
```

```
Huawei Integrated Access Software (MA5600T).
```

```
Copyright(C) Huawei Technologies Co., Ltd. 2002-2013. All rights reserved.
```

```
-----
User last login information:
-----
```

```
Access Type : Serial
IP-Address  : --
Login  Time : 2018-06-04 17:38:25+08:00
Logout Time : 2018-06-04 17:49:42+08:00
-----
```

User root has used a default password. Change the password in time.

```
MA5600T>enable
MA5600T#config
MA5600T(config)#
```

# Chang the device name.

```
MA5600T(config)#sysname huawei
huawei(config)#
```

Configure board information.

# Check a board and ensure that the board runs normally.

huawei(config)#display board 0 // Query the board running status. If the board status is not normal, further check the board status.

```
huawei(config)#board confirm
{ frameid[/slotid]<S><Length 1-15> }:0
Command:
display board 0
```

```
-----
SlotID  BoardName  Status          SubType0 SubType1  Online/Offline
-----
0
1      H802EPBD  Normal
2
3
4
5      H802GPBD  Normal
6      H802EPBD  Normal
7      H805GPBD  Normal
8
9      H801SCUN  Active_normal
10
11
12
13
14
15      H802GPBD  Normal
16
17
18
19      H801GICG  Normal
20
21
22
```

## Step 2 (Optional) Configure a CAR profile.

Traffic control can be implemented on the BRAS, or on the OLT or ONU by using port rate limitation or using a traffic profile to limit the upstream and downstream traffic.

You can query the existing traffic profiles in the system. If the existing traffic profiles in the system do not meet the requirement, you can run the **traffic table ip** command to create a required traffic profile.

# Check whether an appropriate IP traffic profile exists in the system.

```
MA5822S(config)#display traffic table ip
{ from-index<K>|index<K>|name<K> }:from-index
{ row-index<U><0,63> }:0
{ <cr>|to-index<K> }:
```

Command:

```
display traffic table ip from-index 0
```

| TID | CIR<br>(kbps) | CBS<br>(bytes) | PIR<br>(kbps) | PBS<br>(bytes) | Pri | Copy-policy | Pri-Policy |
|-----|---------------|----------------|---------------|----------------|-----|-------------|------------|
| 0   | 1024          | 34768          | 2048          | 69536          | 6   | -           | tag-pri    |
| 1   | 2496          | 81872          | 4992          | 163744         | 6   | -           | tag-pri    |
| 2   | 512           | 18384          | 1024          | 36768          | 0   | -           | tag-pri    |
| 3   | 576           | 20432          | 1152          | 40864          | 2   | -           | tag-pri    |
| 4   | 64            | 4048           | 128           | 8096           | 4   | -           | tag-pri    |
| 5   | 2048          | 67536          | 4096          | 135072         | 0   | -           | tag-pri    |
| 6   | off           | off            | off           | off            | 0   | -           | tag-pri    |

Total Num : 7

# Create a traffic profile. Set the index to 10, CIR to 4096 kbit/s, PIR to 8064 kbit/s, and priority to 5. The priority policy schedules traffic according to the priority configured in the traffic profile.

```
huawei(config)#traffic
{ table<K> }:table
{ ip<K> }:ip
{ cir<K>|index<K>|modify<K>|name<K> }:index
{ row-index<U><0,1023> }:10
{ cir<K>|name<K> }:cir
{ cir<U><64,10240000>|off<K> }:4096
{ cbs<K>|pbs<K>|pir<K>|priority<K> }:pir
{ pir<U><64,10240000> }:8064
{ cbs<K>|pbs<K>|priority<K> }:priority
{ prival<U><0,7>|user-cos<K>|user-inner-cos<K>|user-tos<K> }:user-cos
{ defaultval<U><0,7>|mapping-profile<K> }:5
{ inner-priority<K>|priority-policy<K> }:priority-policy
{ priority-policy<E><Local-Setting,Tag-In-Package> }:local-Setting
```

Command:

```
traffic table ip index 10 cir 4096 pir 8064 priority user-cos 5
priority-policy local-Setting
```

Create traffic descriptor record successfully

```
-----
TD Index          : 10
TD Name           : ip-traffic-table_10
Priority          : 5
Copy Priority     : user-cos
Mapping Index     : 0
CTAG Mapping Priority: -
CTAG Mapping Index : -
CTAG Default Priority: 0
Priority Policy    : local-pri
CIR               : 4096 kbps
CBS               : 133072 bytes
PIR               : 8064 kbps
PBS               : 260048 bytes
CAR Threshold Profile: -
Color policy      : dei
Referenced Status : not used
-----
```

### Step 3 Configure a DBA profile.

Create a DBA profile. A DBA profile describes GPON traffic parameters. A T-CONT is bound to the DBA profile for dynamic bandwidth allocation, improving upstream bandwidth usage.

# Run the **display dba-profile** command to query existing DBA profiles in the system. If existing DBA profiles in the system do not meet the requirement, run the **dba-profile add** command to create a DBA profile. Set the DBA profile ID to 15, bandwidth type to **type4**, and bandwidth to 4096 kbit/s.

```
huawei(config)#dba-profile
{ add<K>|delete<K>|modify<K> }:add
{ profile-id<K>|profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-id
{ profile-id<U><10,512> }:15
{ profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:type4
{ max<K> }:max
{ max-bandwidth<U><128,1235456> }:4096
Command:
    dba-profile add profile-id 15 type4 max 4096
Adding a DBA profile succeeded
Profile ID : 15
Profile name: dba-profile_15
```

# Query the status of the DBA profile.

```
huawei(config)#display dba-profile all
```

```
-----
Profile-ID  type  Bandwidth      Fix      Assure      Max      Bind
              compensation (kbps) (kbps) (kbps) times
-----
          0    3         No          0      8192      20480        1
          1    1         No      5120          0           0        2
          2    1         No     1024          0           0        1
          3    4         No          0          0      32768        0
-----
```

|    |   |    |         |        |       |   |
|----|---|----|---------|--------|-------|---|
| 4  | 1 | No | 1024000 | 0      | 0     | 0 |
| 5  | 1 | No | 32768   | 0      | 0     | 0 |
| 6  | 1 | No | 102400  | 0      | 0     | 0 |
| 7  | 2 | No | 0       | 32768  | 0     | 0 |
| 8  | 2 | No | 0       | 102400 | 0     | 0 |
| 9  | 3 | No | 0       | 32768  | 65536 | 0 |
| 15 | 4 | No | 0       | 0      | 4096  | 0 |

---

#### Step 4 Configure an ONT line profile.

# Set the line profile ID to 10 and the profile name to ftth\_hsi.

```

huawei(config)#ont-lineprofile
{ epon<K>|gpon<K> }:gpon
{ <cr>|profile-id<K>|profile-name<K> }:profile-id
{ profile-id<U><0,8192> }:10
{ <cr>|profile-name<K> }:profile-name
{ profile-name<S><Length 1-32> }:ftth_hsi
Command:
ont-lineprofile gpon profile-id 10 profile-name ftth_hsi
huawei(config-gpon-lineprofile-10)#

```

# Bind T-CONT 10 to DBA profile 15.

```

huawei(config-gpon-lineprofile-10)#tcont
{ tcont-list<S><Length 1-13> }:10
{ <cr>|dba-profile-id<K>|dba-profile-name<K> }:dba-profile-id
{ profile-id<U><0,512> }:15
Command:
tcont 10 dba-profile-id 15

```

# Create GEM port 10 for carrying ETH traffic streams and bind GEM port 10 to T-CONT 10.

```

huawei(config-gpon-lineprofile-10)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:add
{ gem-index<U><0,1023> }:10
{ service-type<E><eth,tdm> }:eth
{ tcont<K> }:tcont
{ tcont-id<U><0,127> }:10
{ <cr>|cascade<K>|encrypt<K>|gem-car<K>|priority-queue<K> }:
Command:
gem add 10 eth tcont 10

```

Configure the service mapping mode from the GEM port to ONT services to VLAN (default), and map traffic streams with VLAN 35 to GEM port 10.

# Set the mapping mode of the GEM port to VLAN.

```

huawei(config-gpon-lineprofile-10)#mapping-mode
{ iptos<K>|port-priority<K>|port-vlan-priority<K>|port-vlan<K>|port<K>|priority<K>|vlan-
priority<K>|vlan<K> }:vlan
Command:

```

```
mapping-mode vlan
```

# Map traffic streams with C-VLAN 35 to GEM port 10. The mapping index is 1.

```
huawei(config-gpon-lineprofile-10)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:mapping
{ gem-index<U><0,1023> }:10
{ mapping-index<U><0,7> }:1
{ e1<K>|eth-bundle<K>|eth<K>|flow-
car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|transparent<K>|vlan<K> }:vlan
{ vlan-id<U><0,4095> }:35
{ <cr>|flow-car<K>|priority<K>|transparent<K> }:
Command:
gem mapping 10 1 vlan 35
```

# After the configuration is completed, run the **commit** command to make the configured parameters take effect.

```
huawei(config-gpon-lineprofile-10)#commit
huawei(config-gpon-lineprofile-10)#quit
```

# Query information about a line profile.

```
huawei(config)#display ont-lineprofile
{ epon<K>|gpon<K> }:gpon
{ all<K>|bound-info<K>|profile-id<K>|profile-name<K> }:profile-id
{ profile-id<U><0,8192> }:10
Command:
display ont-lineprofile gpon profile-id 10
```

|              |          |
|--------------|----------|
| Profile-ID   | :10      |
| Profile-name | :ftt_hsi |
| Access-type  | :GPON    |

```

FEC upstream switch :Disable
OMCC encrypt switch :Off
Qos mode             :PQ
Mapping mode         :VLAN
Tr069 management    :Disable

```

|                |                   |
|----------------|-------------------|
| <T-CONT 0>     | DBA Profile-ID:1  |
| <T-CONT 10>    | DBA Profile-ID:15 |
| <Gem Index 10> |                   |

```

|Serv-Type:ETH |Encrypt:off |Cascade:off |GEM-CAR:-
|Upstream-priority-queue:0 |Downstream-priority-queue:-

```

| Mapping index | VLAN | Priority | Port type | Port ID | Bundle ID | Flow CAR | Transparent |
|---------------|------|----------|-----------|---------|-----------|----------|-------------|
| 1             | 35   | -        | -         | -       | -         | -        | -           |

Binding times :0

## Step 5 Configure a service profile.

A GPON ONT service profile is used to configure the actual ONT capability and service-related parameters, and provides a service configuration channel for the ONT that is managed by using optical network terminal management and control interface (OMCI).

# Set the service profile ID to 10 and profile name to HG824x.

```
huawei(config)#ont-srvprofile
{ epon<K>|gpon<K> }:gpon
{ <cr>|profile-id<K>|profile-name<K> }:profile-id
{ profile-id<U><0,8192> }:10
{ <cr>|profile-name<K> }:profile-name
{ profile-name<S><Length 1-32> }:HG824x
Command:
ont-srvprofile gpon profile-id 10 profile-name HG824x
```

# Configure the ONT port type and quantity: 4 LAN ports, 2 POTS ports, and 1 CATV port.

```
huawei(config-gpon-srvprofile-10)#ont-port
{ catv<K>|eth<K>|moca<K>|pots<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K> }:catv
{ adaptive<K>|catv-port<U><0,8> }:1
{ catv<K>|eth<K>|moca<K>|pots<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K> }:eth
{ adaptive<K>|eth-port<U><0,8> }:4
{ <cr>|catv<K>|moca<K>|pots<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K> }:pots
{ adaptive<K>|pots-port<U><0,8> }:2
{ <cr>|catv<K>|moca<K>|tdm-srvtype<K>|tdm-type<K>|tdm<K> }:  
Command:
ont-port catv 1 eth 4 pots 2
```

# Add ONT port 1 under the Ethernet port in GPON ONT service profile 10 to VLAN 35.

```
huawei(config-gpon-srvprofile-10)#port
{ car<K>|dscp<K>|e1<K>|igmp-forward<K>|priority-policy<K>|q-in-q<K>|vlan<K> }:vlan
{ eth<K>|iphost<K>|moca<K> }:eth
{ ont-portlist<S><Length 1-128> }:1
{ q-in-q<K>|translation<K>|transparent<K>|vlanid<U><0,4095> }:35
{ <cr>|priority<K>|prival<U><0,7>|TLS<K> }:  
Command:
port vlan eth 1 35
Set ONT port(s) VLAN configuration, success: 1, failed: 0
```

# After the configuration is completed, run the **commit** command to make the configured parameters take effect.

```
huawei(config-gpon-srvprofile-10)#commit
huawei(config-gpon-srvprofile-10)#quit
```

# Query the service module status.

```

huawei(config)#display ont-srvprofile
{ epon<K>|gpon<K> }:gpon
{ all<K>|profile-id<K>|profile-name<K> }:profile-id
{ profile-id<U><0,8192> }:10
Command:
display ont-srvprofile gpon profile-id 10

```

```

Profile-ID : 10
Profile-name: HG824x
Access-type : GPON

```

| Port-type | Port-number |
|-----------|-------------|
| POTS      | 2           |
| ETH       | 4           |
| TDM       | 0           |
| MOCA      | 0           |
| CATV      | 1           |

```

TDM port type : E1
TDM service type : TDMoGem
MAC learning function switch : Enable
ONT transparent function switch : Disable
Multicast forward mode : Unconcern
Multicast forward VLAN : -
Multicast mode : Unconcern
Upstream IGMP packet forward mode : Unconcern
Upstream IGMP packet forward VLAN : -
Upstream IGMP packet priority : -
Native VLAN option : Concern
Upstream PQ color policy : -
Downstream PQ color policy : -

```

| Port-type | Port-ID | QinQmode  | PriorityPolicy | Inbound   | Outbound  |
|-----------|---------|-----------|----------------|-----------|-----------|
| ETH       | 1       | unconcern | unconcern      | unconcern | unconcern |
| ETH       | 2       | unconcern | unconcern      | unconcern | unconcern |
| ETH       | 3       | unconcern | unconcern      | unconcern | unconcern |
| ETH       | 4       | unconcern | unconcern      | unconcern | unconcern |

| Port-type | Port-ID | Dscp-mapping-table-index |
|-----------|---------|--------------------------|
| ETH       | 1       | 0                        |
| ETH       | 2       | 0                        |
| ETH       | 3       | 0                        |
| ETH       | 4       | 0                        |
| IPHOST    | 1       | 0                        |

| Port type | Port ID | Service-type | Index | S-VLAN | S-PRI | C-VLAN | C-PRI | ENCAP | S-PRI POLICY |
|-----------|---------|--------------|-------|--------|-------|--------|-------|-------|--------------|
| ETH       | 1       | Translation  | 1     | 35     | -     | 35     | -     | -     | -            |

Notes: \* indicates transparent attribute of the vlan

| Port-type         | Port-ID | IGMP-mode | IGMP-VLAN | IGMP-PRI | Max-MAC-Count |
|-------------------|---------|-----------|-----------|----------|---------------|
| ETH               | 1       | -         | -         | -        | Unlimited     |
| ETH               | 2       | -         | -         | -        | Unlimited     |
| ETH               | 3       | -         | -         | -        | Unlimited     |
| ETH               | 4       | -         | -         | -        | Unlimited     |
| Binding times : 0 |         |           |           |          |               |

### Step 6 Enable the automatic discovery function of a PON port.

Enter the PON board view, and enable the automatic discovery function of a PON port in the network topology.

# Enter the view of PON board 0/5, and enable the automatic discovery function of GPON port 0/5/0.

```

huawei(config)#interface gpon 0/5
huawei(config-if-gpon-0/5)#port
{ optical-alarm-profile<K>|portid<U><0,7> } : 0
{ fec<K>|laser-switch<K>|ont-auto-find<K>|ont-password-renew<K>|range<K> } : ont-auto-find
{ switch<E><enable,disable> } : enable
Command:
port 0 ont-auto-find enable

```

### Step 7 Query the ONTs that are automatically discovered by the PON port.

```

huawei(config-if-gpon-0/5)#display ont autofind
{ portid<U><0,7> } : 0
Command:
display ont autofind 0

```

|                     |                             |
|---------------------|-----------------------------|
| Number              | : 1                         |
| F/S/P               | : 0/5/0                     |
| Ont SN              | : 48575443E1F0A409          |
| Password            | : 0x00000000000000000000    |
| Loid                | :                           |
| Checkcode           | :                           |
| VendorID            | : HWTC                      |
| Ont Version         | : 120D4600                  |
| Ont SoftwareVersion | : V1R006C00S210             |
| Ont EquipmentID     | : 247                       |
| Ont autofind time   | : 2018-04-10 14:35:44+08:00 |

### Step 8 Register an ONT.

# Register an ONT, set the authentication mode to SN authentication and management type to OMCI, and bind line profile 10 and service profile 10 to the ONT.

```

huawei(config-if-gpon-0/5)# ont

```

```
{ activate<K>|add<K>|alarm-policy<K>|alarm-profile<K>|cancel<K>|confirm<K>|cos-queue-map-
profile<K>|deactivate<K>|delete<K>|factory-setting-
restore<K>|gemport<K>|ipconfig<K>|modify<K>|optical-alarm-profile<K>|port<K>|power-shedding-
config<K>|re-discovery<K>|re-register<K>|remote-ping<K>|reset<K>|snmp-profile<K>|snmp-
route<K>|statistics<K>|tdm-vcl<K>|tr069-server-config<K>|voip<K> }:confirm
{ portid<U><0,7> }:0
{ all<K>|loid-auth<K>|ontid<K>|password-auth<K>|sn-auth<K> }:ontid
{ ontid<U><0,127> }:3
{ loid-auth<K>|password-auth<K>|sn-auth<K> }:sn-auth
{ sn<S><Length 13-16> }:48575443E1F0A409
{ omci<K>|password-auth<K>|snmp<K> }:omci
{ <cr>|desc<K>|ont-lineprofile-id<K>|ont-lineprofile-name<K>|ont-srvprofile-id<K>|ont-srvprofile-
name<K> }:ont-lineprofile-id
{ profile-id<U><0,8192> }:10
{ <cr>|desc<K>|ont-srvprofile-id<K>|ont-srvprofile-name<K> }:ont-srvprofile-id
{ profile-id<U><0,8192> }:10
{ <cr>|desc<K> }:  
Command:  
ont confirm 0 ontid 3 sn-auth 48575443E1F0A409 omci ont-lineprofile-id 10 ont-
srvprofile-id 10  
Number of ONTs that can be added: 1, success: 1  
PortID :0, ONTID :1
```

# Query the ONT status.

```
huawei(config-if-gpon-0/5)#display ont info
{ portid<U><0,7> }:0
{ all<K>|ontid<U><0,127> }:all
Command:  
display ont info 0 all
```

| F/S/P  | ONT<br>ID | SN                 | Control<br>flag | Run<br>state | Config<br>state | Match<br>state | Protect<br>side |
|--------|-----------|--------------------|-----------------|--------------|-----------------|----------------|-----------------|
| 0/ 5/0 | 3         | 48575443E1F0A409   | active          | online       | normal          | match          | no              |
| F/S/P  | ONT-ID    | Description        |                 |              |                 |                |                 |
| 0/ 5/0 | 3         | ONT_NO_DESCRIPTION |                 |              |                 |                |                 |

In port 0/ 5/0 , the total of ONTs are: 1, online: 1

# To query detailed information about an ONT, run the following command: **display ont info 0 3**

**Step 9** Configure a default VLAN for the ETH 1 port of an ONT.

# Configure VLAN 35 as the default VLAN for the ETH 1 port of ONT 3 on PON 0/5/0. When the ETH 1 port receives an untagged packet from a user, it adds a VLAN 35 tag to the packet. When data is sent from this port to a user, the port strips the VLAN 35 tag.

```
huawei(config-if-gpon-0/5)#ont
```

```
{ activate<K>|add<K>|alarm-policy<K>|alarm-profile<K>|cancel<K>|confirm<K>|cos-queue-map-
profile<K>|deactivate<K>|delete<K>|factory-setting-
restore<K>|gemport<K>|ipconfig<K>|modify<K>|optical-alarm-profile<K>|port<K>|power-shedding-
config<K>|re-discovery<K>|re-register<K>|remote-ping<K>|reset<K>|snmp-profile<K>|snmp-
route<K>|statistics<K>|tdm-vcl<K>|tr069-server-config<K>|voip<K> }:port
{ attribute<K>|car<K>|igmp-forward-mode<K>|multicast-forward<K>|native-vlan<K>|vlan<K> }:native-
vlan
{ portid<U><0,7> }:0
{ ontid<U><0,127> }:3
{ eth<K>|iphost<K>|moca<K> }:eth
{ ont-portid<U><1,8> }:1
{ priority<K>|vlan<K> }:vlan
{ vlanid<U><0,4095> }:35
{ <cr>|priority<K> }:  
Command:  
ont port native-vlan 0 3 eth 1 vlan 35
```

## Step 10 Configure a broadband service VLAN.

# Configure data service VLAN 2000 and set its type to smart.

```
huawei(config)#vlan
{ attrib<K>|bind<K>|desc<K>|forwarding<K>|loop<K>|packet-policy<K>|priority<K>|reserve<K>|service-
profile<K>|vlan-list<S><Length 1-256>|vlanid<U><2,4093> }:2000
{ <cr>|to<K>|vlantype<E><mux,standard,smart,super> }:smart
Command:  
vlan 2000 smart
```

# Set the attribute of VLAN 2000 to QinQ.

```
huawei(config)#vlan
{ attrib<K>|bind<K>|desc<K>|forwarding<K>|loop<K>|packet-policy<K>|priority<K>|reserve<K>|service-
profile<K>|vlan-list<S><Length 1-256>|vlanid<U><2,4093> }:attrib
{ bydesc<K>|vlan-list<S><Length 1-256>|vlanid<U><2,4093> }:2000
{ common<K>|q-in-q<K>|stacking<K>|to<K> }:q-in-q
Command:  
vlan attrib 2000 q-in-q
```

# Add a data VLAN upstream port that allows VLAN 2000 to pass through.

```
huawei(config)#port
{ desc<K>|mode<K>|vlan<K> }:vlan
{ vlan-list<S><Length 1-256>|vlanid<U><1,4093> }:2000
{ frame/slot<S><Length 1-15>|to<K> }:0/19
{ portlist<S><Length 1-256> }:0
Command:  
port vlan 2000 0/19 0
```

# Query VLAN 2000.

```
huawei(config)#display vlan
{ all<K>|aoe<K>|desc<K>|loop<K>|number<K>|reserve<K>|service-profile<K>|vlan-list<S><Length 1-
256>|vlanid<U><1,4093> }:2000
```

```
{ <cr>|to<K> }:
```

Command:

```
display vlan 2000
VLAN ID: 2000
VLAN name: VLAN_2000
VLAN type: smart
VLAN attribute: QinQ
VLAN description:
VLAN forwarding mode in control board: VLAN-MAC
VLAN forwarding mode: VLAN-MAC
VLAN broadcast packet forwarding policy: forward
VLAN unknown multicast packet forwarding policy: forward
VLAN unknown unicast packet forwarding policy: forward
VLAN bind service profile ID: -
VLAN bind RAIO profile index: -
VLAN priority: -
```

| F / S / P  | Native VLAN | State |
|------------|-------------|-------|
| 0 / 19 / 0 | 1           | up    |

Standard port number: 1  
Service virtual port number: 0

### Step 11 Create a GPON service port for VLAN 2000.

# Create service port 10, bind it to port 0/5/0, set the C-VLAN ID to 35, and bind traffic profile 10 to the port.

```
huawei(config)#service-port 10
{ desc<K>|index<U><0,32767>|remote-desc<K>|source<K>|vlan<K> }:vlan
{ aoe<K>|vlanid<U><1,4093> }:2000
{ adsl<K>|atm<K>|epon<K>|eth<K>|gpon<K>|port<K>|shdsl<K>|vdsl<K> }:gpon
{ frameid/slotid/portid<S><Length 1-15> }:0/5/0
{ ont<K> }:ont
{ ontid<U><0,127> }:3
{ eth<K>|gempport<K>|iphost<K> }:gempport
{ gemindex<U><0,1023> }:10
{ <cr>|bundle<K>|inbound<K>|multi-service<K>|rx-cttr<K>|tag-transform<K> }:multi-service
{ user-8021p<K>|user-encap<K>|user-vlan<K> }:user-vlan
{ other-all<K>|priority-tagged<K>|untagged<K>|user-vlanid<U><1,4095> }:35
{ <cr>|bundle<K>|inbound<K>|rx-cttr<K>|tag-transform<K>|user-encap<K> }:rx-cttr
{ rx-index<U><0,1023> }:10
{ tx-cttr<K> }:tx-cttr
{ tx-index<U><0,1023> }:10
Command:
service-port 10 vlan 2000 gpon 0/5/0 ont 3 gempport 10 multi-service
user-vlan 35 rx-cttr 10 tx-cttr 10
```

The following table describes common commands.

| Common Command | Function |
|----------------|----------|
|----------------|----------|

|                          |                                                                                                                                                                                                                                                                 |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| display dba-profile      | Queries information about DBA profiles in the system.                                                                                                                                                                                                           |
| display traffic table ip | Queries parameter settings of a specified traffic table. The settings include the CAR, priority, and priority policy.                                                                                                                                           |
| display ont-lineprofile  | Queries information about a GPON ONT line profile that has taken effect. The information includes the following: binding relationship between a T-CONT and a DBA profile, QoS mode of a traffic stream, and mapping between a GEM port and services on the ONT. |
| display ont-srvprofile   | Queries information about a GPON ONT service profile.                                                                                                                                                                                                           |
| display ont autofind     | Queries the basic information of an automatically discovered ONT in the system.                                                                                                                                                                                 |
| display ont info         | Queries ONT information, including the current ONT status, ONT settings, and ONT T-CONT information.                                                                                                                                                            |
| display vlan             | Queries information about a VLAN.                                                                                                                                                                                                                               |
| display service-port     | Queries information about a service port.                                                                                                                                                                                                                       |

## 3.3 Result Verification

### 3.3.1 Accessing the Internet Through PPPoE Dial-up

# Connect the FE port of an ONT with HSI services configured to a PC.

# On the PC, use PPPoE dial-up software to dial up or DHCP. If an IP address is obtained, the configuration is successful.

## 3.4 Configuration Reference

### 3.4.1 Configurations on the OLT Side

```
>>User name:huawei
>>User password:huawei123
#
enable
config
sysname huawei
#
display board 0
board confirm 0
#
```

```

display traffic table ip from-index 0
traffic table ip index 10 cir 4096 pir 8064 priority user-cos 5 priority-policy local-Setting
#
dba-profile add profile-id 15 type4 max 4096
display dba-profile all
#
ont-lineprofile gpon profile-id 10 profile-name ftth_hsi
tcont 10 dba-profile-id 15
gem add 10 eth tcont 10
mapping-mode vlan
gem mapping 10 1 vlan 35
commit
quit
#
display ont-lineprofile gpon profile-id 10
#
ont-srvprofile gpon profile-id 10 profile-name HG824x
ont-port catv 1 eth 4 pots 2
port vlan eth 1 35
commit
quit
#
display ont-srvprofile gpon profile-id 10
#
interface gpon 0/5
port 0 ont-auto-find enable
display ont autofind 0
ont confirm 0 ontid 3 sn-auth 48575443E1F0A409 omci ont-lineprofile-id 10 ont-srvprofile-id 10
display ont info 0 all
#
ont port native-vlan 0 3 eth 1 vlan 35
vlan 2000 smart
vlan attrib 2000 q-in-q
port vlan 2000 0/19 0
display vlan 2000
#
service-port 10 vlan 2000 gpon 0/5/0 ont 3 gemport 10 multi-service user-vlan 35 rx-cttr 10
tx-cttr 10

```

## 3.5 Quiz

In this experiment, the BIAS service is configured. According to the data plan, packets in a traffic stream carry double VLAN tags, and the tags are terminated after the packets reach the broadband access server (BAS). If the topology is changed and packets in a traffic stream are required to carry single VLAN tags, what are the required configurations?

# 4 GPON ONU BIAS Configuration

---

## 4.1 Experiment Introduction

### 4.1.1 About This Experiment

In this experiment, you can configure OLT data and MDU data to provision the dial-up access service for a user using an MDU.

### 4.1.2 Objectives

- Understand the FTTB HSI service process.
- Understand the functions of an MDU VLAN and an OLT VLAN.

- Understand the function of a service port.
- Be able to perform data configuration based on a network solution and data plan as guided.

### 4.1.3 Networking

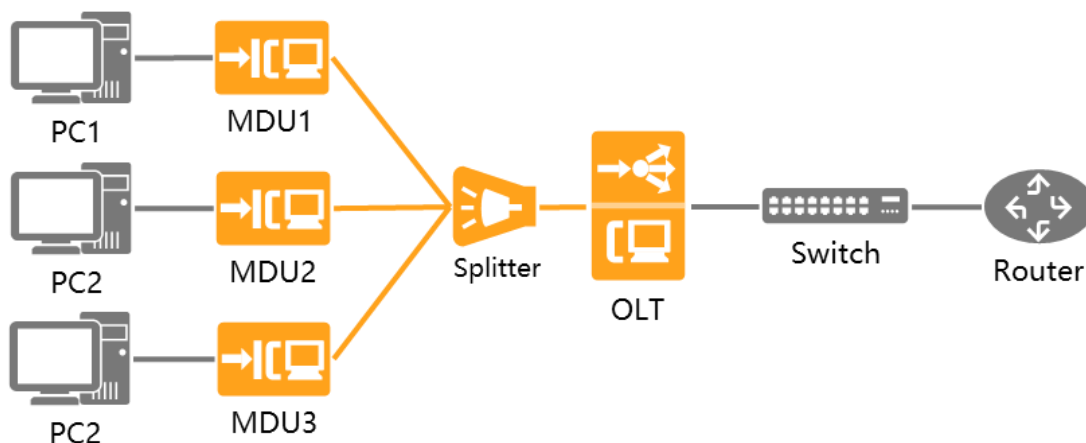


Figure 4-1 Topology for configuring the GPON ONU BIAS service

### 4.1.4 Data Plan

*The following table provides a data plan reference. The data varies depending on PCs to avoid a data conflict. Some parameters such as the port ID and profile ID need to be planned based on actual situations.*

Table 4-1 Data plan for the GPON ONU BIAS service

| ONT Data Plan   | ONU              |
|-----------------|------------------|
| Device type     | MA5822S          |
| OLT PON port    | 0/5/0            |
| ONT_ID          | 4                |
| ONT SN          | 48575443F444CB04 |
| C-VLAN          | 36               |
| User ETH port   | ETH1             |
| Inner VLAN      | 36               |
| SVLAN           | 2001             |
| Management mode | SNMP             |

|                              |                                                                                                                                                                                                   |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dba-profile                  | Profile ID: 25 <ul style="list-style-type: none"> <li>Profile name: fttb_hsi</li> <li>Bandwidth type: Type3</li> <li>CIR: 20480 kbit/s</li> <li>PIR: 40960 kbit/s</li> </ul>                      |
| ont-lineprofile              | <ul style="list-style-type: none"> <li>Profile ID: 20</li> <li>Profile name: MA5822S</li> </ul>                                                                                                   |
|                              | Inband management: <ul style="list-style-type: none"> <li>GEM port 0 buffered to T-CONT 0</li> <li>GEM port 0 mapped to VLAN 500</li> </ul>                                                       |
| Inband management IP address | Broadband service: <ul style="list-style-type: none"> <li>T-CONT ID 20 bound to DBA profile 25</li> <li>GEM port ID 20 buffered to T-CONT 20</li> <li>GEM port ID 20 mapped to VLAN 36</li> </ul> |
|                              | <ul style="list-style-type: none"> <li>OLT management VLAN: 500</li> <li>Upstream port: 0/19/0</li> <li>Static IP address: 10.175.192.36/22</li> </ul>                                            |
|                              | <ul style="list-style-type: none"> <li>ONT management VLAN: 500</li> <li>Static IP address: 10.175.192.42/22</li> </ul>                                                                           |

## 4.2 Experiment Configuration

### 4.2.1 Configuration Tasks

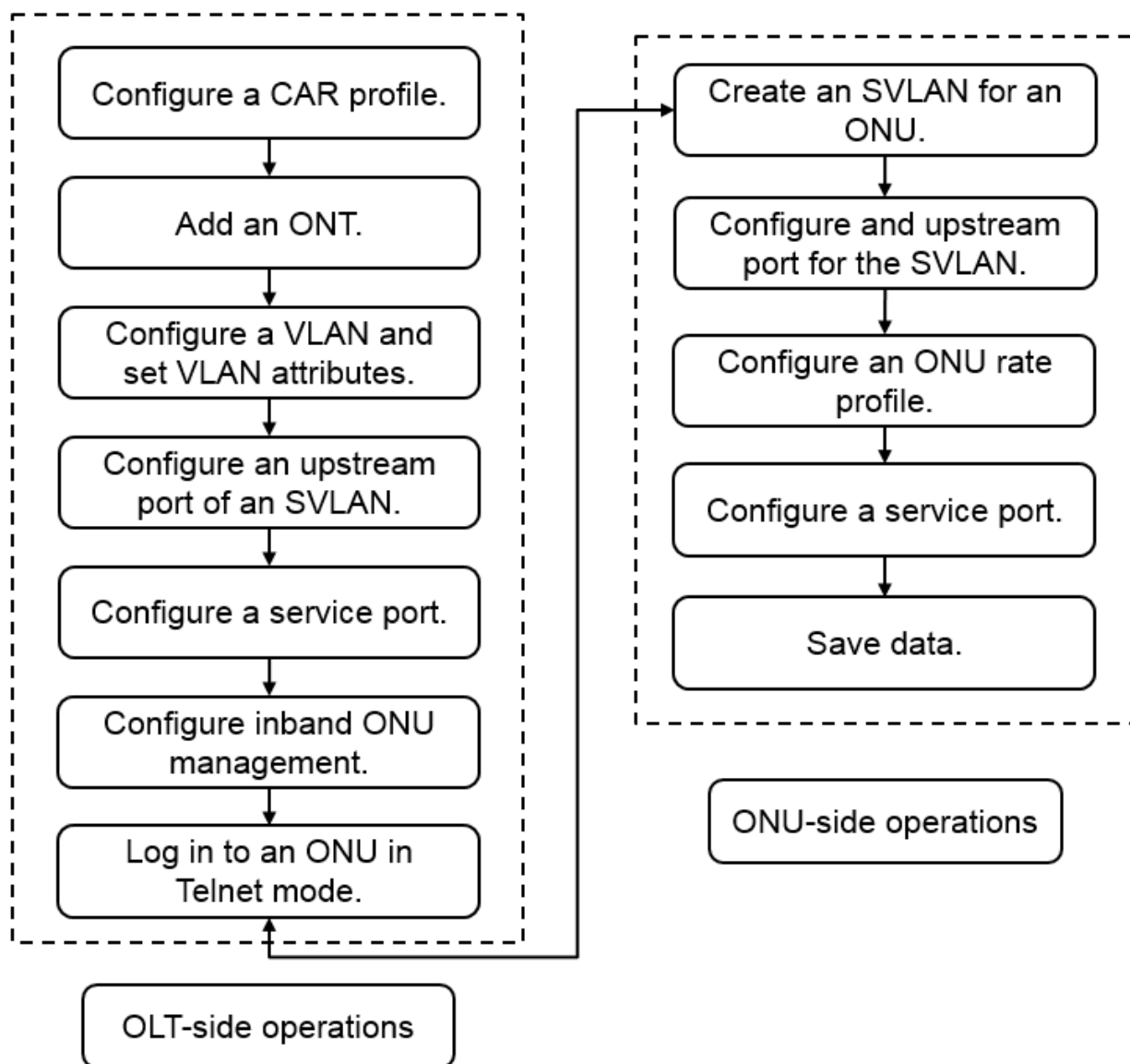
Table 4-2 Tasks for configuring the GPON ONU BIAS service

| Task                            | Requirement                                                                                                                                             | Completion Status |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Add an MDU.                     | An MDU is automatically or manually added. Record the following MDU status data: control flag, running flag, configuration status, and matching status. |                   |
| Create an SVLAN for an OLT.     | According to experiment data, create a VLAN and record the VLAN ID.                                                                                     |                   |
| Set VLAN attributes for an OLT. | Set the attributes so that double VLAN tags are carried. Record VLAN                                                                                    |                   |

|                                                     |                                                                                                                                           |  |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--|
|                                                     | attributes.                                                                                                                               |  |
| Configure an upstream port of the VLAN for an OLT.  | Record the upstream port and VLAN IDs that are allowed to pass through the upstream port.                                                 |  |
| Configure a service port of a VLAN.                 | Create mapping between the inner VLAN ID and outer VLAN ID of a specified ONT, and record the inner VLAN ID and outer VLAN ID separately. |  |
| Log in to the MDU.                                  | Log in to the MDU in Telnet mode. Record the inband management IP address of the MDU.                                                     |  |
| Create traffic entries.                             | CIR: 2 Mbit/s<br>PIR: 4 Mbit/s<br>Record the index of a traffic entry.                                                                    |  |
| Create a C-VLAN for the MDU.                        | According to experiment data, create a VLAN and record the VLAN ID.                                                                       |  |
| Configure an upstream port of the VLAN for the MDU. | Record the upstream port and VLAN IDs that are allowed to pass through the upstream port.                                                 |  |
| Configure a service port of a VLAN for the MDU.     | Create mapping between a user port on the specified MDU and a C-VLAN, and record the user port ID and C-VLAN ID.                          |  |

## 4.2.2 Service Configuration Process

Figure 4-2 Flow for configuring the GPON ONU BIAS service



## 4.2.3 Configuration Procedure (on the OLT Side)

Step 1 Configure a DBA profile.

A DBA profile describes GPON traffic parameters. A T-CONT is bound to a DBA profile for dynamic bandwidth allocation, improving upstream bandwidth utilization.

# Set the line profile ID to 25, profile name to fttb\_hsi, bandwidth type to Type3, CIR to 20480 kbit/s, and PIR to 40960 kbit/s.

```

huawei(config)#dba-profile
{ add<K>|delete<K>|modify<K> }:add
{ profile-id<K>|profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-id
{ profile-id<U><10,512> }:25
{ profile-name<K>|type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:profile-name
{ profile-name<S><Length 1-33> }:fttb_hsi
{ type1<K>|type2<K>|type3<K>|type4<K>|type5<K> }:type3
{ assure<K> }:assure
{ assure-bandwidth<U><128,1235456> }:20480
{ max<K> }:max
{ max-bandwidth<U><128,1235456> }:40960
Command:
    dba-profile add profile-id 25 profile-name fttb_hsi type3 assure
20480 max 40960

```

## Step 2 Configure an ONT line profile.

# Set the line profile ID to 20 and profile name to MA5822S.

```

huawei(config)#ont-lineprofile
{ epon<K>|gpon<K> }:gpon
{ <cr>|profile-id<K>|profile-name<K> }:profile-id
{ profile-id<U><0,8192> }:20
{ <cr>|profile-name<K> }:profile-name
{ profile-name<S><Length 1-32> }:MA5822S
Command:
    ont-lineprofile gpon profile-id 20 profile-name MA5822S

```

Configure inband management channel parameters.

# Configure GEM port 0 (for management) to carry Ethernet services, and bind the GEM port to T-CONT 0.

```

huawei(config-gpon-lineprofile-20)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:add
{ gem-index<U><0,1023> }:0
{ service-type<E><eth,tdm> }:eth
{ tcont<K> }:tcont
{ tcont-id<U><0,127> }:0
{ <cr>|cascade<K>|encrypt<K>|gem-car<K>|priority-queue<K> }:
Command:
    gem add 0 eth tcont 0

```

# Configure the mapping mode from the GEM port to traffic streams to VLAN, and map management services with VLAN 500 to GEM port 0.

```

huawei(config-gpon-lineprofile-20)#mapping-mode
{ iptos<K>|port-priority<K>|port-vlan-priority<K>|port-vlan<K>|port<K>|priority<K>|vlan-iptos<K>|vlan-
priority<K>|vlan<K> }:vlan
Command:
    mapping-mode vlan
huawei(config-gpon-lineprofile-20)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:mapping
{ gem-index<U><0,1023> }:0

```

```
{ mapping-index<U><0,7> }:0
{ e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K> }:vlan
{ untag<K>|vlan-id<U><0,4095> }:500
{ <cr>|flow-car<K>|priority<K>|transparent<K> }:
Command:
gem mapping 0 0 vlan 500
```

Configure broadband service channel parameters.

# Configure GEM port 20 for broadband services to carry Ethernet services, bind the GEM port to T-CONT 20, and bind T-CONT 20 to DBA profile 25 for bandwidth scheduling.

```
huawei(config-gpon-lineprofile-20)#tcont
{ tcont-list<S><Length 1-13> }:20
{ <cr>|dba-profile-id<K>|dba-profile-name<K> }:dba-profile-id
{ profile-id<U><0,512> }:25
Command:
tcont 20 dba-profile-id 25
huawei(config-gpon-lineprofile-20)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:add
{ gem-index<U><0,1023> }:20
{ service-type<E><eth,tdm> }:eth
{ tcont<K> }:tcont
{ tcont-id<U><0,127> }:20
{ <cr>|cascade<K>|downstream-priority-queue<K>|encrypt<K>|gem-car<K>|priority-queue<K> }:
Command:
gem add 20 eth tcont 20
```

# Configure mapping from broadband service VLAN 36 to GEM port 20, and set the mapping relationship index to 20.

```
huawei(config-gpon-lineprofile-20)#gem
{ add<K>|delete<K>|mapping<K>|modify<K> }:mapping
{ gem-index<U><0,1023> }:20
{ mapping-index<U><0,7> }:20
{ e1<K>|eth-bundle<K>|eth<K>|flow-car<K>|iphost<K>|ippath<K>|moca<K>|priority<K>|tdm-
vcl<K>|transparent<K>|vlan<K> }:vlan
{ untag<K>|vlan-id<U><0,4095> }:36
{ <cr>|flow-car<K>|priority<K>|transparent<K> }:
Command:
gem mapping 20 20 vlan 36
```

# After completing the configuration, run the **commit** command to make the configuration take effect.

```
huawei(config-gpon-lineprofile-20)#commit
huawei(config-gpon-lineprofile-20)#quit
```

**Step 3** Configure the function of automatic MDU discovery for a PON port.

# (Optional, and ignorable if the configurations have been completed) Enter the PON board view, and enable the automatic discovery function of GPON port 0/5/0.

```

huawei(config)#interface
{ adsl<K>|atm<K>|csp<K>|edt<K>|emu<K>|epon<K>|eponnni<K>|eth<K>|giu<K>|gpon<K>|gponnni<K>|
h248<K>|loopback<K>|meth<K>|null<K>|opf<K>|opg<K>|scu<K>|shl<K>|top-
stm1<K>|top<K>|tunnel<K>|vdsl<K>|vlanif<K> }:gpon
{ frameid/slotid<S><Length 3-15> }:0/5
Command:
interface gpon 0/5
huawei(config-if-gpon-0/5)#port 0
{ fec<K>|laser-switch<K>|ont-auto-find<K>|ont-password-renew<K>|range<K> }:ont-auto-find
{ switch<E><enable,disable> }:enable
Command:
port 0 ont-auto-find enable

```

# Display an MDU that is automatically discovered by GPON port 0/5/0, and copy the corresponding SN.

```

huawei(config-if-gpon-0/5)#display ont autofind 0
-----
Number          : 1
F/S/P           : 0/5/0
Ont SN          : 48575443F444CB04
Password        : 0x30303030303030303030(0000000000)
Loid            : 0000000000
Checkcode       :
VendorID        : HWTC
Ont Version     : MA5822S VER.B
Ont SoftwareVersion : V8R312 C00
Ont EquipmentID : SmartAX MA5822S
Ont autofind time : 2018-04-30 14:35:44+08:00
-----

```

#### Step 4 Register an MDU.

# Register an MDU. Set the MDU ID to 4, authentication mode to SN authentication, and management type to SNMP. Bind line profile 20.

```

huawei(config-if-gpon-0/5)#ont confirm
{ portid<U><0,7> }:0
{ all<K>|loid-auth<K>|ontid<K>|password-auth<K>|sn-auth<K> }:ontid
{ ontid<U><0,127> }:4
{ loid-auth<K>|password-auth<K>|sn-auth<K> }:sn-auth
{ sn<S><Length 13-16> }:48575443F444CB04
{ omci<K>|password-auth<K>|snmp<K> }:snmp
{ <cr>|desc<K>|ont-lineprofile-id<K>|ont-lineprofile-name<K> }:ont-lineprofile-id
{ profile-id<U><0,8192> }:20
{ <cr>|desc<K> }:
Command:
ont confirm 0 ontid 4 sn-auth 48575443F444CB04 snmp ont-lineprofile-id 20
Number of ONTs that can be added: 1, success: 1
PortID :0, ONTID :4

```

# Check the MDU status. Check whether the running status and configuration status are normal, and check whether the parameter settings match actual device functions.

```
huawei(config-if-gpon-0/5)#display ont info
```

```
{ portid<U><0,7> }:0
```

```
{ all<K>|ontid<U><0,127> }:all
```

```
Command:
```

```
display ont info 0 all
```

| F/S/P  | ONT ID | SN               | Control flag | Run state | Config state | Match state | Protect side |
|--------|--------|------------------|--------------|-----------|--------------|-------------|--------------|
| 0/ 5/0 | 3      | 48575443E1F0A409 | active       | online    | normal       | match       | no           |
| 0/ 5/0 | 4      | 48575443F444CB04 | active       | online    | normal       | match       | no           |

| F/S/P  | ONT-ID | Description        |
|--------|--------|--------------------|
| 0/ 5/0 | 3      | ONT_NO_DESCRIPTION |
| 0/ 5/0 | 4      | ONT_NO_DESCRIPTION |

In port 0/ 5/0 , the total of ONTs are: 2, online: 2

```
huawei(config-if-gpon-0/5)#quit
```

## Step 5 Configure a management VLAN.

To log in to the MDU in Telnet mode from the OLT and then configure the MDU, the management VLAN of the OLT and that of the MDU must be the same, and the management IP address of the OLT and that of the MDU must be in the **same network segment**.

# Configure management VLAN 500, and add upstream port 0/19/0 to VLAN 500.

```
huawei(config)#vlan
```

```
{ attrib<K>|bind<K>|desc<K>|forwarding<K>|loop<K>|name<K>|packet-policy<K>|priority<K>|reserve<K>|service-profile<K>|vlan-list<S><Length 1-256>|vlanid<U><2,4093> }:500
```

```
{ <cr>|to<K>|vlantype<E><mux,standard,smart,super> }:smart
```

```
Command:
```

```
vlan 500 smart
```

```
huawei(config)#port
```

```
{ desc<K>|mode<K>|vlan<K> }:vlan
```

```
{ name<K>|vlan-list<S><Length 1-256>|vlanid<U><1,4093> }:500
```

```
{ frameid/slotid<S><Length 3-15>|inner-vlan-list<K>|to<K> }:0/19
```

```
{ portlist<S><Length 1-256> }:0
```

```
Command:
```

```
port vlan 500 0/19 0
```

# Enter the view of the VLANIF 500 interface, and set the management IP address of the OLT to 10.175.192.36/22.

```
huawei(config)#interface vlanif 500
```

```
huawei(config-if-vlanif500)#ip address 10.175.192.36/22
```

```
{ integer<U><0,32>|ip_addr<I><X.X.X.X> }:22
```

```
{ <cr>|description<K>|sub<K> }:
```

```
Command:
```

```
ip address 10.175.192.36 22
```

```
huawei(config-if-vlanif500)#quit
```

## Step 6 Configure a service channel for management VLAN 500.

# In global config mode, create service channel service-port 0 for management VLAN 500. Set the VLAN ID to 500, ONT ID to 4, GEM port ID to 0, and C-VLAN ID to 500.

```

huawei(config)#service-port
{ desc<K>|index<U><0,32767>|remote-desc<K>|source<K>|uplink-port<K>|vlan<K> }:0
{ adminstatus<K>|inbound<K>|modify<K>|outbound<K>|source<K>|tag-transform<K>|uplink-
port<K>|vlan<K> }:vlan
{ aoe<K>|name<K>|vlanid<U><1,4093> }:500
{ adsl<K>|atm<K>|epon<K>|eth<K>|gpon<K>|port<K>|shdsl<K>|vdsl<K> }:gpon
{ frameid/slotid/portid<S><Length 5-18> }:0/5/0
{ ont<K> }:ont
{ ontid<U><0,127> }:4
{ eth<K>|gemport<K>|iphost<K> }:gemport
{ gemindex<U><0,1023> }:0
{ <cr>|bundle<K>|inbound<K>|multi-service<K>|rx-cttr<K>|tag-transform<K> }:multi-service
{ user-8021p<K>|user-encap<K>|user-vlan<K> }:user-vlan
{ other-all<K>|priority-tagged<K>|untagged<K>|user-vlanid<U><1,4095> }:500
{ <cr>|bundle<K>|inbound<K>|rx-cttr<K>|tag-transform<K>|user-encap<K> }:  

Command:
service-port 0 vlan 500 gpon 0/5/0 ont 4 gemport 0 multi-service user-vlan 500

```



### NOTE

When service-port is not bound to a traffic profile, the system does not limit the rate of the traffic stream and the packet **priority is set to 0**.

## Step 7 Configure an inband management VLAN and IP address for the MDU.

In GPON mode, run the **ont ipconfig** command to configure the inband NMS address of ONT 4. The system automatically sends the address to ONT 4 through the OMCI channel.

# In PON port 0/5/0 mode, set the **static** IP address of ONT 4 to 10.175.192.42/22, gateway address to 10.175.192.1, and management VLAN to 500 (same as the management VLAN of the OLT).

```

huawei(config)#interface gpon 0/5
huawei(config-if-gpon-0/5)#ont
{ activate<K>|add<K>|alarm-policy<K>|alarm-profile<K>|cancel<K>|confirm<K>|cos-queue-map-
profile<K>|deactivate<K>|delete<K>|factory-setting-
restore<K>|gemport<K>|ipconfig<K>|modify<K>|optical-alarm-profile<K>|port<K>|power-shedding-
config<K>|re-discovery<K>|re-register<K>|remote-ping<K>|reset<K>|snmp-profile<K>|snmp-
route<K>|statistics<K>|tdm-vcl<K>|tr069-server-config<K>|voip<K> }:ipconfig
{ portid<U><0,7> }:0
{ ontid<U><0,127> }:4
{ dhcp<K>|dscp-mapping-table<K>|ip-index<K>|pppoe<K>|static<K> }:static
{ <cr>|gateway<K>|ip-address<K>|pri-dns<K>|slave-dns<K>|vlan<K> }:ip-address
{ ip-address<I><X.X.X.X> }:10.175.192.42
{ mask<K> }:mask
{ mask<M><X.X.X.X> }:255.255.252.0
{ <cr>|gateway<K>|pri-dns<K>|slave-dns<K>|vlan<K> }:gateway
{ gateway<I><X.X.X.X> }:10.175.192.1
{ <cr>|pri-dns<K>|slave-dns<K>|vlan<K> }:vlan

```

```
{ vlanid<U><0,4095> }:500
{ <cr>|pri-dns<K>|priority<K>|slave-dns<K> }:  
  Command:  
    ont ipconfig 0 4 static ip-address 10.175.192.42 mask 255.255.252.0 gateway  
10.175.192.1 vlan 500  
huawei(config-if-gpon-0/5)#quit
```

### Step 8 Test the connectivity between the OLT and the MDU.

Check whether the management channel (namely, the management traffic stream) between the OLT and the ONT is available.

# Run the **ping 10.175.192.42** command on the OLT to check the connectivity between the OLT and ONT 4. It is expected that the OLT can receive response packets from ONT 4.

```
huawei(config)#ping 10.175.192.42  
PING 10.175.192.42: 56 data bytes, press CTRL_C to break  
  Reply from 10.175.192.42: bytes=56 Sequence=1 ttl=255 time=19 ms  
  Reply from 10.175.192.42: bytes=56 Sequence=2 ttl=255 time=9 ms  
  Reply from 10.175.192.42: bytes=56 Sequence=3 ttl=255 time=9 ms  
  Reply from 10.175.192.42: bytes=56 Sequence=4 ttl=255 time=9 ms  
  Reply from 10.175.192.42: bytes=56 Sequence=5 ttl=255 time=8 ms  
--- 10.175.192.42 ping statistics ---  
  5 packet(s) transmitted  
  5 packet(s) received  
  0.00% packet loss  
  round-trip min/avg/max = 8/10/19 ms
```

### Step 9 Configure a broadband service VLAN.

# Configure broadband service VLAN 2001, set the VLAN attribute to common, and create upstream port 0/19/0 to the SVLAN.

```
huawei(config)#vlan  
{ attrib<K>|bind<K>|desc<K>|forwarding<K>|loop<K>|name<K>|packet-  
policy<K>|priority<K>|reserve<K>|service-profile<K>|vlan-list<S><Length 1-  
256>|vlanid<U><2,4093> }:2001  
{ <cr>|to<K>|vlantype<E><mux,standard,smart,super> }:smart  
  Command:  
    vlan 2001 smart
```



#### NOTE

By default, when a smart VLAN is created, its VLAN attribute is common.

```
huawei(config)#port  
{ desc<K>|mode<K>|vlan<K> }:vlan  
{ vlan-list<S><Length 1-256>|vlanid<U><1,4093> }:2001  
{ frame/slot<S><Length 1-15>|to<K> }:0/19  
{ portlist<S><Length 1-256> }:0  
  Command:  
    port vlan 2001 0/19 0
```

### Step 10 Configure a service port of the broadband VLAN.

# Create traffic stream 20 connected to ONT 4. Set the SVLAN ID to 2001, GEM port ID to 20, and C-VLAN ID to 36. (When a user flow with VLAN 36 is received, VLAN 36 is translated to VLAN 2001.)

```

huawei(config)#service-port
{ desc<K>|index<U><0,32767>|remote-desc<K>|source<K>|uplink-port<K>|vlan<K> }:20
{ adminstatus<K>|inbound<K>|modify<K>|outbound<K>|source<K>|tag-transform<K>|uplink-
port<K>|vlan<K> }:vlan
{ aoe<K>|name<K>|vlanid<U><1,4093> }:2001
{ adsl<K>|atm<K>|lepon<K>|eth<K>|gpon<K>|port<K>|shdsl<K>|vdsl<K> }:gpon
{ frameid/slotid/portid<S><Length 5-18> }:0/5/0
{ ont<K> }:ont
{ ontid<U><0,127> }:4
{ eth<K>|gemport<K>|iphost<K> }:gemport
{ gemindex<U><0,1023> }:20
{ <cr>|bundle<K>|inbound<K>|multi-service<K>|rx-cttr<K>|tag-transform<K> }:multi-service
{ user-8021p<K>|user-encap<K>|user-vlan<K> }:user-vlan
{ other-all<K>|priority-tagged<K>|untagged<K>|user-vlanid<U><1,4095> }:36
{ <cr>|bundle<K>|inbound<K>|rx-cttr<K>|tag-transform<K>|user-encap<K> }:
Command:
service-port 20 vlan 2001 gpon 0/5/0 ont 4 gemport 20 multi-service user-vlan 36

```

## Step 11 Query information about SVLAN 2001.

```

huawei(config)#display vlan 2001
{ <cr>|inner-vlan<K>|to<K> }:
Command:
display vlan 2001
VLAN ID: 2001
VLAN name: VLAN_2001
VLAN type: smart
VLAN attribute: common // By default, the attribute of a smart VLAN is common.
VLAN description:
VLAN forwarding mode in control board: VLAN-MAC
VLAN forwarding mode: VLAN-MAC
VLAN broadcast packet forwarding policy: forward
VLAN unknown multicast packet forwarding policy: forward
VLAN unknown unicast packet forwarding policy: forward
VLAN bind service profile ID: -
VLAN bind RAIO profile index: -
VLAN priority: -
-----
F /S /P      Native VLAN  State
-----
0 /19/0      1          up
-----
Standard port number: 1
-----
INDEX  TYPE  STATE F /S /P  VPI  VCI  FLOWTYPE FLOWPARA
-----
20    gpon  up    0 /5 /0  4    20   vlan     36
-----
Service virtual port number: 1

```

Note: F--Frame, S--Slot, P--Port,  
VPI indicates ONT ID for PON, VCI indicates GEM index for GPON,  
v/e--vlan/encap, pritag--priority-tagged

## Step 12 Save data.

# Save configuration data, and log in to the MDU for subsequent configuration.

```
huawei(config)#save
{ <cr>|configuration<K>|data<K> }:
Command:
save
```

It will take several minutes to save configuration file, please wait...  
Configuration file had been saved successfully  
Note: The configuration file will take effect after being activated  
The data is being saved, please wait a moment...

## 4.2.4 Configuration Procedure (on the MDU Side)

### Step 1 Log in to the MDU.

Before logging in to the MDU in Telnet mode, ensure that the management link between the OLT and the MDU is connected. That is, ping 10.175.192.42. If the connection fails, check and rectify the fault.

```
huawei(config)#telnet 10.175.192.42
{ <cr>|service-port<U><1,65535> }:
Command:
telnet 10.175.192.42
Press CTRL_] to quit telnet mode
Trying 10.175.192.42 ...
Connected to 10.175.192.42 ...
>>User name:root
>>User password: // The default password is mduadmin. This input information has no
output.
Huawei Integrated Access Software (MA5822S).
Copyright(C) Huawei Technologies Co., Ltd. 2002-2012. All rights reserved.
MA5822S>
User root has used a default password. Change the password in time.
MA5822S>
MA5822S>enable
MA5822S#config
MA5822S(config)#
```

### Step 2 Create an SVLAN and add an upstream port to the SVLAN.

# Create SVLAN 36 and add an upstream port of the MDU to VLAN 36.

```
MA5822S (config)#vlan
{ attrib<K>|bind<K>|desc<K>|priority<K>|reserve<K>|service-profile<K>|vlan-list<S><Length 1-
256>|vlanid<U><2,4093> }:36
{ <cr>|to<K>|vlantype<E><mux,standard,smart> }:smart
Command:
```

```

vlan 36 smart
MA5822S(config)#port
{ desc<K>|mode<K>|vlan<K> }:vlan
{ vlan-list<S><Length 1-256>|vlanid<U><1,4093> }:36
{ frame/slot<S><Length 1-15>|to<K> }:0/0
{ portlist<S><Length 1-256> }:1
Command:
port vlan 36 0/0 1

```

#### NOTE

The SVLAN ID must be the same as the C-VLAN ID configured on the OLT.

### Step 3 Query a traffic profile of the system.

# Check whether an appropriate traffic profile exists in the system.

```

MA5822S(config)#display traffic table
{ ip<K> }:ip
{ from-index<K>|index<K>|name<K> }:from-index
{ row-index<U><0,63> }:0
{ <cr>|to-index<K> }::
Command:
display traffic table ip from-index 0

```

| TID | CIR<br>(kbps) | CBS<br>(bytes) | PIR<br>(kbps) | PBS<br>(bytes) | Pri | Copy-policy | Pri-Policy |
|-----|---------------|----------------|---------------|----------------|-----|-------------|------------|
| 0   | 512           | 18384          | 1024          | 36768          | 6   | -           | tag-pri    |
| 1   | 1024          | 34768          | 2048          | 69536          | 0   | -           | tag-pri    |
| 2   | 2048          | 67536          | 4096          | 135072         | 0   | -           | tag-pri    |
| 3   | 4096          | 133072         | 8192          | 266144         | 4   | -           | tag-pri    |
| 4   | 8192          | 264144         | 16384         | 528288         | 4   | -           | tag-pri    |
| 5   | 16384         | 526288         | 32768         | 1024000        | 4   | -           | tag-pri    |
| 6   | off           | off            | off           | off            | 0   | -           | tag-pri    |
| 10  | off           | off            | off           | off            | 4   | -           | tag-pri    |

Total Num : 8

### Step 4 Add a service port.

# Create a traffic stream. Generally, one port corresponds to one user. For other users, modify the SVLAN and the port.

```

MA5822S(config)#service-port
{ desc<K>|index<U><0,511>|remote-desc<K>|vlan<K> }:20
{ adminstatus<K>|inbound<K>|outbound<K>|vlan<K> }:vlan
{ transparent<K>|vlanid<U><1,4093> }:36
{ eth<K> }:eth
{ frameid/slotid/portid<S><Length 1-15> }:0/1/1
{ multi-service<K> }:multi-service
{ user-8021p<K>|user-encap<K>|user-vlan<K> }:user-vlan
{ other-all<K>|priority-tagged<K>|untagged<K>|user-vlanid<U><1,4095> }:untagged
{ bundle<K>|inbound<K>|rx-cttr<K>|user-encap<K> }:rx-cttr

```

```
{ rx-index<U><0,63> }:6
{ tx-cttr<K> }:tx-cttr
{ tx-index<U><0,63> }:6
Command:
service-port 20 vlan 36 eth 0/1/1 multi-service user-vlan untagged rx-cttr 6 tx-cttr
6
```

## Step 5 Save configuration data.

# Save data and exit the MDU.

```
MA5822S(config)#save
{ <cr>|configuration<K>|data<K> }:
Command:
save
It will take several minutes to save configuration file, please wait...
Configuration file had been saved successfully
Note: The configuration file will take effect after being activated
The data is being saved, please wait a moment...
MA5822S(config)#quit
MA5822S#quit
Check whether system data has been changed. Please save data before logout.
Are you sure to log out? (y/n)[n]:y
Configuration console exit, please retry to log on
```

## 4.3 Result Verification

### 4.3.1 Accessing the Internet Through PPPoE Dial-up

# Connect the FE port of an ONT with HSI services configured to a PC.

# On the PC, use PPPoE dial-up software to dial up or DHCP. If an IP address is obtained, the configuration is successful.

## 4.4 Configuration Reference

### 4.4.1 Configurations on the OLT Side

```
#
dba-profile add profile-id 25 profile-name fttb_hsi type3 assure 20480 max 40960
#
nt-lineprofile gpon profile-id 20 profile-name MA5822S
tcont 20 dba-profile-id 25
gem add 0 eth tcont 0
gem add 20 eth tcont 20
mapping-mode vlan
gem mapping 0 0 vlan 500
gem mapping 20 20 vlan 36
```

```

commit
quit
#
interface gpon 0/5
port 0 ont-auto-find enable
display ont autofind 0
ont confirm 0 ontid 4 sn-auth 48575443F444CB04 snmp ont-lineprofile-id 20
display ont info 0 all
quit
#
vlan 500 smart
vlan 2001 smart
port vlan 500 0/19 0
port vlan 2001 0/19 0
#
interface vlanif 500
ip address 10.175.192.36/22
quit
#
service-port 0 vlan 500 gpon 0/5/0 ont 4 gemport 0 multi-service user-vlan 500
#
interface gpon 0/5
ont ipconfig 0 4 static ip-address 10.175.192.42 mask 255.255.252.0 gateway 10.175.192.1 vlan
500
quit
#
ping 10.175.192.42
service-port 20 vlan 2001 gpon 0/5/0 ont 4 gemport 20 multi-service user-vlan 36
display vlan 2001
save
#
telnet 10.175.192.42
root
mduadmin

```

## 4.4.2 Configurations on the MDU Side

```

#
enable
config
switch language-mode
vlan 36 smart
port vlan 36 0/0 1
#
display traffic table ip from-index 0
service-port 20 vlan 36 eth 0/1/1 multi-service user-vlan untagged rx-cttr 6 tx-cttr 6
save
quit
quit
y

```

## 4.5 Quiz

In this experiment, the FTTB BIAS service is configured. According to the data plan, packets in a traffic stream carry single VLAN tags. If packets in a traffic stream are required to carry double VLAN tags, what are the required configurations?

# 5 GPON MDU xDSL BIAS Configuration

---

## 5.1 Experiment Introduction

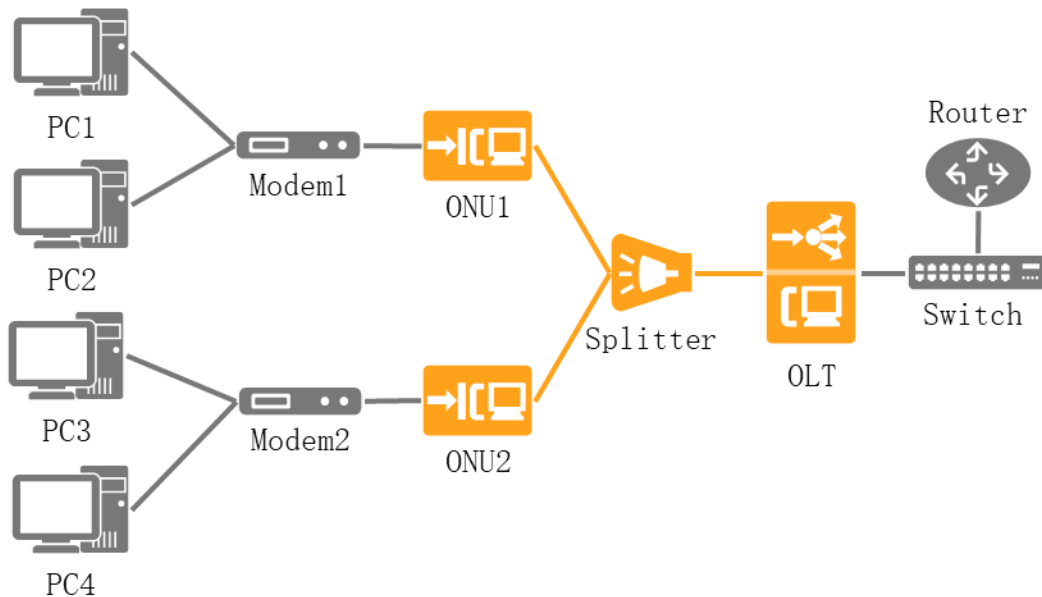
### 5.1.1 About This Experiment

A DSLAM is connected to a remote ONU through a GPON port to provide the high-speed Internet access service. ONUs that support ADSL2+ access include the MA5616 and MA5603T. This chapter uses the MA5616 and OLT using a GPBC board as an example. The MA5616 can leverage the wide coverage, flexible networking, and low cost features of an EPON/GPON, interwork with an OLT to provide high-bandwidth service access, and expand the access capacity of an OLT.

### 5.1.2 Objectives

- Understand basic GPON principles.
- Understand the technical principles of ADSL and VDSL.

### 5.1.3 Networking



**Figure 5-1 Topology for configuring the xDSL BIAS service**

In GPON networking shown in the preceding figure, the OLT is an optical line terminal, and ONU1 and ONU2 are optical network units. In this example, the MA5616 is used, modem1 and modem 2 provide ADSL and VDSL access, and PC1, PC2, PC3, and PC4 are user terminals.

For data services, a user PC uses the PPPoE dialup mode to connect to ONU1 and ONU2 through the ADSL2+ and VDSL2 ports respectively, and then connects to the OLT through ONU1 and ONU2 to implement high-speed Internet access.

### 5.1.4 Data Plan

The high-speed Internet access service is identified by double precisely-bound VLAN tags. On an ONU, each user is allocated with a C-VLAN; on the OLT, each slot is allocated with an SVLAN.

The high-speed Internet access service adopts an assured bandwidth mode with the maximum bandwidth 100 Mbit/s as the DBA profile and performs the 4 Mbit/s rate limitation in both the upstream and downstream directions.

The VDSL2 mode is **Normal**. (You can run the **display vdsl mode** command in privilege mode to query the VDSL mode.)

**Table 5-1 Data plan for the xDSL BIAS service**

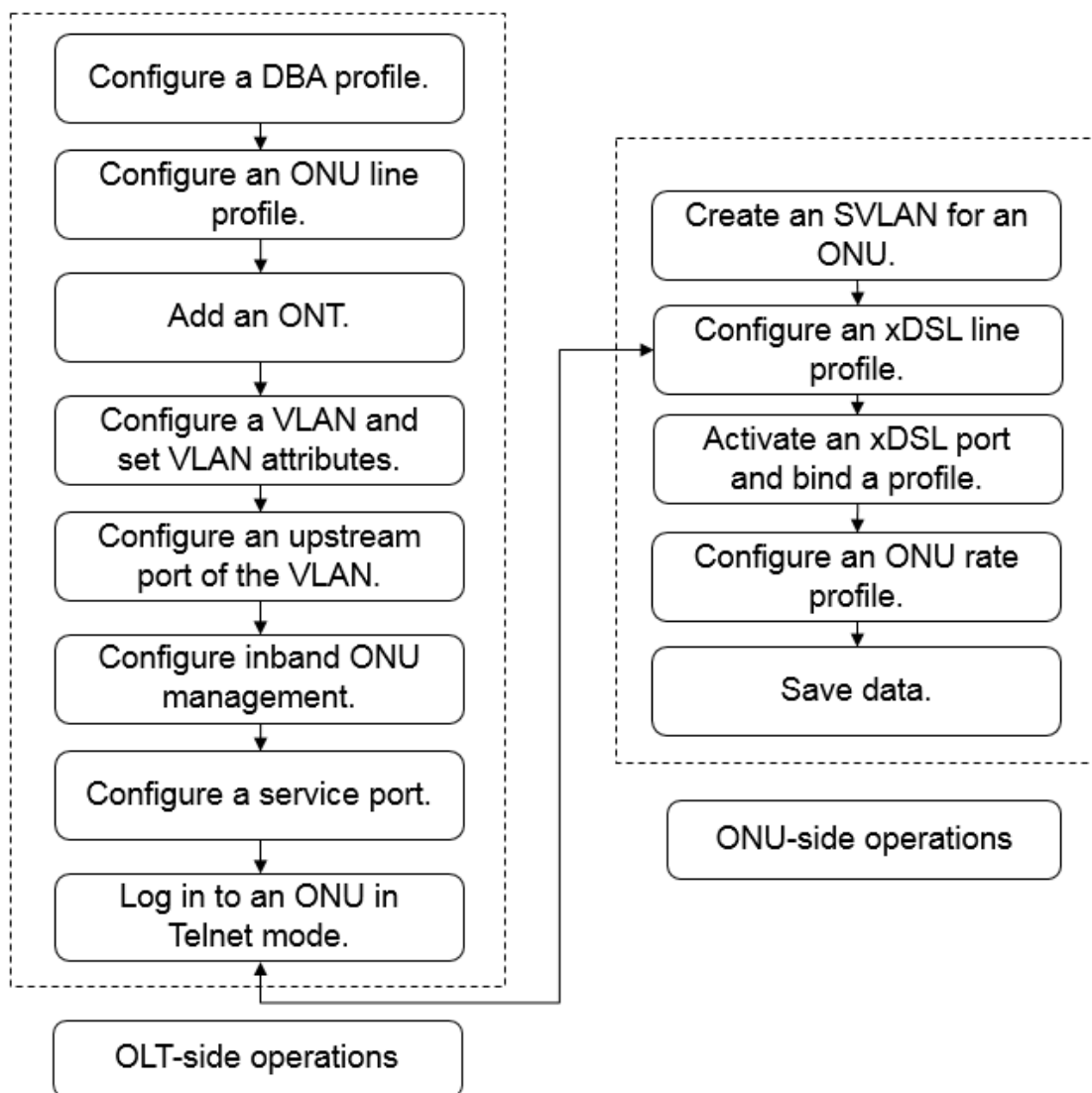
| Service Type    | Item | Detailed Data       | Remarks                                    |
|-----------------|------|---------------------|--------------------------------------------|
| Networking data | xDSL | OLT PON port: 0/2/1 | In the current network, ONT and MDU can be |

|                   |                                         |                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|-----------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   |                                         | ONT IDs: 1 and 2                                                                             | implemented on the same OLT. Generally, different slots are used for the implementation. L                                                                                                                                                                                                                                                                                           |
| Device management | Inband NMS IP address of the OLT        | 10.175.192.36/22<br>(IP address planned for a lab)                                           | PVID:1<br>Allow-pass: VLAN 10 20                                                                                                                                                                                                                                                                                                                                                     |
|                   | Management VLAN of the OLT              | 500                                                                                          | To log in to the MDU in Telnet mode from the OLT and then configure the MDU, the management VLAN of the OLT and that of the MDU must be the same, and the management IP address of the OLT and that of the MDU must be in the same network segment.                                                                                                                                  |
|                   | Inband management IP address of the MDU | ONU 1:<br>10.175.192.51/22<br>ONU 2:<br>10.175.192.52/22<br>(IP addresses planned for a lab) |                                                                                                                                                                                                                                                                                                                                                                                      |
|                   | Management VLAN of the MDU              | 500                                                                                          | In GPON access mode, the management protocol of the MDU is SNMP.                                                                                                                                                                                                                                                                                                                     |
| SVLAN             | HSI service                             | ONU VLANs: 2–199<br>OLT VLANs:<br>C-VLANs (using the ONU VLANs): 2–199<br>SVLANs: 2000–2199  | For the Internet access service, you can use double precisely-bound VLAN tags to extend VLANs and identify users.<br><br>On an ONU, each user is allocated with a C-VLAN. On the OLT, an SVLAN can be allocated for each OLT, each slot of the OLT, or each PON port.<br><br>The ONU VLANs of the same OLT must be planned in a unified manner, and each ONU VLAN ID must be unique. |
| QoS (priority)    | HSI service                             | Priority: 1; queue scheduling: WRR                                                           | Generally, the QoS priorities of various services in descending order are as follows: NMS service and IP voice service, private line service, IPTV service, and                                                                                                                                                                                                                      |

|           |             |                                                                                                                 |                                                                                                                                                                                                                                                                                                                                             |
|-----------|-------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |             |                                                                                                                 | <p>Internet access service.</p> <p>The priorities are configured on an ONU, and the OLT inherits priorities on the ONU.</p>                                                                                                                                                                                                                 |
| QoS (DBA) | HSI service | <p>Profile name: PPPoE</p> <p>Profile type: type 4</p> <p>Maximum bandwidth: 100 Mbit/s</p> <p>T-CONT ID: 4</p> | <p>A DBA profile is used to control the upstream bandwidth of an ONU. DBA profiles are bound to T-CONTs. Different T-CONTs are provided with different types of bandwidths.</p> <p>High-priority services use fixed bandwidth or assured bandwidth. Low-priority services use maximum bandwidth or BE bandwidth.</p>                        |
| QoS (CAR) | HSI service | <p>Upstream/Downstream bandwidth: 4 Mbit/s</p>                                                                  | <p>Traffic control can be implemented on the BRAS, or on the OLT or ONU by using port rate limitation or using a traffic profile to limit the upstream and downstream traffic.</p> <p>Generally, in the FTTH scenario, rate limitation is implemented on the OLT; in the FTTB/FTTC scenario, rate limitation is implemented on the ONU.</p> |

## 5.2 Configuration Task: ADSL2+ Access

### 5.2.1 Configuration Process



### 5.2.2 Configuration Procedure (on the OLT Side)

Step 1 Create an SVLAN and add an upstream port to it.

Set the SVLAN ID to 2000, type to smart VLAN, and attribute to QinQ. Add upstream port 0/19/0 to VLAN 2000.

```

huawei(config)#vlan 2000 smart
huawei(config)#vlan attrib 2000 q-in-q
huawei(config)#port vlan 2000 0/19 0
    
```

## Step 2 (Optional) Configure upstream link aggregation.

In this example, a single upstream port is used. In the case of multiple upstream ports, upstream link aggregation can be configured.

Configure upstream ports 0/19/0 and 0/19/1 as an aggregation group. Each member port in the aggregation group distributes packets according to the source MAC address and works in LACP static aggregation mode.

```
huawei(config)#link-aggregation 0/19 0-1 ingress workmode lacp-static
```

### NOTE:

The types of the 2 ports to be aggregated must be the same.

An aggregation group can be configured between an ETH board and an SCU board.

An aggregation group can be configured between an ETH board and a GIU slot.

Inter-board aggregation is supported between 2 GIU slots.

When only one control board is configured, inter-board aggregation is supported between an SCUN board and a GIU slot.

To query information about an aggregation group, run the **display link-aggregation** command .

## Step 3 Configure a DBA profile.

Configure GPON ONU profiles, including a DBA profile, line profile, service profile, and alarm profile.

- DBA profile: It describes the traffic parameters of GPON. A T-CONT is bound to a DBA profile to dynamically allocate the bandwidth, improving the upstream bandwidth utilization.
- Line profile: It describes the binding relationship between a T-CONT and a DBA profile, the QoS mode of a traffic stream, and the mapping between a GEM port and an ONU-side service.
- Service profile: It provides a service configuration channel for the ONUs managed through OMCI.
- Alarm profile: It configures a series of alarm threshold parameters to monitor the performance of an activated ONT line. When statistics reach an alarm threshold, the system notifies the NE and sends alarm information to the log NE and the NMS.

To query existing DBA profiles in the system, run the **display dba-profile** command. If existing DBA profiles in the system do not meet the requirement, run the **dba-profile add** command to create a DBA profile.

Configure a DBA profile. Set the name to PPPoE, type to type4, and maximum bandwidth to 100 Mbit/s.

```
huawei(config)#dba-profile add profile-name PPPoE type4 max 102400
```

## Step 4 Configure an ONU line profile.

Create GPON ONU line profile 10 and bind T-CONT 4 to the DBA profile named PPPoE.

```
huawei(config)#ont-lineprofile gpon profile-id 10
huawei(config-gpon-lineprofile-10)#tcont 4 dba-profile-name PPPoE
```

**NOTE:**

The ID of the line profile to be created must not exist in the system. In addition, you need to create different line profiles according to different services. In this example, line profile 10 is used.

Add GEM port 0 to carry management traffic streams and GEM port 1 to carry Internet access traffic streams. Bind GEM ports 0 and 1 to T-CONT 4. The QoS mode is **priority-queue** (default).

```
huawei(config-gpon-lineprofile-10)#gem add 0 eth tcont 4
huawei(config-gpon-lineprofile-10)#gem add 1 eth tcont 4
```

**NOTE:**

- To modify the default QoS mode of the system, run the `qos-mode` command to set the QoS mode to `gem-car` or `flow-car`, and run the `gem add` command to configure an ID of a traffic profile bound to a GEM port.
- When the QoS mode is PQ, the default queue priority is 0. When the QoS mode is flow-car, the default ID of a bound traffic profile is 6 (no rate limitation). When the QoS mode is gem-car, the default ID of a bound traffic profile is 6 (no rate limitation).
- If you run the `multi-service-port` command to create traffic streams in batches, ensure that the number of GEM ports is the same as the number of C-VLANs. Therefore, you need to create GEM ports based on the number of C-VLANs. In this example, 24 GEM ports need to be created.
- If you run the `service-port` command to create traffic streams one by one, note that a maximum of 8 traffic streams can be bound to 1 GEM port. Therefore, create sufficient GEM ports according to the number of traffic streams. In this example, 3 GEM ports need to be created. This example uses this command to create 1 GEM port. For different traffic streams on the same GEM port, only the mapping index mapping-index needs to be replaced, and the mapped VLAN needs to be replaced with the corresponding C-VLAN.

Configure mapping mode between a GEM port and an ONU service as VLAN (default), map a management traffic stream (C-VLAN 500) to GEM port 0, and map an Internet access traffic stream (C-VLAN 101) to GEM port 1.

```
huawei(config-gpon-lineprofile-10)#mapping-mode vlan
huawei(config-gpon-lineprofile-10)#gem mapping 0 0 vlan 500
huawei(config-gpon-lineprofile-10)#gem mapping 1 1 vlan 101
```

After completing the configuration, run the **commit** command to make the configuration take effect.

```
huawei(config-gpon-lineprofile-10)#commit
huawei(config-gpon-lineprofile-10)#quit
```

### Step 5 (Optional) Configure an alarm threshold profile.

The ID of the default GPON alarm threshold profile is 1. The thresholds of all the alarm parameters in the default alarm profile are 0, which indicates that no alarm is reported.

In this example, the default alarm threshold profile is used, and therefore the configuration of the alarm profile is not required.

To configure alarm threshold parameters for monitoring the performance statistics of an activated ONU line, run the **gpon alarm-profile add** command to configure a GPON alarm threshold profile.

#### Step 6 Add an ONU on the OLT.

ONU 1 and ONU 2 are connected to GPON port 0/5/0 through an optical splitter. The IDs of ONU 1 and ONU 2 are 1 and 2 respectively. The SN of ONU 1 is 32303131B39FD641, the SN of ONU 2 is 32303131B39FD642, the management mode is SNMP, and the bound line profile ID is 10.

#Add an ONT offline.

If the password or SN of an ONT is obtained, run the **ont add** command to add the ONT offline.

```
huawei(config)#interface gpon 0/5
huawei(config-if-gpon-0/5)#ont add 1 ontid 1 sn-auth 32303131B39FD641 snmp ont-lineprofile-id 10
huawei(config-if-gpon-0/5)#ont add 1 ontid 2 sn-auth 32303131B39FD642 snmp ont-lineprofile-id 10
```

#Automatically discover an ONT.

If the password or SN of an ONT is unknown, run the **port portid ont-auto-find** command in GPON mode to enable automatic ONT discovery of a GPON port. Then, run the **ont confirm** command to confirm the ONT.

```
huawei(config)#interface gpon 0/5
huawei(config-if-gpon-0/5)#port 1 ont-auto-find enable
huawei(config-if-gpon-0/5)#display ont autofind 1
// The ONU type varies according to the access mode. In this example, an ONU in xDSL access mode
is used as an example. In the case of LAN access mode, ONU information of the corresponding
version is displayed.
```

```
-----
Number          : 1
F/S/P           : 0/5/1
Ont SN          : 32303131B39FD641
Password        :
VenderID        : HWTC
Ont Version     :
Ont SoftwareVersion : V8R308 C00
Ont EquipmentID  : SmartAX MA5616
Ont autofind time : 2019-06-21 16:51:45
-----
```

```
-----
Number          : 2
F/S/P           : 0/5/1
Ont SN          : 32303131B39FD642
Password        :
VenderID        : HWTC
Ont Version     :
Ont SoftwareVersion : V8R308 C00
Ont EquipmentID  : SmartAX MA5616
Ont autofind time : 2019-06-21 16:51:45
-----
```

```

huawei(config-if-gpon-0/5)#ont confirm 1 ontid 1 sn-auth 32303131B39FD641 snmp ont-lineprofile-id 10
huawei(config-if-gpon-0/5)#ont confirm 1 ontid 2 sn-auth 32303131B39FD642 snmp ont-lineprofile-id 10

```

#(Optional) Bind an alarm threshold profile.

In this example, the default profile (profile 1) is used.

```

huawei(config-if-gpon-0/5)#ont alarm-profile 1 1 profile-id 1
huawei(config-if-gpon-0/5)#ont alarm-profile 1 2 profile-id 1

```

### Step 7 Confirm that the ONU goes online successfully.

In this step, query the status of ONU 1 as an example. The same method is applied for querying the status of ONU 2.

After adding an ONU, run the **display ont info** command to query the ONU status. Ensure that the values of **Control flag**, **Run State**, **Config state**, and **Match state** of the ONU are **active**, **online**, **normal**, and **match** respectively.

```

huawei(config-if-gpon-0/5)#display ont info 1 1
-----
F/S/P           : 0/5/1
ONT-ID          : 0
Control flag    : active    // The ONU is activated.
Run state      : online    // The ONU is online.
Config state   : normal    // The ONU configuration status is normal.
Match state    : match     // The capability profile bound to the ONU is consistent with
the actual capability of the ONU.
...//The rest of the response information is omitted.

```

#### NOTE

When the ONU configuration status indicates a failure, the ONU cannot go online, or an ONU match fails, you are advised to check the ONU status with reference to the preceding description.

- If **Control flag** is **deactive**, run the **ont activate** command in GPON mode to activate the ONU.
- If the ONU fails to go online, that is, **Run state** is **offline**, the physical line may be disconnected or an optical module may be damaged. In this case, check the materials and lines.
- If the ONU configuration status indicates a failure, that is, **Config state** is **failed**, run the **display ont failed-configuration** command in diagnose mode to check the configuration failure items and causes and modify the configurations according to the actual situation.
- If the ONU does not match, that is, **Match state** is **mismatch**, run the **display ont capability** command to query the actual capability of the ONU, and choose one of the following methods to modify the ONU configurations.
- Create an appropriate ONU profile according to the actual capability of the ONU, and run the **ont modify** command to modify the ONU configuration data.
- Modify the ONU profile according to the actual capability of the ONU and save the modification. The ONU automatically restores the configurations.

### Step 8 Configure a management channel from the OLT to an ONU.

#Configure the inband management VLAN and IP address of the OLT.

To log in to an ONU in Telnet mode from the OLT and then configure the ONU, you must configure the inband management VLANs and IP addresses of the OLT and the ONU on the OLT.

Set the management VLAN ID to 500, add the upstream port to VLAN 500, and set the inband management IP address to 10.175.192.36/22.

```
huawei(config)#vlan 500 smart
huawei(config)#port vlan 500 0/19 0
huawei(config)#interface vlanif 500
huawei(config-if-vlanif500)#ip address 10.175.192.36 22
huawei(config-if-vlanif500)#quit
```

#Configure the inband management VLAN and IP address of the ONU.

Set the static IP address of ONU 1 to 10.175.192.51/22, the static IP address of ONU 2 to 10.175.192.52/22, the gateway to 10.175.192.1, and the management VLAN ID to 500 (same as the management VLAN of the OLT).

```
huawei(config-if-gpon-0/5)#ont ipconfig 1 1 static ip-address 10.175.192.51 mask 255.255.252.0
gateway 10.175.192.1 vlan 500
huawei(config-if-gpon-0/5)#ont ipconfig 1 2 static ip-address 10.175.192.52 mask 255.255.252.0
gateway 10.175.192.1 vlan 500
huawei(config-if-gpon-0/5)#quit
```

#Configure an inband management traffic stream.

Set the ID of the traffic stream for managing ONU 1 to 1, the ID of the traffic stream for managing ONU 2 to 2, the management VLAN ID to 500, the GEM port ID to 0, and the C-VLAN ID to 500. The OLT does not limit the rate of an inband management traffic stream. Therefore, use the default traffic profile (profile 6). If you need to limit the rate of a service port, run the **traffic table ip** command to create a traffic profile and then bind the traffic profile to the service port.

```
huawei(config)#service-port 1 vlan 500 gpon 0/5/1 ont 1 gemport 0 multi-service user-vlan 500 rx-  
cttr 6 tx-cttr 6  
huawei(config)#service-port 2 vlan 500 gpon 0/5/1 ont 2 gemport 0 multi-service user-vlan 500 rx-  
cttr 6 tx-cttr 6
```

**Step 9** Check whether the management channel between the OLT and the ONU is available.

#You can run the **ping 10.175.192.51** and **ping 10.175.192.52** commands on the OLT to verify the connectivity between the OLT and ONU 1 and that between the OLT and ONU 2 respectively. It is expected that the OLT can receive an ICMP ECHO-REPLY response packet from each ONU.

#You can run the **telnet 10.175.192.51** and **telnet 10.175.192.52** commands to log in to the ONU 1 and ONU 2 respectively to configure the ONUs.

**Step 10** Create a traffic stream.

Create traffic stream 101 connected to ONU 1 and traffic stream 102 connected to ONU 2. Set the SVLAN ID to **2000**, GEM port ID to **1**, and C-VLAN ID to **101**. The upstream traffic and downstream traffic are limited on an ONU but not on the OLT. Therefore, use traffic profile 6 (default). To limit the rate of a service port, run the **traffic table ip** command to add a traffic profile and bind it to the service port.

```
huawei(config)#service-port 101 vlan 2000 gpon 0/5/1 ont 1 gemport 0 multi-service user-vlan 101  
rx-cttr 6 tx-cttr 6  
huawei(config)#service-port 102 vlan 2000 gpon 0/5/1 ont 2 gemport 0 multi-service user-vlan 101  
rx-cttr 6 tx-cttr 6
```

#### NOTE

- The C-VLAN must be the same as the upstream VLAN of an ONU.
- Run the **service-port** command to create traffic streams one by one. The following uses the creation of one traffic stream as an example. Pay attention to the mapping between a GEM port created when an ONU line profile is configured and the C-VLAN.
- You can also run the **multi-service-port** command to create traffic streams in batches. In the GPON access mode, you need to specify **ontid** and **gemindex** to specify a traffic stream. In addition, the number of GEM ports must be the same as the number of C-VLANs.

**Step 11** Configure queue scheduling.

Use the 3PQ+5WRR queue scheduling. Queues 0–4 adopt the WRR mode, with the weights of 10, 10, 20, 20, and 40 respectively; queues 5–7 adopt the PQ mode. Configure the priority of the Internet access service to 1 and adopt the WRR mode.

#### NOTE

- Queue scheduling is globally configured and needs to be configured only once on the OLT. After the configuration is completed, the configuration takes

effect globally. You do not need to configure other services when configuring other services.

For boards that support only 4 queues, the mapping between 802.1p priorities and queue IDs is as follows: Priorities 0 and 1 are mapped into queue 1. Priorities 2 and 3 are mapped into queue 2. Priorities 4 and 5 are mapped into queue 3. Priorities 6 and 7 are mapped into queue 4.

```
huawei(config)#queue-scheduler wrr 10 10 20 20 40 0 0 0
```

Configure the mapping between queues and 802.1p priorities. Priorities 0– 7 is mapped into queues 0–7 respectively.

```
huawei(config)#cos-queue-map cos0 0 cos1 1 cos2 2 cos3 3 cos4 4 cos5 5 cos6 6 cos7 7
```

#### Step 12 Save data.

```
huawei(config)#save
```

### 5.2.3 Configuration Procedure (on the ONU Side)

#### Step 1 Log in to an ONU.

On the OLT, visit the management IP address of the ONU in Telnet mode and log in to the ONU. The user name is **root**, and the password is **mduadmin**.

```
huawei(config)#telnet 10.175.192.51
{ <cr>|service-port<U><0,4294967295> }:
Command:
    telnet 10.175.192.51
Press CTRL_] to quit telnet mode
Trying 10.175.192.51 ...
Connected to 10.175.192.51 ...
>>User name:root
>>User password:
```

#### Step 2 Configure a traffic profile.

You can run the **display traffic table ip** command to query information about the traffic profiles existing in the system. If the existing traffic profile in the system does not meet the requirement, run the **traffic table ip** command to add a traffic profile.

Configure traffic profile 8 and ensure that the information rate is 4 Mbit/s and the priority is 1. The priority policy is based on the priority of packets.

```
huawei(config)#traffic table ip index 8 cir 4096 priority 1 priority-policy tag-In-Package
```

#### Step 3 Configure an SVLAN and add an upstream port to the VLAN.

Configure the SVLAN. The VLAN ID ranges from 101 to 124. Add upstream port 0/0/1. (If the ONU is the MA5603T, the corresponding upstream port is 0/8/0.)

```
huawei(config)#vlan 101-124 smart
huawei(config)#port vlan 101-124 0/0 1
```

#### Step 4 Configure an ADSL2+ line profile.

You can run the **display adsl line-profile** command to query a configured ADSL2+ line profile in the system. Directly use a profile if it meets the requirement.

If no ADSL2+ line profile in the system meets the requirements, create an ADSL2+ line profile by running the **adsl line-profile quickadd** command.

The data in an ADSL2+ line profile must be configured according to the actual line conditions.

Configure ADSL2+ line profile 4 and use default parameter settings.

```
huawei(config)#adsl line-profile quickadd 4
```

#### Step 5 Activate an ADSL2+ port and bind a line profile to the ADSL2+ port.

Activate ADSL2+ port 0/1/0 and bind it to line profile 3.

Activate other ports using the following method. To activate all ports in batches on a board, run the **activate all** command.

```
huawei(config)#interface adsl 0/1
huawei(config-if-adsl-0/1)#deactivate 0
huawei(config-if-adsl-0/1)#activate 0 line-index 3
```

#### Step 6 (Optional) Bind an ADSL2+ alarm profile.

Bind default ADSL2+ alarm profile 1. To meet actual requirements, run the **adsl alarm-template quickadd** command to add an ADSL2+ alarm profile.

```
huawei(config-if-adsl-0/1)#alarm-config 0 1
```

#### Step 7 Configure a service port.

Service port 101 is an example. Set the SVLAN ID to 101, VPI to 8, VCI to 35, and C-VLAN to untagged. Bind traffic profile 8. You can also add other traffic streams by changing the C-VLAN ID and ADSL2+ port ID.

The VLAN must be the same as the C-VLAN configured on the OLT.

```
huawei(config)#service-port 101 vlan 101 adsl 0/1/0 vpi 8 vci 35 multi-service user-vlan untagged rx-
cttr 8 tx-cttr 8
```

#### NOTE

- To enable services in batches, run the **multi-service-port** command.

#### Step 8 Save data.

```
huawei(config)#save
```

Step 9 Verify the configuration results.

On a PC, you can use the high-speed Internet access service in PPPoE dial-up mode.

## 5.3 Configuration Reference

### 5.3.1 Configurations on the OLT Side

```
#
vlan 2000 smart
vlan attrib 2000 q-in-q
port vlan 2000 0/19 0
#
dba-profile add profile-name PPPoE type4 max 102400
#
ont-lineprofile gpon profile-id 10
tcont 4 dba-profile-name PPPoE
gem add 0 eth tcont 4
gem add 1 eth tcont 4
mapping-mode vlan
gem mapping 0 0 vlan 500
gem mapping 1 1 vlan 101
commit
quit
#
interface gpon 0/5
port 1 ont-auto-find enable
display ont autofind 1
ont confirm 1 ontid 1 sn-auth 32303131B39FD641 snmp ont-lineprofile-id 10
ont confirm 1 ontid 2 sn-auth 32303131B39FD642 snmp ont-lineprofile-id 10
#
ont alarm-profile 1 1 profile-id 1
ont alarm-profile 1 2 profile-id 1
display ont info 1 1
#
vlan 500 smart
port vlan 500 0/19 0
interface vlanif 500
ip address 10.175.192.36 22
quit
#
interface gpon 0/5
ont ipconfig 1 1 static ip-address 10.175.192.51 mask 255.255.252.0 gateway 10.175.192.1 vlan 500
ont ipconfig 1 2 static ip-address 10.175.192.52 mask 255.255.252.0 gateway 10.175.192.1 vlan 500
quit
#
service-port 1 vlan 500 gpon 0/5/1 ont 1 gemport 0 multi-service user-vlan 500 rx-cttr 6 tx-cttr 6
service-port 2 vlan 500 gpon 0/5/1 ont 2 gemport 0 multi-service user-vlan 500 rx-cttr 6 tx-cttr 6
service-port 101 vlan 2000 gpon 0/5/1 ont 1 gemport 0 multi-service user-vlan 101 rx-cttr 6 tx-cttr 6
service-port 102 vlan 2000 gpon 0/5/1 ont 2 gemport 0 multi-service user-vlan 101 rx-cttr 6 tx-cttr 6
#
```

```
queue-scheduler wrr 10 10 20 20 40 0 0 0
cos-queue-map cos0 0 cos1 1 cos2 2 cos3 3 cos4 4 cos5 5 cos6 6 cos7 7
save
#
telnet 10.175.192.51
root
mduadmin
```

## 5.3.2 Configurations on the ONU side

```
#
traffic table ip index 8 cir 4096 priority 1 priority-policy tag-In-Package
#
vlan 101-124 smart
port vlan 101-124 0/0 1
#
adsl line-profile quickadd 4
adsl channel-profile quickadd 4
adsl line-template quickadd 3
#
interface adsl 0/1
deactivate 0
activate 0 template-index 3
alarm-config 0 1
#
service-port 101 vlan 101 adsl 0/1/0 vpi 8 vci 35 multi-service user-vlan untagged rx-cttr 8 tx-cttr 8
save
```

## 5.4 Configuration Task: VDSL2 Access

### 5.4.1 Experiment Introduction

The MA5600T/MA5603T is connected to a remote ONU through a GPON port to provide the high-speed Internet access service. ONUs that support the VDSL2 access mode are as follows: MA5616, MA5652G, and MA5603T. This chapter uses the MA5616 and OLT using a GPBC board as an example.

### 5.4.2 Configuration Procedure (on the OLT Side)

**Step 1** Create an SVLAN and add an upstream port to it.

Set the SVLAN ID to **2000**, type to smart VLAN, and attribute to QinQ. Add upstream port 0/19/0 to VLAN 2000.

```
huawei(config)#vlan 2000 smart
huawei(config)#vlan attrib 2000 q-in-q
huawei(config)#port vlan 2000 0/19 0
```

**Step 2** (Optional) Configure upstream link aggregation.

In this example, a single upstream port is used. In the case of multiple upstream ports, upstream link aggregation can be configured. For details, see step 2 in section 5.2.2.

### Step 3 Configure a DBA profile.

GPON ONU profiles include the DBA profile, line profile, service profile, and alarm profile.

To query existing DBA profiles in the system, run the **display dba-profile** command. If existing DBA profiles in the system do not meet the requirement, run the **dba-profile add** command to create a DBA profile.

Configure a DBA profile. Set the name to PPPoE, type to type4, and maximum bandwidth to 100 Mbit/s.

```
huawei(config)#dba-profile add profile-name PPPoE type4 max 102400
```

### Step 4 Configure an ONU line profile.

Create GPON ONU line profile 10 and bind T-CONT 4 to the DBA profile named PPPoE.

```
huawei(config)#ont-lineprofile gpon profile-id 10
huawei(config-gpon-lineprofile-10)#tcont 4 dba-profile-name PPPoE
```

#### NOTE

The ID of the line profile to be created must not exist in the system. In addition, you need to create different line profiles according to different services. In this example, line profile 10 is used.

Add GEM port 0 to carry management traffic streams and GEM port 1 to carry Internet access traffic streams. Bind GEM ports 0 and 1 to T-CONT 4. The QoS mode is **priority-queue** (default).

```
huawei(config-gpon-lineprofile-10)#gem add 0 eth tcont 4
huawei(config-gpon-lineprofile-10)#gem add 1 eth tcont 4
```

#### NOTE

- To modify the default QoS mode of the system, run the **qos-mode** command to set the QoS mode to **gem-car** or **flow-car**, and run the **gem add** command to configure an ID of a traffic profile bound to a GEM port.
- When the QoS mode is PQ, the default queue priority is 0. When the QoS mode is **flow-car**, the default ID of a bound traffic profile is 6 (no rate limitation). When the QoS mode is **gem-car**, the default ID of a bound traffic profile is 6 (no rate limitation).
- If you run the **multi-service-port** command to create traffic streams in batches, ensure that the number of GEM ports is the same as the number of C-VLANs. Therefore, you need to create GEM ports based on the number of C-VLANs. In this example, 24 GEM ports need to be created.
- If you run the **service-port** command to create traffic streams one by one, note that a maximum of 8 traffic streams can be bound to 1 GEM port. Therefore, create sufficient GEM ports according to the number of traffic

streams. In this example, 3 GEM ports need to be created. This example uses this command to create 1 GEM port. For different traffic streams on the same GEM port, only the mapping index **mapping-index** needs to be replaced, and the mapped VLAN needs to be replaced with the corresponding C-VLAN.

Configure mapping mode between a GEM port and an ONU service as VLAN (default), map a management traffic stream (C-VLAN 500) to GEM port 0, and map an Internet access traffic stream (C-VLAN 101) to GEM port 1.

```
huawei(config-gpon-lineprofile-10)#mapping-mode vlan
huawei(config-gpon-lineprofile-10)#gem mapping 0 0 vlan 500
huawei(config-gpon-lineprofile-10)#gem mapping 1 1 vlan 101
```

After completing the configuration, run the **commit** command to make the configuration take effect.

```
huawei(config-gpon-lineprofile-10)#commit
huawei(config-gpon-lineprofile-10)#quit
```

#### Step 5 (Optional) Configure an alarm threshold profile.

- The ID of the default GPON alarm threshold profile is 1. The thresholds of all the alarm parameters in the default alarm profile are 0, which indicates that no alarm is reported.
- In this example, the default alarm threshold profile is used, and therefore the configuration of the alarm profile is not required.
- To configure alarm threshold parameters for monitoring the performance statistics of an activated ONU line, run the **gpon alarm-profile add** command to configure a GPON alarm threshold profile.

#### Step 6 Add an ONU on the OLT.

ONU 1 and ONU 2 are connected to GPON port 0/5/2 through an optical splitter. The IDs of ONU 1 and ONU 2 are 1 and 2 respectively. The SN of ONU 1 is 32303131B39FD641, the SN of ONU 2 is 32303131B39FD642, the management mode is SNMP, and the bound line profile ID is 10.

##### 1. Add an ONT offline.

If the password or SN of an ONT is obtained, run the **ont add** command to add the ONT offline.

```
huawei(config)#interface gpon 0/5
huawei(config-if-gpon-0/5)#ont add 2 ontid 1 sn-auth 32303131B39FD641 snmp ont-lineprofile-id 10
huawei(config-if-gpon-0/5)#ont add 2 ontid 2 sn-auth 32303131B39FD642 snmp ont-lineprofile-id 10
```

##### 2. Automatically discover an ONT.

If the password or SN of an ONT is unknown, run the **port portid ont-auto-find** command in GPON mode to enable automatic ONT discovery of a GPON port. Then, run the **ont confirm** command to confirm the ONT.

```

huawei(config)#interface gpon 0/5
huawei(config-if-gpon-0/5)#port 2 ont-auto-find enable
huawei(config-if-gpon-0/5)#display ont autofind 2
// The ONU type varies according to the access mode. In this example, an ONU in xDSL access mode
is used as an example. In the case of LAN access mode, ONU information of the corresponding
version is displayed.

```

```

-----
Number          : 1
F/S/P           : 0/5/2
Ont SN          : 32303131B39FD641
Password        :
VenderID        : HWTC
Ont Version     :
Ont SoftwareVersion : V8R308 C00
Ont EquipmentID  : SmartAX MA5616
Ont autofind time : 2018-06-21 16:51:45

```

```

-----
Number          : 2
F/S/P           : 0/5/2
Ont SN          : 32303131B39FD642
Password        :
VenderID        : HWTC
Ont Version     :
Ont SoftwareVersion : V8R308 C00
Ont EquipmentID  : SmartAX MA5616
Ont autofind time : 2018-06-21 16:51:45

```

```

huawei(config-if-gpon-0/5)#ont confirm 2 ontid 1 sn-auth 32303131B39FD641 snmp ont-lineprofile-
id 10
huawei(config-if-gpon-0/5)#ont confirm 2 ontid 2 sn-auth 32303131B39FD642 snmp ont-lineprofile-
id 10

```

## Step 7 (Optional) Bind an alarm threshold profile.

In this example, the default profile (profile 1) is used.

```

huawei(config-if-gpon-0/5)#ont alarm-profile 2 1 profile-id 1
huawei(config-if-gpon-0/5)#ont alarm-profile 2 2 profile-id 1

```

## Step 8 Confirm that the ONU goes online successfully.

In this step, query the status of ONU 1 as an example. The same method is applied for querying the status of ONU 2.

After adding an ONU, run the **display ont info** command to query the ONU status. Ensure that the values of **Control flag**, **Run State**, **Config state**, and **Match state** of the ONU are **active**, **online**, **normal**, and **match** respectively.

```

huawei(config-if-gpon-0/5)#display ont info 2 1
-----
F/S/P           : 0/5/2
ONT-ID          : 0
Control flag    : active    // The ONU is activated.
Run state       : online   // The ONU is online.

```

Config state : **normal** // The ONU configuration status is normal.  
 Match state : **match** // The capability profile bound to the ONU is consistent with the actual capability of the ONU.  
 ...//The rest of the response information is omitted.

#### NOTE

When the ONU configuration status indicates a failure, the ONU cannot go online, or an ONU match fails, you are advised to check the ONU status with reference to the preceding description.

- If **Control flag** is **deactive**, run the **ont activate** command in GPON mode to activate the ONU.
- If the ONU fails to go online, that is, **Run state** is **offline**, the physical line may be disconnected or an optical module may be damaged. In this case, check the materials and lines.
- If the ONU configuration status indicates a failure, that is, **Config state** is **failed**, run the **display ont failed-configuration** command in diagnose mode to check the configuration failure items and causes and modify the configurations according to the actual situation.
- If the ONU does not match, that is, **Match state** is **mismatch**, run the **display ont capability** command to query the actual capability of the ONU, and choose one of the following methods to modify the ONU configurations.
- Create an appropriate ONU profile according to the actual capability of the ONU, and run the **ont modify** command to modify the ONU configuration data.
- Modify the ONU profile according to the actual capability of the ONU and save the modification. The ONU automatically restores the configurations.

### Step 9 Configure a management channel from the OLT to an ONU.

#### 1. Configure the inband management VLAN and IP address of the OLT.

To log in to an ONU in Telnet mode from the OLT and then configure the ONU, you must configure the inband management VLANs and IP addresses of the OLT and the ONU on the OLT.

Set the management VLAN to 500, add the upstream port to VLAN 500, and set the inband management IP address to 10.175.192.36/22.

```
huawei(config)#vlan 500 smart
huawei(config)#port vlan 500 0/19 0
huawei(config)#interface vlanif 500
huawei(config-if-vlanif500)#ip address 10.175.192.36 22
huawei(config-if-vlanif500)#quit
```

#### 2. Configure the inband management VLAN and IP address of the ONU.

Set the static IP address of ONU 1 to 10.175.192.51/22, the static IP address of ONU 2 to 10.175.192.52/22, the gateway to 10.175.192.1, and the management VLAN to 500 (same as the management VLAN of the OLT).

```
huawei(config-if-gpon-0/5)#ont ipconfig 2 1 static ip-address 10.175.192.51 mask 255.255.252.0
gateway 10.175.192.1 vlan 500
huawei(config-if-gpon-0/5)#ont ipconfig 2 2 static ip-address 10.175.192.52 mask 255.255.252.0
gateway 10.175.192.1 vlan 500
huawei(config-if-gpon-0/5)#quit
```

### 3. Configure an inband management traffic stream.

Set the ID of the traffic stream for managing ONU 1 to 1, the ID of the traffic stream for managing ONU 2 to 2, the management VLAN ID to 500, the GEM port ID to 0, and the C-VLAN ID to 500. The OLT does not limit the rate of an inband management traffic stream. Therefore, use the default traffic profile (profile 6). If you need to limit the rate of a service port, run the **traffic table ip** command to create a traffic profile and then bind the traffic profile to the service port.

```
huawei(config)#service-port 1 vlan 500 gpon 0/5/2 ont 1 gemport 0 multi-service user-vlan 500 rx-
cttr 6 tx-cttr 6
huawei(config)#service-port 2 vlan 500 gpon 0/5/2 ont 2 gemport 0 multi-service user-vlan 500 rx-
cttr 6 tx-cttr 6
```

### Step 10 Check whether the management channel between the OLT and the ONU is available.

You can run the **ping 10.175.192.51** and **ping 10.175.192.52** commands on the OLT to verify the connectivity between the OLT and ONU 1 and between the OLT and ONU 2 respectively. It is expected that the OLT can receive an ICMP ECHO-REPLY response packet from each ONU.

You can run the **telnet 10.175.192.51** and **telnet 10.175.192.52** commands to log in to the ONU 1 and ONU 2 respectively to configure the ONUs.

### Step 11 Create a traffic stream.

Create traffic stream 101 connected to ONU 1 and traffic stream 102 connected to ONU 2. Set the SVLAN to 100, GEM port ID to 1, and C-VLAN to 101. The upstream traffic and downstream traffic are limited on an ONU but not on the OLT. Therefore, use traffic profile 6 (default). To limit the rate of a service port, run the **traffic table ip** command to add a traffic profile and bind it to the service port.

```
huawei(config)#service-port 101 vlan 2000 gpon 0/5/2 ont 1 gemport 0 multi-service user-vlan 101
rx-cttr 6 tx-cttr 6
huawei(config)#service-port 102 vlan 2000 gpon 0/5/2 ont 2 gemport 0 multi-service user-vlan 101
rx-cttr 6 tx-cttr 6
```



#### NOTE

**The C-VLAN must be the same as the upstream VLAN of an ONU.**

- Run the **service-port** command to create traffic streams one by one. The following uses the creation of one traffic stream as an example. Pay attention to the mapping between a GEM port created when an ONU line profile is configured and the C-VLAN.

- You can also run the **multi-service-port** command to create traffic streams in batches. In the GPON access mode, you need to specify **ontid** and **gemindex** to specify a traffic stream. In addition, the number of GEM ports must be the same as the number of C-VLANs.

### Step 12 Configure queue scheduling.

Use the 3PQ+5WRR queue scheduling. Queues 0–4 adopt the WRR mode, with the weights of 10, 10, 20, 20, and 40 respectively; queues 5–7 adopt the PQ mode. Configure the priority of the Internet access service to 1 and adopt the WRR mode.

Queue scheduling is a global configuration. You need to configure queue scheduling only once on the OLT, and then the configuration takes effect globally. In the subsequent phases, you do not need to configure queue scheduling repeatedly when configuring other services.

For the service board that supports only four queues, the mapping between 802.1p priorities and queue IDs is as follows: priorities 0 and 1 map queue 1; priorities 2 and 3 map queue 2; priorities 4 and 5 map queue 3; priorities 6 and 7 map queue 4.

```
huawei(config)#queue-scheduler wrr 10 10 20 20 40 0 0 0
```

Configure the mapping between queues and 802.1p priorities. Priorities 0–7 is mapped into queues 0–7 respectively.

```
huawei(config)#cos-queue-map cos0 0 cos1 1 cos2 2 cos3 3 cos4 4 cos5 5 cos6 6 cos7 7
```

### Step 13 Save data.

```
huawei(config)#save
```

## 5.4.3 Configuration Procedure (on the ONU Side)

### Step 1 Log in to an ONU.

On the OLT, visit the management IP address of the ONU in Telnet mode and log in to the ONU. The user name is **root**, and the password is **mduadmin**.

```
huawei(config)#telnet 10.175.192.51
{ <cr>|service-port<U><0,4294967295> }:  
Command:  
telnet 10.175.192.51  
Press CTRL_] to quit telnet mode  
Trying 10.175.192.51 ...  
Connected to 10.175.192.51 ...  
>>User name:root  
>>User password:
```

### Step 2 Configure a traffic profile.

You can run the **display traffic table ip** command to query information about the traffic profiles existing in the system. If the existing traffic profile in the system does not meet the requirement, run the **traffic table ip** command to add a traffic profile.

Configure traffic profile 8, and set the CIR to 4 Mbit/s.

```
huawei(config)#traffic table ip index 8 cir 4096 priority 1 priority-policy tag-In-Packag
```

**Step 3** Configure an SVLAN and add an upstream port to the VLAN.

Configure the SVLAN. The VLAN ID ranges from 101 to 124. Add upstream port 0/0/1.

(If the ONU is the MA5603T, the corresponding upstream port is 0/8/0.)

```
huawei(config)#vlan 101-124 smart
huawei(config)#port vlan 101-124 0/0 1
```

**Step 4** Configure a VDSL2 line profile.

 **NOTE**

The data in a VDSL2 line profile must be configured according to the actual line conditions.

- You can run the **display vdsl line-profile** command to query a configured VDSL2 line profile in the system. Directly use a profile if it meets the requirement.
- If no VDSL2 line profile in the system meets the requirements, create a VDSL2 line profile. Run the **vdsl line-profile quickadd** command to quickly add a VDSL2 line profile.

Quickly add VDSL2 line profile 4 and use default parameter settings.

```
huawei(config)#vdsl line-profile quickadd 4
```

**Step 5** Configure a VDSL2 channel profile.

 **NOTE**

- You can run the **display vdsl channel-profile** command to query a configured VDSL2 channel profile in the system. Directly use a profile if it meets the requirement.
- If no VDSL2 channel profile in the system meets the requirement, create a VDSL2 channel profile by running the **vdsl channel-profile quickadd** command.

The data in the VDSL2 channel profile must be configured according to the actual channel conditions. In this example, the traffic profile is used to limit the subscriber access rate; therefore, when the VDSL2 channel profile is configured, the line rate parameters do not need to be configured.

Quickly add VDSL2 channel profile 4 and use default parameter settings.

```
huawei(config)#vdsl channel-profile quickadd 4
```

**Step 6** Configure a VDSL2 line profile.

 **NOTE**

- You can run the **display vdsl line-template** command to query a configured VDSL2 line profile in the system. Directly use a profile if it meets the requirement.
- If no VDSL2 line profile in the system meets the requirements, create a VDSL2 line profile by running the **vdsl line-template quickadd** command.

Bind the VDSL2 line profile configured in step 4 and the VDSL2 channel profile configured in step 5. Set the downstream rate adaptation ratio to 100% and the upstream rate adaptation ratio to 100%.

```
huawei(config)#vdsl line-template quickadd 3 line 4 channel1 4 100 100
```

**Step 7** Activate VDSL2 port 0/1/0 and bind it to line profile 3.

Activate other ports using the following method. To activate all ports in batches on a board, run the **activate all** command.

```
huawei(config)#interface vdsl 0/1
huawei(config-if-vdsl-0/1)#deactivate 0
huawei(config-if-vdsl-0/1)#activate 0 template-index 3
```

**Step 8** (Optional) Bind a VDSL2 alarm profile.

In this example, bind the default VDSL2 alarm profile (profile 1). To meet actual requirements, run the **vdsl alarm-template add** command to add a VDSL2 alarm profile.

```
huawei(config-if-vdsl-0/1)#alarm-config 0 1
```

**Step 9** Configure a service port.

The following uses service port 101 as an example. Set the SVLAN ID to 101, the mode to PTM, and the C-VLAN to untagged. You can also add other traffic streams by changing the C-VLAN ID and VDSL2 port ID.

The SVLAN ID must be the same as the C-VLAN ID configured on the OLT.

```
huawei(config)#service-port 101 vlan 101 vdsl mode ptm 0/1/0 multi-service user-vlan untagged rx-
cttr 8 tx-cttr 8
```

In the case of batch service provisioning, run the **multi-service-port** command.

**Step 10** Save data.

```
huawei(config)#save
```

**Step 11** Verify the configuration results.

On a PC, you can use the high-speed Internet access service in PPPoE dial-up mode.

## 5.5 Configuration Reference

### 5.5.1 Configurations on the OLT Side

```
#
vlan 2000 smart
vlan attrib 2000 q-in-q
port vlan 2000 0/19 0
#
dba-profile add profile-name PPPoE type4 max 102400
#
ont-lineprofile gpon profile-id 10
tcont 4 dba-profile-name PPPoE
gem add 0 eth tcont 4
gem add 1 eth tcont 4
mapping-mode vlan
gem mapping 0 0 vlan 500
gem mapping 1 1 vlan 101
commit
quit
#
interface gpon 0/5
port 2 ont-auto-find enable
display ont autofind 2
ont confirm 2 ontid 1 sn-auth 32303131B39FD641 snmp ont-lineprofile-id 10
ont confirm 2 ontid 2 sn-auth 32303131B39FD642 snmp ont-lineprofile-id 10
#
ont alarm-profile 2 1 profile-id 1
ont alarm-profile 2 2 profile-id 1
display ont info 2 1
#
vlan 500 smart
port vlan 500 0/19 0
interface vlanif 500
ip address 10.175.192.36 22
quit
#
interface gpon 0/5
ont ipconfig 2 1 static ip-address 10.175.192.51 mask 255.255.252.0 gateway 10.175.192.1 vlan 500
ont ipconfig 2 2 static ip-address 10.175.192.52 mask 255.255.252.0 gateway 10.175.192.1 vlan 500
#
service-port 1 vlan 500 gpon 0/5/2 ont 1 gemport 0 multi-service user-vlan 500 rx-cttr 6 tx-cttr 6
service-port 2 vlan 500 gpon 0/5/2 ont 2 gemport 0 multi-service user-vlan 500 rx-cttr 6 tx-cttr 6
service-port 101 vlan 2000 gpon 0/5/2 ont 1 gemport 0 multi-service user-vlan 101 rx-cttr 6 tx-cttr 6
service-port 102 vlan 2000 gpon 0/5/2 ont 2 gemport 0 multi-service user-vlan 101 rx-cttr 6 tx-cttr 6
#
queue-scheduler wrr 10 10 20 20 40 0 0 0
cos-queue-map cos0 0 cos1 1 cos2 2 cos3 3 cos4 4 cos5 5 cos6 6 cos7 7
save
#
telnet 10.175.192.51
root
```

mduadmin

## 5.5.2 Configurations on the ONU side

```
#
traffic table ip index 8 cir 4096 priority 0 priority-policy tag-In-Packag
#
vlan 101-124 smart
port vlan 101-124 0/0 1
#
vdsl line-profile quickadd 4
vdsl channel-profile quickadd 4
vdsl line-template quickadd 3 line 4 channel1 4 100 100
#
interface vdsl 0/1
deactivate 0
activate 0 template-index 3
alarm-config 0 1
#
service-port 101 vlan 101 vdsl mode ptm 0/1/0 multi-service user-vlan untagged rx-cttr 8 tx-cttr 8
save
```

# 6 Vectoring Configuration

## 6.1 Experiment Introduction

### 6.1.1 About This Experiment

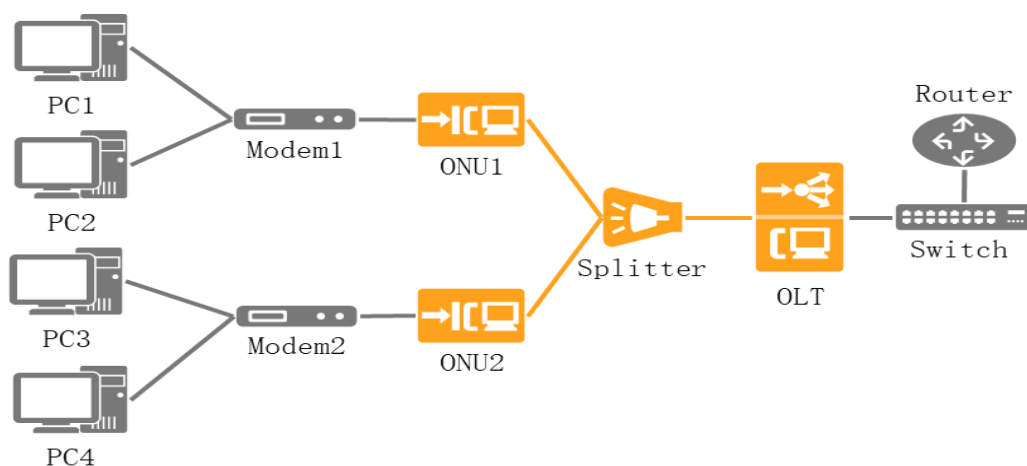
A DSLAM is connected to a remote ONU through a GPON port to provide the high-speed Internet access service. ONUs that support ADSL2+ access include the MA5616 and MA5603T. This chapter uses the MA5616 and OLT using a GPBC board as an example. The MA5616 can leverage the wide coverage, flexible networking, and low cost features of an EPON/GPON, interwork with an OLT to provide high-bandwidth service access, and expand the access capacity of an OLT. This chapter describes how to configure the Vectoring feature in the FTTx network scenario. On a fiber to the building (FTTB) or fiber to the curb (FTTC) network, an ONU has the vectoring function enabled and provides the Point-to-Point Protocol over Ethernet (PPPoE) Internet access service for G.fast users. This topic describes how to configure Internet access service on such an ONU.

### 6.1.2 Objectives

- Understand the vectoring principles.
- Understand the functions of vectoring.
- Master vectoring feature configuration.

### 6.1.3 Networking

Figure 6-1 Networking diagram of the Vectoring



## 6.1.4 Data Plan

Before the configuration, properly plan the data to avoid data conflicts during the configuration. Perform the experiment based on the data plan. Collect the information listed in the following table based on the actual situation of devices in the equipment room.

**Table 6-1 Table 1 Key parameters of other configurations**

| Parameter                                           | Data |
|-----------------------------------------------------|------|
| Dialup mode for Internet access                     |      |
| Anti-theft and roaming of user account through PITP |      |
| VDSL2 mode                                          |      |
| VLAN                                                |      |
| Service port index                                  |      |

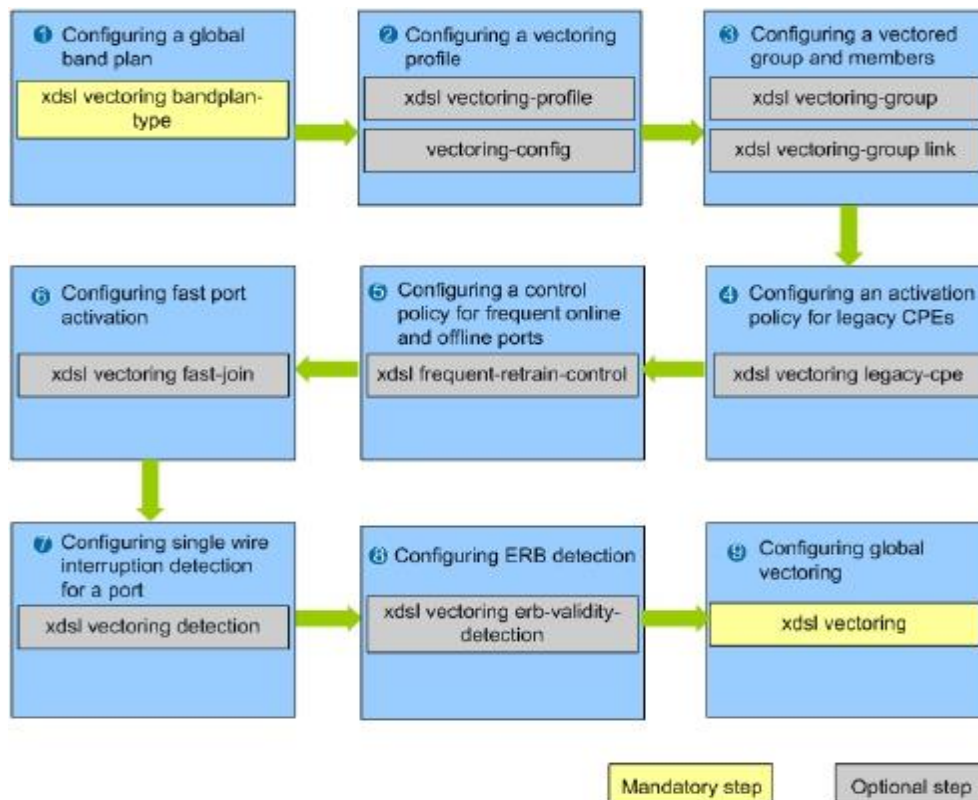
**Table 6-2 Table 2 Key parameters of vectoring configurations**

| Parameter                                                   | Data |
|-------------------------------------------------------------|------|
| Global vectoring configurations                             |      |
| Global band plan                                            |      |
| Policy for activating a CPE                                 |      |
| Policy for controlling frequent online and offline on ports |      |
| Vectoring group                                             |      |

## 6.2 Experiment Task Configuration

### 6.2.1 Service Configuration Process

Figure 6-2 Service Configuration Process



### 6.2.2 Configuration Procedure

#### 6.2.2.1 Basic Service Configuration on an OLT and ONU

1. Add an ONU to a GPON port.
2. Configure the OLT service VLAN, upstream port, and downstream port.
3. Configure the HSI service

#### 6.2.2.2 Vectoring Configuration on an ONU

Step 1 Create a service VLAN (SVLAN) and add an upstream port to the SVLAN.

In global config mode Create Smart SVLAN 50 and add upstream port 0/0 to SVLAN 50.

Step 2 Configure a VDSL2 access mode.

In global config mode, Configure a VDSL2 profile. In interface config mode, activate the VDSL2 port.

### Step 3 Create a service port

In global config mode, Run the service-port command to create a service port on user port.

### Step 4 Configure a security mode for the user account.

In global config mode, The Policy Information Transfer Protocol (PITP) P mode can be

used to protect user accounts against theft and roaming. The relay agent info option

(RAIO) mode can be customized based on site requirements. This procedure uses the

common mode as an example.

### Step 5 Configure the vectoring function.

## 6.2.2.3 Configuration Reference

The following configuration takes the MA5818 connected to GPON port 0/3/0 as an example. XDSL port ID on the ONT is 0/1/1, service VLAN ID is 50.

### Step 1 Create a service VLAN (SVLAN) and add an upstream port to the SVLAN..

```
Huawei(config)#vlan 50 smart
Huawei(config)#port vlan 50 0/0 0
```

### Step 2 Configure a VDSL2 profile..

```
Huawei(config)#vdsl line-profile quickadd 3 snr 60 0 300 60 0 300
Huawei(config)#vdsl channel-profile quickadd 3 path-mode ptm interleaved-delay 8 2 inp 4 2
Huawei(config)#vdsl line-template quickadd 3 line 3 channel1 3 100 100
```

### Step 3 Activate VDSL2 port.

```
Huawei(config)#interface vdsl 0/1
Huawei(config-if-vdsl-0/1)#deactivate 1
Huawei(config-if-vdsl-0/1)#activate 1 template-index 3
Huawei(config-if-vdsl-0/1)#alarm-config 1 1
Huawei(config-if-vdsl-0/1)#quit
```

### Step 4 Check the traffic table.

```
Huawei(config)#display traffic table ip from-index 0
{ <cr>|to-index<K> }:
```

Command:

```
display traffic table ip from-index 0
```

-----

| TID | CIR<br>(kbps) | CBS<br>(bytes) | PIR<br>(kbps) | PBS<br>(bytes) | Pri | Copy-policy | Pri-Policy |
|-----|---------------|----------------|---------------|----------------|-----|-------------|------------|
| 0   | 1024          | 34768          | 2048          | 69536          | 6   | -           | tag-pri    |
| 1   | 2496          | 81872          | 4992          | 163744         | 6   | -           | tag-pri    |
| 2   | 512           | 18384          | 1024          | 36768          | 0   | -           | tag-pri    |
| 3   | 576           | 20432          | 1152          | 40864          | 2   | -           | tag-pri    |
| 4   | 64            | 4048           | 128           | 8096           | 4   | -           | tag-pri    |
| 5   | 2048          | 67536          | 4096          | 135072         | 0   | -           | tag-pri    |
| 6   | off           | off            | off           | off            | 0   | -           | tag-pri    |

**Step 5** Create a service port.

```
Huawei(config)#service-port 3 vlan 50 vdsl mode ptm 0/1/1 multi-service user-vlan untagged
inbound traffic-table index 6 outbound traffic-table index 6
```

```
Huawei(config)#service-port desc 3 description Vlanid:50/vdsl
```

**Step 6** Configure a security mode for the user account.

```
Huawei(config)#pitp enable pmode
```

```
Huawei(config)#raio-mode common pitp-pmode
```

**Step 1** Configure a global bandplan. This example sets it to the default values.

**Step 2** Use the default vectoring group (group 1) to cancel the crosstalk on all frequency bands.

```
Huawei(config)#display xdsl vectoring-group 1
```

```
-----
Vectoring group index : 1
Lines in a vectoring group:
0/1:0-x
FEXT cancellation not required frequency bands downstream: 33,100-700,1216-1961
Vectoring lines protection switch downstream      : Enable
FEXT cancellation not required frequency bands upstream : -
Vectoring lines protection switch upstream        : Disable
-----
```

**Step 3** Run the display xdsl vectoring-profile command to query the default vectoring profile (profile 1).

```
Huawei(config)#display xdsl vectoring-profile 1
```

```
-----
Profile name : DEFVAL
FEXT cancellation control upstream      : Enable
FEXT cancellation control downstream    : Enable
.....
-----
```

**Step 4** Configure the vectoring legacy CPE activation policy to no-limit in consideration that the vectoring function is currently in the beginning phase of applications.

```
Huawei(config)#xds1 vectoring legacy-cpe activate-policy no-limit
```

Step 5 Enable the global vectoring function.

```
Huawei(config)#xds1 vectoring enable
```

## 6.3 Result Verification

Step 1 Log in to the network rate test website to test the rate. It is found that the upstream and downstream rates are 90% higher than the rates when the vectoring function is not enabled on the device.

## 6.4 Quiz

How to check the xds1 vectoring profile?

# 7 G.fast Feature Configuration

## 7.1 Experiment Introduction

### 7.1.1 About This Experiment

This chapter describes the detailed configuration methods and procedures of G.fast configuration.

### 7.1.2 Objectives

- Understand the G.fast principles.
- Understand the functions of G.fast.
- Master G.fast feature configuration.

### 7.1.3 Networking

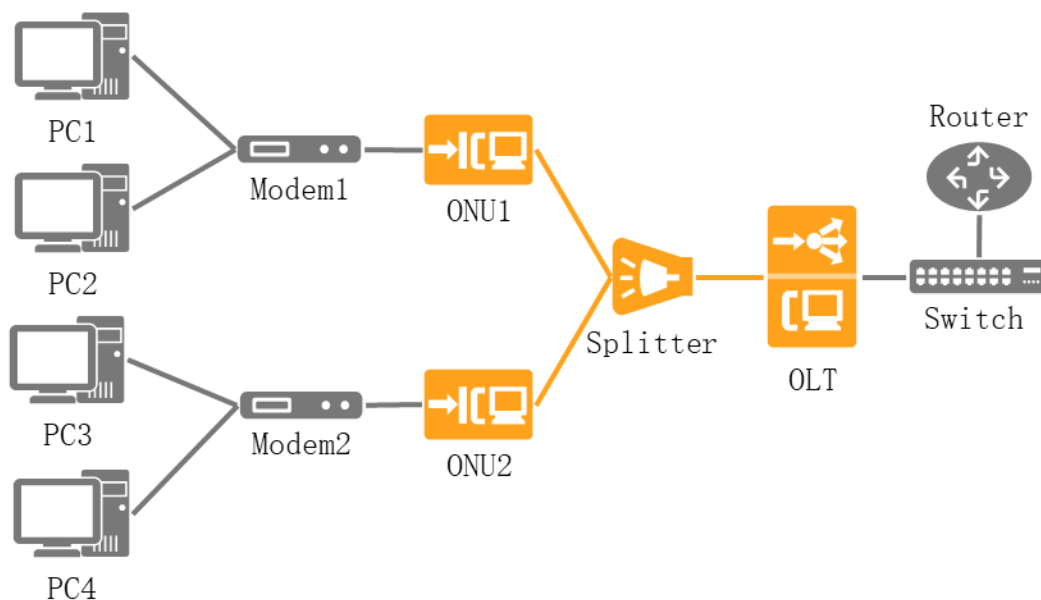


Figure 7-1 Networking diagram of the G.fast

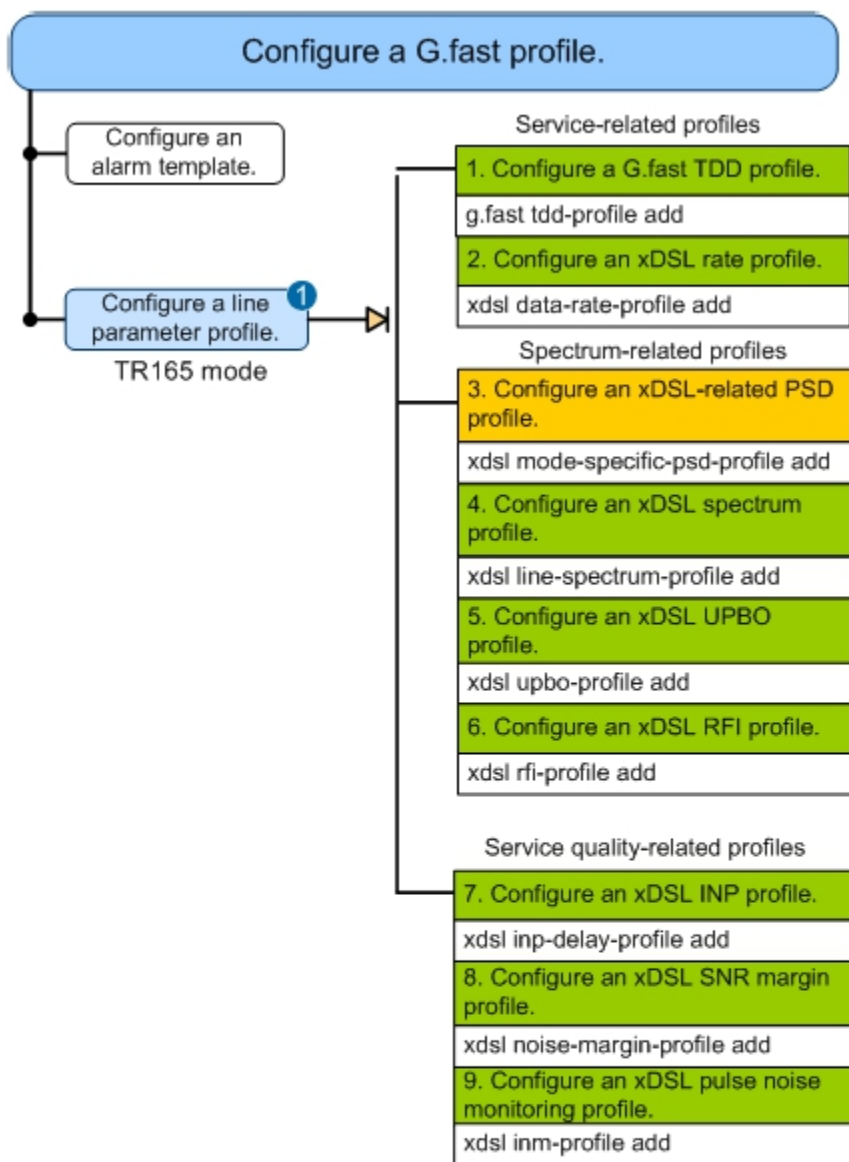
## 7.1.4 Data Plan

Before the configuration, properly plan the data to avoid data conflicts during the configuration. Perform the experiment based on the data plan. Collect the information listed in the following table based on the actual situation of devices in the equipment room.

| Parameter             | Value |
|-----------------------|-------|
| G.fast mode           |       |
| Traffic profile       |       |
| G.fast line profile   |       |
| G.fast alarm template |       |

## 7.2 Experiment Task Configuration

### 7.2.1 Service Configuration Process



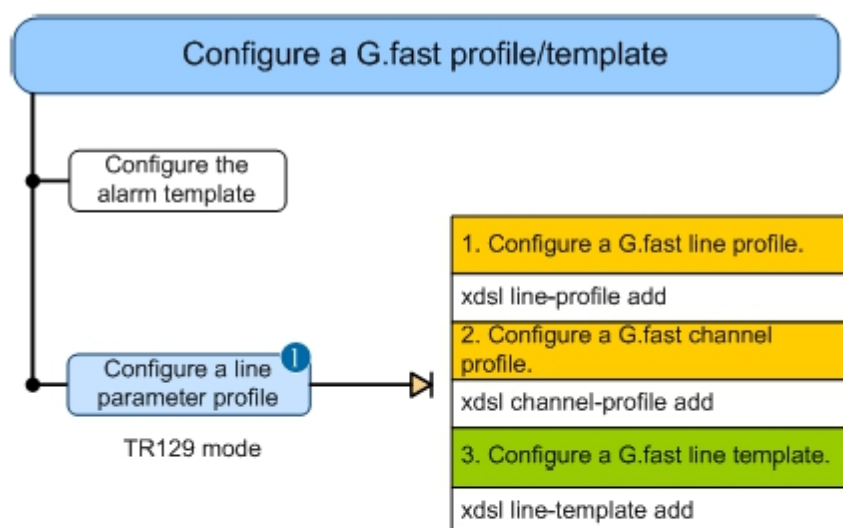
<sup>1</sup> Note: Each G.fast line parameter profile has a default profile, which is numbered 1. If the default profile can meet service requirements, use the default one.

  The profile cannot be bound to a G.fast port.

  The profile can be bound to a G.fast port.

Steps marked in the same color do not need to be performed in a certain sequence. Numbers before the steps do not indicate the sequence.  
Steps marked in   must be performed prior to steps marked in  .

Figure 7-2 Flowchart for configuring a G.fast line parameter profile in TR165 mode



**1** Note: Each line profile/template type has a default profile, which is numbered 1. If the default profile/template can meet the actual requirements, use the default one.

- The profile cannot be directly bound to a G.fast port.
  - The profile/template can be directly bound to a G.fast port.
- Steps marked in the same color do not need to be performed in a certain sequence. Numbers before the steps do not indicate the sequence. Steps marked in   must be performed prior to steps marked in  .

**Figure 7-3 Flowchart for configuring a G.fast line parameter profile in TR129 mode**

## 7.2.2 Configuration Procedure

### 7.2.2.1 Basic Service Configuration on an OLT and ONU

1. Add an ONU to a GPON port.
2. Configure the OLT service VLAN, upstream port, and downstream port.
3. Configure the HSI service

### 7.2.2.2 G.fast feature Configuration on an ONU

1. Configuring G.fast Profiles.
2. Configuring a G.fast Alarm Template.
3. Configuring a G.fast Line Parameter Profile.
4. Configuring G.fast User Ports

### 7.2.2.3 Configuration Reference

7.2.2.3.1 The following configuration takes the MA5818 connected to GPON port 0/3/0 as an example. XDSL port ID on the ONT is 0/1/0, service VLAN ID is 1001.

**Step 1** Configure a traffic profile.

```
Huawei(config)#traffic table ip index 16 cir 122880 priority 0 priority-policy local-setting
```

**Step 1** Create S-VLANs.

```
Huawei(config)#vlan 1001-1016 smart
Huawei(config)#port vlan 1001-1016 0/0 1
```

**Step 2** Add a service port.

```
Huawei(config)#service-port 101 vlan 1001 g.fast 0/1/0 multi-service user-vlan untagged rx-cttr 16
tx-cttr 16
```

**Step 3** Configure a G.fast line profile.

```
Huawei(config)#g.fast tdd-profile quickadd
{ <cr>|name<K>|profile-index<U><2,4294967294>|symbol-parameter<K> } :3
{ <cr>|name<K>|symbol-parameter<K> } :symbol-parameter
{ symbol-num-per-tdd-frame<E><23,36> } :36 //The default G.9701 setting is used
as an example.
{ symbol-num-per-tdd-frame-ds<U><6,32> } :27 //The ratio of the upstream rate to
the downstream rate is 1:3.
{ <cr>|name<K>|operation-interval<K> } :operation-interval
{ ttr-ds<U><1,32> } :5 //This parameter can be configured but does not take
effect. Use the default value.
```

```
{ ttr-us<U><1,25> }:2          //This parameter can be configured but does not take
effect. Use the default value.
{ tbudget-ds<U><1,32> }:8      //This parameter can be configured but does not take
effect. Use the default value.
{ tbudget-us<U><1,25> }:2      //This parameter can be configured but does not take
effect. Use the default value.enable
```

#### Step 4 Configure a G.fast alarm template.

This example uses the default G.fast alarm template (ID: 1).

#### Step 5 Bind the line profile and alarm template and activate the G.fast port.

```
Huawei huawei(config)#g.fast tdd-profile-config 0 profile-index 3
Huawei(config)#interface g.fast 0/1
Huawei(config-if-g.fast-0/1)#deactivate 0
Huawei(config-if-g.fast-0/1)#activate 0 prof-idx ds-rate 9 us-rate 10 spectrum
2 inp-delay 5
Huawei(config-if-g.fast-0/1)#alarm-config 0 1
```

## 7.3 Result Verification

Step 1 Log in to the network rate test website to test the rate. It is found that the upstream and downstream rates are higher than the rates when the G.fast function is not enabled on the device.

## 7.4 Quiz

How to query existing traffic profiles?